

Ultra Efficient Dual Sources Energy Manager with Ratio/Constant Voltage Regulation, Regulated Buck Output and 5 V Charger

Features and Benefits

Dual sources inputs

- Boost operation optimized for each source.
- Conversion efficiency above 90 % on each source.
- Harvests from 120 mV after cold start.
- Simultaneous harvesting from both sources.
- Up to 135 mA current extracted from the harvester.

Maximum Power Point Tracking

- Both sources configurable to constant voltage mode or to open-circuit voltage ratio mode.
- Optimal harvesting from a multiple combinations of harvesters (PV cells, RF, vibration, pulsed sources...).

Cold start from 275 mV / 1.5 μ W input

- Startup at ultra-low power from each harvesting source input.

Configurable overdischarge & overcharge protection

- Supports various types of rechargeable storage elements (LiC, Li-ion, LiPo, Li-ceramic pouch...).

Regulated output for application circuit

- Buck regulator conversion efficiency up to 96 %.
- Selectable output voltage between 0.6 V and 2.5 V.
- Output current up to 100 mA.

Thermal monitoring

- Storage element protection against over-temperature and under-temperature during charging and discharging, independently.

Average Power Monitoring

- Provides data to determine how much energy has been transferred to the storage from each source and from the 5 V charger, as well as the energy drained from the storage to supply the application circuit.

System configuration by GPIO or I²C communication

- All settings dynamically configurable through GPIO or I²C (Fast Mode Plus).
- System data available through I²C.

Shipping mode

- Storage element charge and discharge disabling during shipment.

External 5 V charging capability

- Extra charging input for 5 V power supplies.
- Configurable CC and CV modes (max. 135 mA).
- Provides a fast charging alternative when no source is available for a long time.

Applications

Smart home	Industrial sensor
Smart building	Retail
Edge IoT	PC accessories

Description

The AEM13920 is a fully integrated and compact power management circuit that extracts DC power from two harvesting sources to store energy in a rechargeable storage element and supply an application circuit. A 5 V input can also be used to charge the storage element (e.g., if it gets depleted). This compact and ultra-efficient PMIC allows for extending battery lifetime and eliminating the primary energy storage in a large range of applications.

Both sources implement Maximum Power Point Tracking (MPPT) based on open-circuit voltage ratio as well as constant source voltage regulation features, allowing for harvesting the maximum power available from each source to charge the storage element.

With its unique cold-start circuit, it can start operating with an input voltage as low as 275 mV (min. 1.5 μ W power).

The configurable protection thresholds prevent overcharging and overdischarging the storage element. No external components are required to set those thresholds. The thermal monitoring prevents charging and discharging the storage element outside a configurable temperature range.

The Average Power Monitoring (APM) allows the application circuit to get an estimate of the energy harvested from each source to the battery and from the battery to the application circuit. A shipping mode is available to avoid charging and discharging the storage element during shipping or storage.

A buck regulator with selectable output voltage allows an application circuit to be supplied with high efficiency.

I²C communication allows users to control every setting of the AEM13920 from the application circuit MCU.

Device Information

Part Number	Package	Body size
10AEM13920J0001	QFN 40-pin	5 x 5 mm

Evaluation Board

Part number
2AAEM13920J002

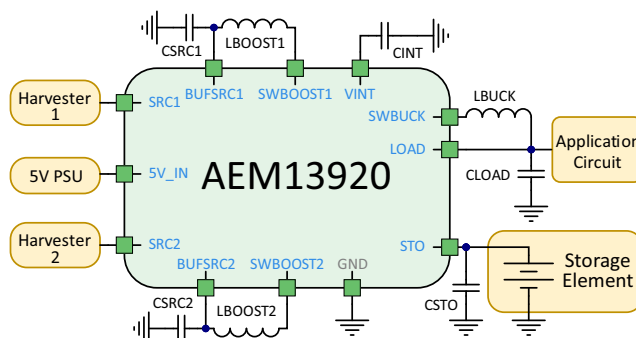


Table of Contents

1. Introduction	9
2. Pin Configuration and Functions	10
3. Specifications	13
3.1. Absolute Maximum Ratings	13
3.2. ESD Ratings	13
3.3. Thermal Resistance	13
3.4. Electrical Characteristics at 25 °C	14
3.5. Recommended Operating Conditions	17
3.5.1. External Inductors Information	17
3.5.2. External Capacitors Information	17
3.6. Typical Characteristics	18
3.6.1. Boost Converters Conversion Efficiency	18
3.6.2. Buck Converter Conversion Efficiency	19
4. Functional Block Diagram	20
5. Theory of Operation	21
5.1. Cold-Start Circuits	21
5.2. Boost Converters	21
5.2.1. Operation Principle	21
5.2.2. Maximum Power Point Tracking	22
5.2.2.1. Open-Circuit Voltage Ratio	22
5.2.2.2. ZMPP	22
5.2.3. Source Constant Voltage Regulation	23
5.2.4. Automatic High-Power Mode	23
5.2.5. Using Both Boost Converters in Parallel	23
5.3. Buck Converter	24
5.4. Thermal Monitoring	25
5.5. Average Power Monitoring	25
5.5.1. SRCx and LOAD APM	25
5.5.2. 5 V Charger APM	25
5.5.3. APM IRQ	26
5.5.4. APM Accumulator	26
5.6. IRQ Pin	26
5.7. 5 V Charger	26
5.7.1. CC Mode	26
5.7.2. CC/CV Mode	26
5.8. Shipping Mode	27
5.9. State Machine Description	28
5.9.1. RESET STATE	28
5.9.2. SENSE STO STATE	28
5.9.3. START STATE	29
5.9.4. SUPPLY STATE	29
5.9.5. OVDIS STATE	29
5.9.6. SLEEP STATE	29
6. System Configuration	31

6.1. Configuration Pins and I ² C	31
6.1.1. Configuration Pins	31
6.1.2. Configuration by I ² C	31
6.2. Source Constant Voltage Regulation	32
6.3. Maximum Power Point Tracking	33
6.4. Storage Element Protection Thresholds	34
6.5. Boost Converter Timings	35
6.6. Buck Converter	35
6.6.1. Load Voltage	35
6.6.2. Buck Converter Timings	35
6.7. Thermal Monitoring	35
6.8. 5 V Charger	36
6.9. Shipping Mode	36
7. I²C Serial Interface Protocol	37
8. Register Map	39
9. Registers Configuration	42
9.1. I ² C Control (CTRL)	42
9.2. Version Register (VERSION)	42
9.3. Source Regulation (SRCxREGUx)	43
9.3.1. SRCxREGU0	44
9.3.2. SRCxREGU1	44
9.3.3. Constant Voltage Configuration	45
9.3.4. MPPT Configuration	46
9.4. Source Low Threshold (SRCLOW)	47
9.5. Storage Element Threshold Voltages (VOVDIS / VCHRDY / VOVCH)	48
9.5.1. Overdischarge Voltage (VOVDIS)	48
9.5.2. Charge Ready Voltage (VCHRDY)	49
9.5.3. Overcharge Voltage (VOVCH)	50
9.6. Boost Converters (BSTxCFG)	51
9.7. Buck Converter (BUCKCFG)	52
9.8. 5 V Charger (CHG5V)	53
9.9. STO Charge Temperature Protection (TEMPCOLDCH and TEMPHOTCH)	54
9.9.1. TEMPCOLDCH	54
9.9.2. TEMPHOTCH	54
9.10. STO Discharge Temperature Protection (TEMPCOLDDIS and TEMPHOTDIS)	55
9.10.1. TEMPCOLDDIS	55
9.10.2. TEMPHOTDIS	55
9.11. Thermal Protection Monitoring (TEMPPROTECT)	56
9.12. Average Power Monitoring (APM)	57
9.13. APM Accumulator (APMACC)	57
9.14. IRQ Enable (IRQENx)	58
9.14.1. IRQEN0	58
9.14.2. IRQEN1	59
9.15. IRQ Flags (IRQFLGx)	60
9.15.1. IRQFLG0	60
9.15.2. IRQFLG1	61
9.16. Status (STATUSx)	63

9.16.1. STATUS0	63
9.16.2. STATUS1	64
9.17. APM Data Summary (APMxSRCx, APMxCHG5V and APMxLOAD)	65
9.17.1. APMxSRCx and APMxLOAD Summary	65
9.17.2. APMxCHG5V Summary	66
9.18. SRCx APM Data (APMxSRCx)	67
9.18.1. APM0SRCx	67
9.18.2. APM1SRCx	67
9.18.3. APM2SRCx	67
9.19. LOAD APM Data (APMxLOAD)	68
9.19.1. APM0LOAD	68
9.19.2. APM1LOAD	68
9.19.3. APM2LOAD	68
9.20. CHG5V APM Data (APMxCHG5V)	69
9.20.1. APM0CHG5V	69
9.20.2. APM1CHG5V	69
9.21. APM Error (APMERR)	70
9.22. Temperature Monitoring Data (TEMP)	71
9.23. Storage Element Voltage Data (STO)	71
9.24. Sources Voltage Data (SRCx)	72
9.25. Part Number (PNx)	73
10. Typical Application Circuits	74
10.1. Example Circuit 1	74
10.2. Example Circuit 2	76
10.3. Example Circuit 3	78
11. Circuit Behavior	81
11.1. Start Up from SRCx	81
11.1.1. Configuration	81
11.1.2. Observations	82
11.2. Shutdown	83
11.2.1. Configuration	83
11.2.2. Observations	84
12. Minimum BOM	85
13. Layout	86
13.1. Guidelines	86
13.2. Example	87
14. Package Information	88
14.1. Moisture Sensitivity Level	88
14.2. RoHS Compliance	88
14.3. REACH Compliance	88
14.4. Tape and Reel Dimensions	88
14.5. Package Dimensions	89
14.6. Board Layout	90
15. Glossary	91
16. Revision History	94

List of Figures

Figure 1: Simplified schematic view	9
Figure 2: Pinout diagram	10
Figure 3: SRCx to STO efficiency vs. V_{SRCx} with $L_{BOOSTx} = 33 \mu H$ (Coilcraft LPS4018-333MRB), BSTxCFG.TMULT = 0x02 (x3)	18
Figure 4: STO to LOAD efficiency vs. I_{LOAD} with $L_{BUCK} = 10 \mu H$ (TDK VLS252012CS-100M-1), BUCKCFG.TMULT = 0x01 (x2)	19
Figure 5: Functional block diagram	20
Figure 6: Simplified schematic view of the boost converters	21
Figure 7: MPPT evaluation behavior (with $T_{MPPT, WAIT}$ of 1.8 ms)	22
Figure 8: ZMPP connection with both boost converters used in parallel	23
Figure 9: Simplified schematic view of the buck converter	24
Figure 10: Average Power Monitoring in the power chain	25
Figure 11: State machine	28
Figure 12: TH_REF and TH_MON connections	35
Figure 13: I ² C transmission frame	37
Figure 14: Read and write transmission	38
Figure 15: Typical application circuit 1	74
Figure 16: Typical application circuit 2	76
Figure 17: Typical application circuit 3	78
Figure 18: Behavior at start up	82
Figure 19: Behavior at shutdown	84
Figure 20: Schematic	85
Figure 21: Layout example	87
Figure 22: Tape and reel dimensions	88
Figure 23: QFN 40-pin 5x5mm drawing (all dimensions in mm)	89
Figure 24: Recommended board layout for QFN40 package (all dimensions in mm)	90

List of Tables

Table 1: Pins description (part 1)	10
Table 2: Pins description (part 2)	11
Table 3: Pins description (part 3)	12
Table 4: Absolute maximum ratings	13
Table 5: ESD ratings	13
Table 6: Thermal data	13
Table 7: Electrical characteristics (part 1)	14
Table 8: Electrical characteristics (part 2)	15
Table 9: Electrical characteristics (part 3)	16
Table 10: Recommended external components	17
Table 11: Configuration of the source constant voltage regulation with SRCx_CFG[4:0] pins	32
Table 12: MPPT ratio configuration with SRCx_CFG[2:0] pins	33
Table 13: MPPT timing configuration with SRCx_CFG[4:3] pins	33
Table 14: Storage element thresholds configuration with STO_CFG[2:0] pins	34
Table 15: Default temperature thresholds depending on STO_CFG[2:0] configuration with the recommended RDIV and RTH	34
Table 16: Configuration of LOAD voltage with LOAD_CFG[2:0]	35
Table 17: Typical resistor values for setting 5 V charger max. current	36
Table 18: Register map	39
Table 19: CTRL register	42
Table 20: VERSION register	42
Table 21: Summary of SRCxREGUx register fields	43
Table 22: SRCxREGU0 register	44
Table 23: SRCxREGU1 register	44
Table 24: SRCx constant voltage values configured by SRCxREGUx (SRCxREGU0.MODE = 0)	45
Table 25: SRCx MPPT ratio/ZMPP configured by SRCxREGUx (SRCxREGU0.MODE = 1)	46
Table 26: SRCx MPPT wait time configured by SRCxREGUx (SRCxREGU0.MODE = 1)	46
Table 27: SRCx MPPT period configured by SRCxREGUx (SRCxREGU0.MODE = 1)	46
Table 28: SRCLOW register	47
Table 29: V_{SRCLOW} thresholds as configured by SRCLOW register	47
Table 30: VOVDIS register	48
Table 31: Storage element V_{OVDIS} configuration by VOVDIS register	48

Table 32: VCHRDY register	49
Table 33: Storage element V_{CHRDY} configuration by VCHRDY register	49
Table 34: VOVCH register	50
Table 35: Storage element V_{OVCH} configuration by VOVCH register	50
Table 36: BSTxCFG registers	51
Table 37: Timing multiplier configured by TMULT field with corresponding boost inductor value	51
Table 38: BUCKCFG register	52
Table 39: Buck inductor values according to buck timing	52
Table 40: V_{LOAD} settings by BUCKCFG.VLOAD register	52
Table 41: CHG5V register	53
Table 42: $V_{5V,STOP}$ configuration by CHG5V register	53
Table 43: TEMPCOLDCH register	54
Table 44: TEMPHOTCH register	54
Table 45: TEMPCOLDDIS register	55
Table 46: TEMPHOTDIS register	55
Table 47: TEMPPROTECT register	56
Table 48: APM register	57
Table 49: APMACC register	57
Table 50: IRQEN0 register	58
Table 51: IRQEN1 register	59
Table 52: IRQFLG0 register	60
Table 53: IRQFLG1 register	61
Table 54: STATUS0 register	63
Table 55: STATUS1 register	64
Table 56: Summary of APMxSRCx and APMxLOAD register fields	65
Table 57: Summary of APMxCHG5V register fields	66
Table 58: APM0SRCx registers	67
Table 59: APM1SRCx registers	67
Table 60: APM2SRCx registers	67
Table 61: APM0LOAD register	68
Table 62: APM1LOAD register	68
Table 63: APM2LOAD register	68
Table 64: APM0CHG5V register	69

Table 65: APM1CHG5V register	69
Table 66: APMERR register	70
Table 67: TEMP register	71
Table 68: STO register	71
Table 69: SRCx register	72
Table 70: Source voltage V_{SRCx} from SRCx.DATA register value (formula)	72
Table 71: Source voltage V_{SRCx} from SRCx.DATA register value (lookup table)	72
Table 72: PN0 register	73
Table 73: PN1 register	73
Table 74: PN2 register	73
Table 75: PN3 register	73
Table 76: PN4 register	73
Table 77: Summary of I ² C register configuration for typical application circuit 3	80
Table 78: Minimum BOM	85
Table 79: Moisture sensitivity level	88
Table 80: Revision history	94

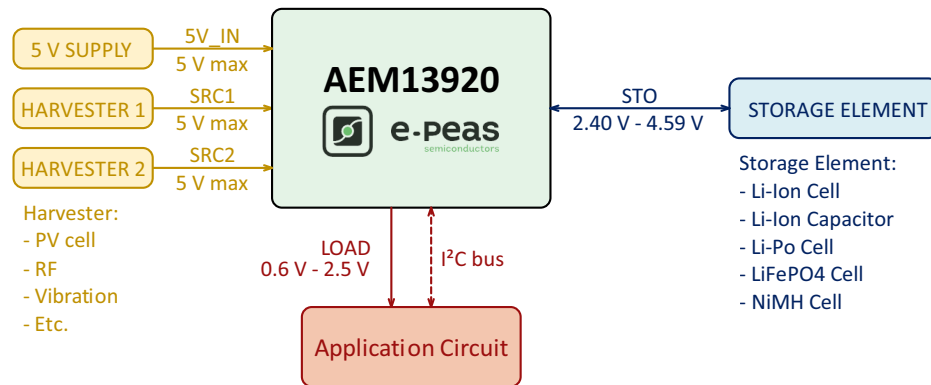


Figure 1: Simplified schematic view

1. Introduction

The AEM13920 is a full-featured energy efficient power management circuit able to harvest energy from two energy sources (connected to **SRC1** and/or **SRC2**) to charge a storage element (connected to **STO**) and to supply an application circuit (connected to **LOAD**). The storage element can also be charged from a 5 V power supply. This is done with a minimal bill of material.

The heart of the AEM13920 is composed of two switching boost converters for energy harvesting and a buck converter for supplying the load. All three have high power conversion efficiency.

The AEM13920 can be configured either by configuration pins or by a set of registers accessed through an I²C bus. Furthermore, some advanced configurations are accessible only through the I²C registers.

A 5 V power input **5V_IN** allows for charging the storage element. This is done using a CC/CV (constant current / constant voltage) method. The CC mode maximum current can be configured between 13.5 mA and 135 mA with an external resistor. The CV mode can be enabled by I²C with a configurable charge stop voltage (**V_{5V,STOP}**).

At first start-up, as soon as a required cold-start voltage of 275 mV and a sparse amount of power of at least 1.5 μW is available on **SRCx** (**SRC1** or **SRC2**), the AEM13920 coldstarts. After the cold start, the AEM13920 extracts the power available from the sources if the working input voltage is above **V_{SRCx,REG}** (constant voltage mode) or **V_{MPP}** (MPPT mode). Cold start can also be done from the 5 V power supply input **5V_IN**.

The storage element protection thresholds are configured through three configuration pins (**STO_CFG[2:0]**), from which the user can select a specific operating mode out of 8 modes that cover most application requirements without any dedicated external component. If none of those 8 modes fits the user's storage element, the protection thresholds can also be configured individually through I²C registers to allow the user to define a mode with custom specifications.

The **ST_STO** status pin provides information about the voltage level of the storage element, and thus, about its readiness to supply an application through the AEM13920 **LOAD** output.

Both **SRCx** inputs of the AEM13920 can work in Maximum Power Point tracking (MPPT) mode or in constant voltage mode. The mode is configured with a dedicated pin **SRCx_MODE** or through the I²C registers.

When in MPPT mode, the Maximum Power Point tracking (MPPT) ratio can be set through three configuration pins (**SRCx_CFG[2:0]**) to ensure an optimum biasing of the harvester and maximize power extraction. Depending on the harvester, it is possible to adapt the timings of the MPP evaluations with the two configuration pins (**SRCx_CFG[4:3]**) that sets the periodicity and the duration of the MPP evaluations. The MPPT ratio and the MPPT timings can also be configured through the I²C registers.

When in constant voltage mode, the source regulation voltage **V_{SRCx,REG}** can be configured thanks to five configuration pins (**SRCx_CFG[4:0]**). The constant voltage can also be configured through I²C registers for higher resolution and extended range of values.

If the storage element is sufficiently charged, the buck converter provides a regulated output voltage on the **LOAD** pin, allowing an application circuit to be supplied. The regulated voltage can be set through the **LOAD_CFG[2:0]** pins or through the I²C registers.

A shipping mode feature can be enabled through the **SHIP_MODE** pin, disabling the boost converters, the buck converter as well as the 5 V input, thus, preventing any charge or discharge of the storage element.

2. Pin Configuration and Functions

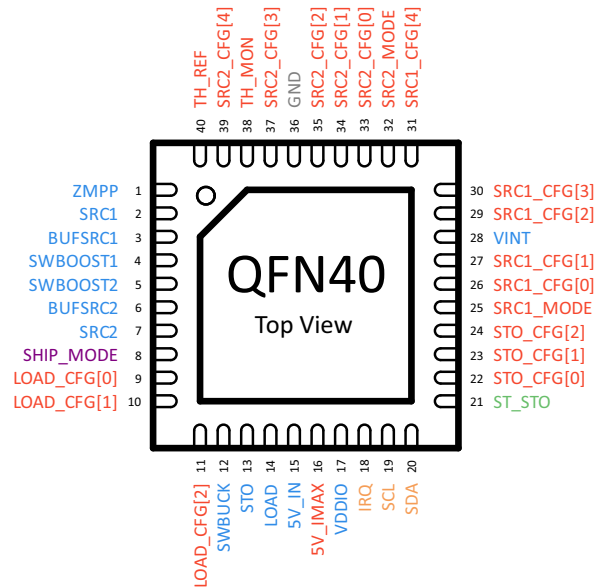


Figure 2: Pinout diagram

NAME	PIN NUMBER	FUNCTION
Power Pins		
SRC1	2	Connection to the energy source harvested by the boost converter #1 and #2 respectively. Connect to GND if not used (typically for single source use of AEM13920).
SRC2	7	
BUFSRC1	3	Connection to an external capacitor buffering the boost converter #1 and #2 inputs respectively. Connect to GND if not used (typically for single source use of AEM13920).
BUFSRC2	6	
SWBOOST1	4	Switching node of the boost converter #1 and #2 respectively. Leave floating if not used (typically for single source use of AEM13920).
SWBOOST2	5	
ZMPP	1	Connection for R_{ZMPP} (see Section 5.2.2.2). Leave floating if not used.
STO	13	Connection to the energy storage element (rechargeable storage element).
SWBUCK	12	Switching node of the buck converter. If not used: - Disable buck converter through LOAD_CFG[2:0] pins or BUCKCFG.VLOAD register field. - Leave the SWBUCK pin floating.
LOAD	14	Output voltage of the buck converter to supply an application circuit. If not used: - Disable buck converter through LOAD_CFG[2:0] pins or BUCKCFG.VLOAD register field. - Leave the LOAD pin floating.
5V_IN	15	Input of the 5 V DC power supply. Leave floating if not used.
VDDIO	17	Voltage reference for the I ² C interface, as well as for the IRQ and ST_STO pins. - If used, connect to a DC power supply. - If not used, connect to GND.
VINT	28	Connection for C_{INT} buffering capacitor. AEM13920 internal power supply (do not connect any external circuit on VINT).

Table 1: Pins description (part 1)



NAME		PIN NUMBER	LOGIC LEVEL		FUNCTION
			LOW	HIGH	
Configuration Pins					
SRC1_MODE		25	GND	VINT	Used to configure SRCx voltage regulation mode (see Section 5.2.1): - LOW: constant voltage mode. - HIGH: MPPT mode (ratio or ZMPP). Read as HIGH if left floating.
SRC2_MODE		32	GND	VINT	
SRC1_CFG[4:0]	SRC1_CFG[4]	31	GND	VINT	Used to configure SRCx regulation voltage. SRCx_MODE = LOW (constant voltage mode, see Section 6.2): - SRCx_CFG[4:0] are used to set SRCx constant regulation voltage. SRCx_MODE = HIGH (MPPT ratio mode, see Section 6.3): - SRCx_CFG[2:0] are used to set SRCx MPPT ratio. - SRCx_CFG[4:3] are used to set SRCx MPPT timings. Read as HIGH if left floating.
	SRC1_CFG[3]	30	GND	VINT	
	SRC1_CFG[2]	29	GND	VINT	
	SRC1_CFG[1]	27	GND	VINT	
	SRC1_CFG[0]	26	GND	VINT	
SRC2_CFG[4:0]	SRC2_CFG[4]	39	GND	VINT	
	SRC2_CFG[3]	37	GND	VINT	
	SRC2_CFG[2]	35	GND	VINT	
	SRC2_CFG[1]	34	GND	VINT	
	SRC2_CFG[0]	33	GND	VINT	
STO_CFG[2:0]	STO_CFG[2]	24	GND	VINT	Used to configure the storage element protection thresholds (see Section 6.4). Read as HIGH if left floating.
	STO_CFG[1]	23	GND	VINT	
	STO_CFG[0]	22	GND	VINT	
LOAD_CFG[2:0]	LOAD_CFG[2]	11	GND	VINT	Used to configure the LOAD output regulation voltage (see Section 6.6). Read as HIGH if left floating.
	LOAD_CFG[1]	10	GND	VINT	
	LOAD_CFG[0]	9	GND	VINT	
5V_IMAX		16	Analog Pin		Connection to an external resistor to set the charging current from the 5V_IN supply to STO (see Section 6.8). Leave floating if the 5V_IN power supply is not used.
TH_REF		40	Analog Pin		Reference voltage for thermal monitoring (see Section 5.4). Leave floating if not used.
TH_MON		38	Analog Pin		Connection for the mid-point of the thermistor voltage divider (see Section 5.4). Connect to VINT if not used.

Table 2: Pins description (part 2)



NAME	PIN NUMBER	LOGIC LEVEL		FUNCTION
		LOW	HIGH	
Control Pin				
SHIP_MODE	8	GND	STO	Used to configure the shipping mode. When HIGH: - Minimum consumption from the storage element. - Storage element charge is disabled (Boost converters are disabled). - Buck (LOAD) is disabled. - VINT is only supplied from SRC1 and SRC2 if energy is available. Read as LOW if left floating.
I ² C Pins				
SCL	19	GND	VDDIO	Unidirectional serial clock for I ² C communication. Connect a pull-up resistor to VDDIO if used (see Section 6.1.2). Connect to GND if not used.
SDA	20	GND	VDDIO	Bidirectional data line for I ² C communication. Connect a pull-up resistor to VDDIO if used (see Section 6.1.2). Connect to GND if not used.
IRQ	18	GND	VDDIO	Logic output signal to indicate AEM13920 events to an external circuit GPIO. Leave floating if not used.
Status Pin				
ST_STO	21	GND	VDDIO	Logic output. - HIGH when in SUPPLY STATE or in SLEEP STATE. - LOW otherwise. Leave floating if not used.
Other pins				
GND	Exposed Pad, 36			Ground connection, each terminal must be strongly tied to the PCB ground plane.

Table 3: Pins description (part 3)

3. Specifications

3.1. Absolute Maximum Ratings

Parameter		Min	Max	Unit
Operating junction temperature T_j		-40	85	°C
Storage temperature T_{stg}		-65	150	°C
Input voltage	SRCx, BUFSRCx, SWBOOSTx, ZMPP, STO, SWBUCK, 5V_IN, VDDIO, SHIP_MODE, 5V_IMAX, LOAD_CFG[2:0], SCL, SDA, IRQ, ST_STO.	-0.3	5.50	V
	VINT, LOAD, SRCx_MODE, SRCx_CFG[4:0], STO_CFG[2:0], TH_REF, TH_MON.	-0.3	2.75	V

Table 4: Absolute maximum ratings

3.2. ESD Ratings

Parameter		Value	Unit
Electrostatic discharge V_{ESD}	Human-Body Model (HBM) ¹	± 2000	V
	Charged-Device Model (CDM) ²	± 1000	V

Table 5: ESD ratings

- ESD Human-Body Model (HBM) value tested according to JEDEC standard JS-001-2024.
- ESD Charged-Device Model (CDM) value tested according to JEDEC standard JS-002-2022.

ESD CAUTION	
	ESD (ELECTROSTATIC DISCHARGE) SENSITIVE DEVICE These devices have limited built-in ESD protection and damage may thus occur on devices subjected to high-energy ESD. Therefore, proper ESD precautions should be taken to avoid performance degradation or loss of functionality

3.3. Thermal Resistance

Package	θ_{JA}	θ_{JC}	Unit
QFN-40	50	5	°C/W

Table 6: Thermal data



3.4. Electrical Characteristics at 25 °C

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
Power conversion						
$P_{SRCx,CS}$	Minimum source power required for cold start.			1.5		μW
$V_{SRCx,CS}$	Minimum source voltage required for cold start.			0.275		V
V_{MPP}	Target regulation voltage on $SRCx$ when extracting power.	$SRCx_MODE = HIGH$.	0.12		V_{STO} ¹	V
$V_{SRCx,REG}$	Target regulation voltage of the source, depending on GPIO or I ² C configuration.	$SRCx_MODE = LOW$, configured by $SRCx_CFG[4:0]$.	0.14		2.10	V
		$SRCx_MODE = LOW$, configured by I ² C register.	0.120		min (4.455, V_{STO}) ²	V
V_{OC}	Open-circuit voltage of the source.		0.00 ³		5	V
V_{5V_IN}	Voltage on the $5V_IN$ pin to allow for charging the storage element.		max (3.60, $V_{STO} + 0.20$) ⁴		5.50	V
$I_{5V,CC}$	Maximum charging current of the 5 V charger ($5V_IN$ input). Configured by the R_{5V_IMAX} resistor.	In constant current (CC) mode when $V_{STO} > V_{OVDIS}$.	13.50		135	mA
$I_{5V,OVDIS,CC}$	Charging current of the 5 V charger ($5V_IN$ input) in overdischarge.	In constant current (CC) mode when $V_{STO} < V_{OVDIS}$.		6.75		mA
V_{VDDIO}	Voltage on $VDDIO$.		1.50		5.00	V

Table 7: Electrical characteristics (part 1)

- V_{MPP} may exceed V_{STO} without causing damage to the AEM13920. However, doing so will reduce the boost converters efficiency.
- The maximum value of $V_{SRCx,REG}$ is determined by the highest configurable $V_{SRCx,REG}$, but it must never be higher than V_{STO} to ensure proper operation.
- When the open-circuit voltage is below the source regulation voltage (MPPT or constant voltage), the AEM13920 does not extract power from the source. Voltages down to GND voltage does not damage the AEM13920 though.
- The 5 V charger is considered connected when the voltage on $5V_IN$ is greater than or equal to 3.60 V and at least 200 mV higher than the voltage on STO . It can be actively charging or not.



Symbol	Parameter	Conditions	Min	Typ	Max	Unit
Timing						
$T_{MPPT, WAIT}$	Wait time before V_{OC} measurement begins during MPP evaluation (see Section 5.2.2).	Configured by SRCx_CFG[4:3] .	1.8		233	ms
		Configured by I ² C.			465	
$T_{MPPT, MEASURE}$	Duration of V_{OC} measurement during MPP evaluation.			1.36		ms
$T_{MPPT, PERIOD}$	Time between two MPP evaluations (see Section 5.2.2).	Configured by SRCx_CFG[4:3] .	0.116		14.895	s
		Configured by I ² C.				
$T_{CRIT, ST}$	Delay for the AEM13920 to notify the application about an overdischarged storage element, a temperature out of discharge temperature range, and to schedule LOAD output disable (see Section 5.3).			1.86		s
T_{CRIT}	Delay for the AEM13920 to go in OVDIS STATE and to disable the LOAD output (see Section 5.9).			2.56		s
$T_{GPIO, MON}$	GPIO monitoring rate.			1.86		s
$T_{STO, MON}$	Storage element voltage monitoring rate.	When the buck converter is disabled and the 5 V charger is not connected ¹ .		116		ms
		When the buck converter is enabled or the 5 V charger is connected ¹ .		15		ms
$T_{TEMP, MON}$	Temperature monitoring rate.			7.45		s
$T_{5V, RISE}$	Minimum rise time from 0 V to 5 V on the 5V_IN pin (see Section 6.8).			50		μs

Table 8: Electrical characteristics (part 2)

1. The 5 V charger is considered connected when the voltage on **5V_IN** is greater than or equal to 3.60 V and at least 200 mV higher than the voltage on **STO**. It can be actively charging or not.



Symbol	Parameter	Conditions	Min	Typ	Max	Unit
Storage element						
V_{STO}	Voltage on the storage element.		2.40 ¹		4.59 ²	V
V_{OVDIS}	Voltage below which the storage element is considered to be fully depleted, and must not be discharged any further (see Section 6.4).	Configured by $STO_CFG[2:0]$.	2.51		3.51	V
		Configured by I^2C . ³	2.400		3.581	V
V_{CHRDY}	In START STATE , voltage required on the storage element to switch to SUPPLY STATE (see Section 6.4).	Configured by $STO_CFG[2:0]$.	2.61		3.60	V
		Configured by I^2C . ³	2.456		4.312	V
V_{OVCH}	Maximum voltage accepted on the storage element before disabling its charging (see Section 6.4).	Configured by $STO_CFG[2:0]$.	3.49		4.13	V
		Configured by I^2C . ³	2.700		4.594	V
$V_{5V,STOP}$	Voltage on STO at which the 5 V charger (5V_IN input) stops charging the storage element in constant voltage (CV) mode (see Section 5.7).		2.65		4.70	V
Internal supply & quiescent current						
V_{INT}	Internal voltage supply.			2.25		V
$V_{INT,RESET}$	Minimum voltage on VINT before switching to RESET STATE (from any other state).			2.00		V
$V_{INT,CS}$	Minimum voltage on VINT to allow the AEM13920 to switch from RESET STATE to SENSE STO STATE .			2.30		V
$I_{Q,SUPPLY}$	AEM13920 internal quiescent current on STO in SUPPLY STATE . ⁴	Buck enabled (LOAD).		645		nA
		Buck disabled (LOAD).		460		nA
$I_{Q,SLEEP}$	AEM13920 internal quiescent current on STO in SLEEP STATE .	Buck enabled (LOAD).		435	1000	nA
		Buck disabled (LOAD).		275	1000	nA
$I_{Q,SHIP}$	AEM13920 internal quiescent current on STO in shipping mode (SHIP_MODE set HIGH).			15		nA
$I_{Q,RESET}$	AEM13920 internal quiescent current on STO in RESET STATE .					

Table 9: Electrical characteristics (part 3)

1. As set by the battery overdischarge threshold configuration.

2. As set by the battery overcharge threshold configuration.

3. Ensure that $V_{OVDIS} < V_{CHRDY} < V_{OVCH}$ with sufficient margin to prevent unintended state changes due to small voltage variation on the storage element.

4. Both boost converters are enabled but not extracting current from **SRCx**. The **LOAD** pin is left floating.

3.5. Recommended Operating Conditions

Symbol	Parameter	Min ¹	Typ	Max ¹	Unit
External components					
L _{BOOSTx}	Boost converters inductors (optional ²).	3.3 ³	33 ⁴		μH
C _{SRCx}	BUFSRCx terminals decoupling capacitors (optional ²).	3.3 ⁵		10 ⁶	μF
L _{BUCK}	Buck converter inductor (optional ⁷).	1.7 ³	10 ⁴		μH
C _{LOAD}	Buck converter decoupling capacitor (optional ⁷).	10	22		μF
C _{INT}	VINT terminal decoupling capacitor (mandatory).	5	10		μF
C _{STO}	STO terminal decoupling capacitor (mandatory).	5	47		μF
C _{5V}	5V_IN terminal decoupling capacitor (optional ⁸).	22	47		μF
R _{5V_IMAX}	Resistor connected to the 5V_IMAX pin for configuring the 5 V charger current when in constant current (CC) mode (optional ⁸).	0.37		3.7	kΩ
R _{SDA}	I ² C interface pull-up resistors (optional ⁹).		1		kΩ
R _{SCL}					
R _{TH}	NTC thermistor used for thermal monitoring operation (optional ¹⁰).	R0	10 ¹¹	250	kΩ
		Beta	3380 ¹¹		K
R _{DIV}	Resistor used to create a resistive divider with R _{TH} for thermal monitoring operation (optional ¹⁰).	4	22 ¹¹	40	kΩ
R _{ZMPP}	Resistor used for the configuration of the ZMPP feature (optional ¹²).	0.033		100	kΩ

Table 10: Recommended external components

1. All minimum and maximum values are effective components values, taking into account tolerances, derating, temperatures, voltages and any operating conditions (special care must be taken with capacitor derating).
2. Mount only if boost converter x is used.
3. Those minimum values are only applicable with minimum timings (see Sections 9.6 and 9.7).
4. L_{BOOSTx} and L_{BUCK} typical values recommended for best trade-off between boost/buck efficiency and current capability.
5. This value may be reduced if the peak current in L_{BOOSTx} is low enough to not cause excessive ripple on BUFSRCx. L_{BOOSTx} peak current depends on the boost timings and L_{BOOSTx} value. To maintain maximum efficiency, the guideline is to keep the peak-to-peak ripple below 10% of the source target regulation voltage.
6. Typically, a 22 μF / 10 V (MLCC, 0603) capacitor can be used as the capacitance DC bias derating lowers the effective capacitance down to ~5 μF at 5 V.
7. Mount only if buck converter is used.
8. Mount only if the 5 V charger is used.
9. Mount only if the I²C interface is used. For more information on how to select the value of these resistors, refer to "Pull-up resistor sizing" section in NXP's UM10204 "I²C-bus specification and user manual".
10. Mount only if the temperature monitoring feature is used.
11. Those values allow for having the default temperature thresholds at startup shown in Table 15.
12. Mount only if the ZMPP feature is used.

3.5.1. External Inductors Information

The AEM13920 operates with three external miniature inductors. All inductors must support a minimum switching frequency of 10 MHz. Using inductors with low equivalent series resistance (ESR) improves the power-conversion efficiency of both the boost and buck converters.

L_{BOOSTx}

With the recommended operating conditions (33 μH inductor, T_{MULT} = x3), the boost inductors L_{BOOSTx} must support a minimum peak current of 135 mA.

L_{BUCK}

With the recommended operating conditions (10 μH inductor, T_{MULT} = x2), the buck inductors L_{BUCK} must support a minimum peak current of 135 mA.

3.5.2. External Capacitors Information

The AEM13920 operates with four external miniature capacitors to ensure stable operation of the boost converters input, buck converter output, storage element output, and internal supply. Each capacitor serves as a local energy buffer that limits voltage fluctuations caused by switching activity or dynamic load transition.

To maintain optimal performances and minimized quiescent current, all capacitors must exhibit a low leakage current and follow the recommended nominal values listed in Table 10, with a tolerance of ± 20 %.

3.6. Typical Characteristics

3.6.1. Boost Converters Conversion Efficiency

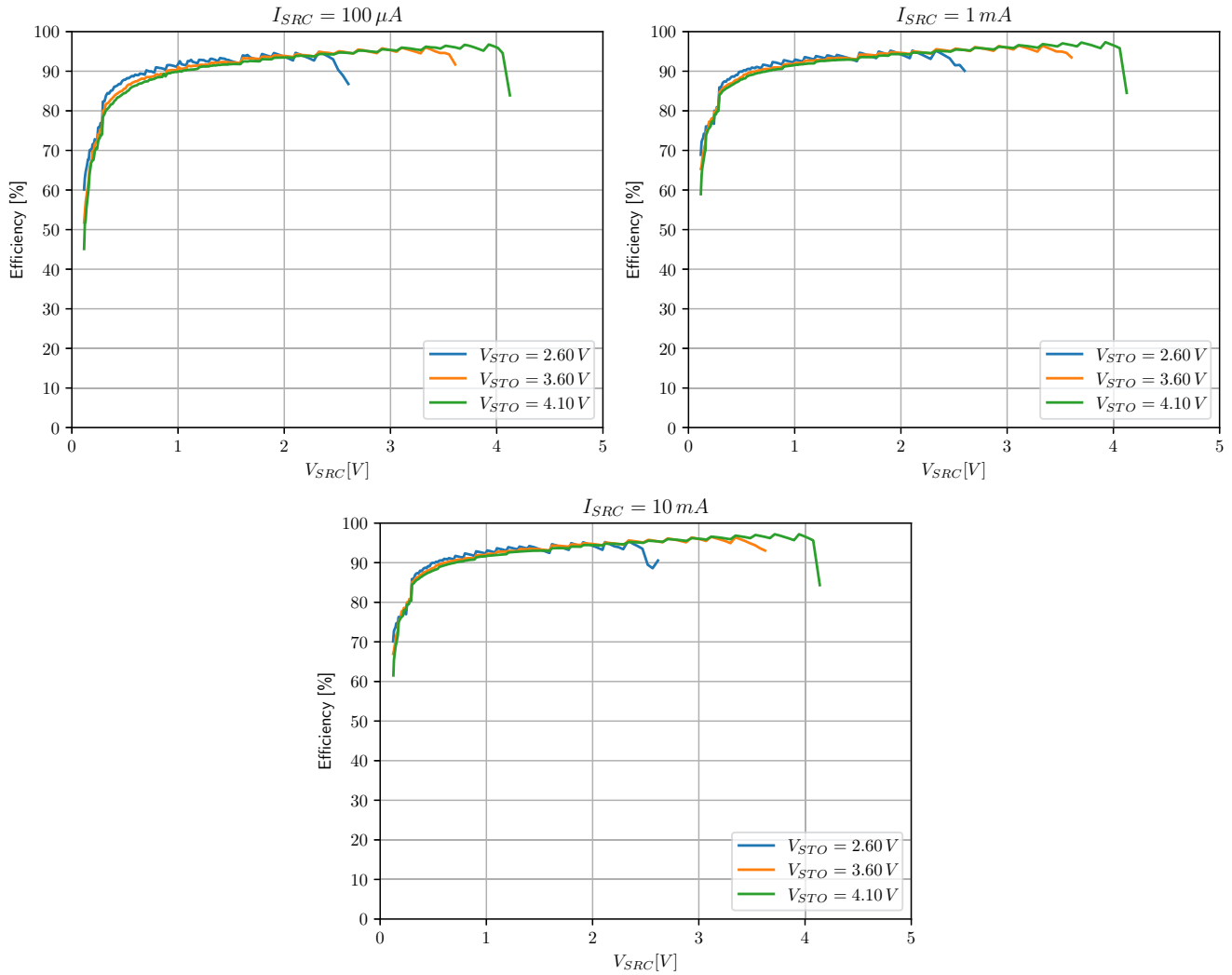


Figure 3: SRCx to STO efficiency vs. V_{SRCx} with $L_{BOOSTx} = 33 \mu H$ (Coilcraft LPS4018-333MRB), $BSTxCFG.TMULT = 0x02$ (x3)

NOTE: The boost efficiency graphs presented in this section include the loss of efficiency due to the AEM13920 quiescent current.

NOTE: The boost efficiency graphs have been measured with the external components recommended in Section 12.

3.6.2. Buck Converter Conversion Efficiency

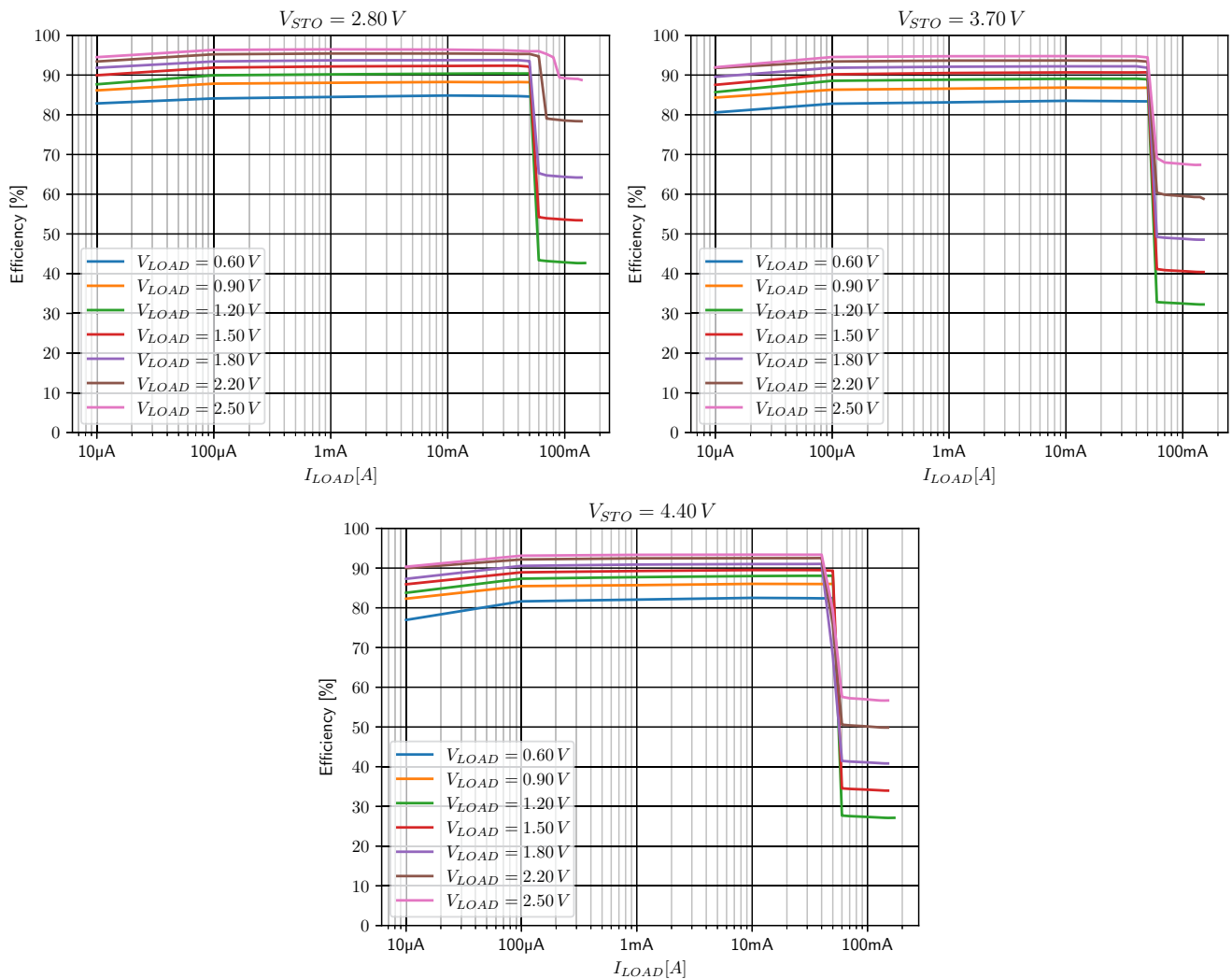


Figure 4: ST0 to LOAD efficiency vs. I_{LOAD} with $L_{BUCK} = 10 \mu H$ (TDK VLS252012CS-100M-1), BUCKCFG.TMULT = 0x01 (x2)

NOTE: The buck efficiency graphs presented in Figure 4 do not include the loss of efficiency due to the AEM13920 quiescent current, as it has already been included in the boost efficiency data shown in Section 3.6.1.

NOTE: The buck efficiency graphs have been measured with the external components recommended in Section 12.

4. Functional Block Diagram

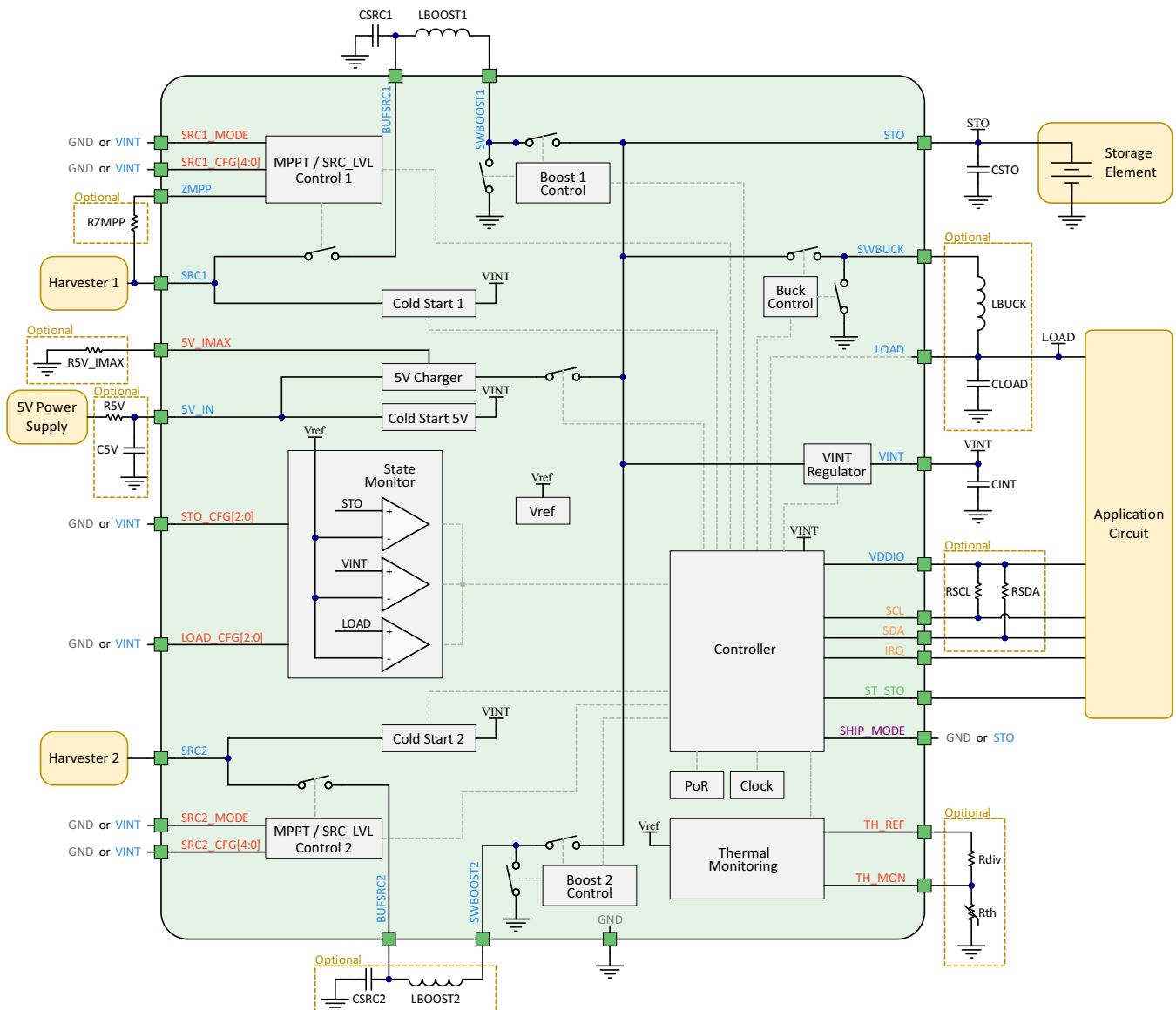


Figure 5: Functional block diagram

5. Theory of Operation

5.1. Cold-Start Circuits

The AEM13920 is able to coldstart from either source ($SRCx$) or from the $5V_IN$ input (see Table 7 for cold-start conditions). The cold-start circuit provides energy to the AEM13920 internal supply ($VINT$) when the device is in **RESET STATE**, **SENSE STO STATE** or **OVDIS STATE**.

See Table 7 for the typical AEM13920 minimum cold-start voltage $V_{SRCx,CS}$ and minimum cold-start power $P_{SRCx,CS}$. Those values have been measured starting with all AEM13920 nodes discharged, except V_{STO} that is charged above V_{CHRDY} . The cold start is considered to be finished when **LOAD** is supplied (buck is enabled), meaning that the AEM13920 has switched to **SUPPLY STATE**.

The time necessary for the AEM13920 to perform a cold start depends on multiple parameters such as:

- I_{SRCx} : the higher the source current, the faster the cold start.
- C_{INT} : the higher the capacitance on $VINT$, the slower the cold start.

Typical cold-start time may vary between a few minutes for very low I_{SRCx} to a few tens of milliseconds for high I_{SRCx} .

5.2. Boost Converters

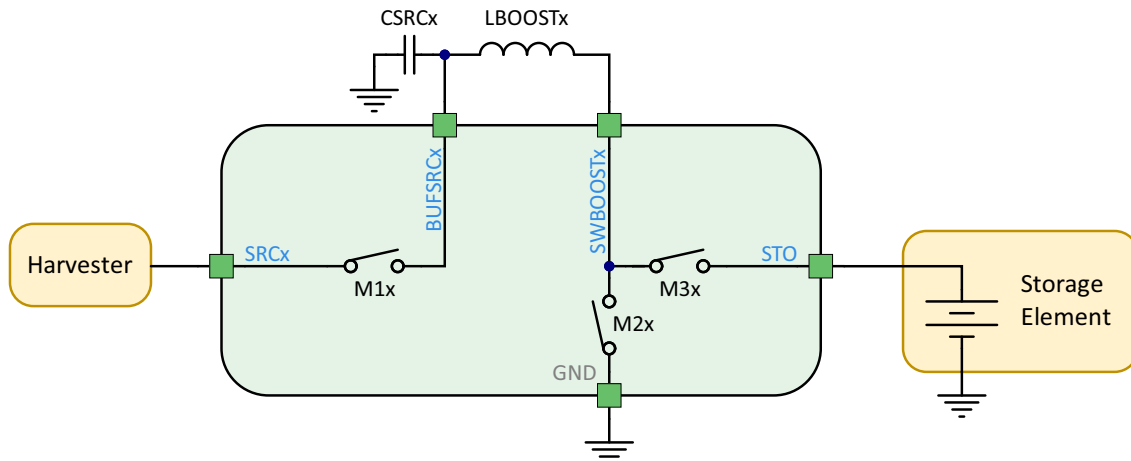


Figure 6: Simplified schematic view of the boost converters

Please note that the following explanations apply to both boost converters #1 and #2.

5.2.1. Operation Principle

The boost (step-up) converter raises the voltage available at $BUFSRCx$ to a level suitable for charging the storage element, in the range of 2.40 V to 4.59 V, according to the system configuration. The switching transistors of the boost converter are M2x and M3x. The reactive power component of this converter is the external inductor L_{BOOSTx} .

M1x allows for disconnecting the $SRCx$ pin from $BUFSRCx$, which happens in the following cases:

- When measuring the source open-circuit voltage, if source mode is MPPT (see Section 5.2.2).
- When the boost converter is disabled through I^2C (see Section 9.6).
- When AEM13920 is in **SLEEP STATE**, **SENSE STO STATE** or **RESET STATE** (see Section 5.9).
- When temperature is out of range (see Section 5.4).
- When a suitable power supply is connected to the $5V_IN$ pin (both boosts are disabled in that case).

When the boost converter is extracting energy from the source, M1x is closed, connecting **SRCx** to **BUFSRCx**.

Target source regulation voltage can be determined by:

- The MPPT module (ratio of open-circuit voltage or target impedance connected to **ZMPP**) when **SRCx_MODE** is HIGH (see Section 5.2.2).
- The constant voltage regulation setting when **SRCx_MODE** is LOW (see Section 5.2.3).

BUFSRCx is decoupled by the capacitor **C_{SRCx}**, which smooths the voltage against the current pulses induced by the boost converter.

The storage element is connected to the **STO** pin. This node is linked to the output of the boost converters.

The maximum current supplied to the **STO** pin depends on the value of both **L_{BOOSTx}** and the boost converters timings for charging and discharging **L_{BOOSTx}**, and thus, the peak current **I_{LBOOST,PEAK}**. The boost timings can be configured thanks to the I²C register fields **BSTxCFG[4:2]**. See Section 6.5 for boost timings multiplier default values and Section 9.6 for further information, as well as typical combinations of **L_{BOOSTx}** inductor value and boost converters timings.

While using an energy source is mandatory, using both boost converters is not: the user might use a single boost converter or even use the AEM13920 only with the 5 V charger as energy source. The following connections must be done if a boost converter is not used:

- **SRCx** and **BUFSRCx** to GND.
- Leave **SWBOOSTx** floating.

5.2.2. Maximum Power Point Tracking

This section describes the AEM13920 behavior when the source regulation mode is MPPT ratio. Switching to this mode is done by setting the **SRCx_MODE** pin HIGH or by setting **SRCxREGU0.MODE** I²C register field bit HIGH (see Sections 6.3 and 9.3).

The MPPT module is active during **START STATE**, **OVDIS STATE** and **SUPPLY STATE**.

5.2.2.1. Open-Circuit Voltage Ratio

In MPPT ratio mode, the AEM13920 MPPT relies on the fact that, for several models of harvesters (typ. solar cells), the ratio between the maximum power point voltage (**V_{MPP}**) and the open-circuit voltage (**V_{OC}**) is constant for a wide range of harvesting conditions. For a solar cell, this means that **V_{MPP} / V_{OC}** is constant for any lighting conditions, even though both voltages increase when luminosity increases. Please note that this is valid for a large variety of harvesters, not only solar cells.

The Maximum Power Point tracking (MPPT) ratio **V_{MPP} / V_{OC}** differs from one harvester model to another. The user must set the MPPT ratio (**R_{MPPT}**) to match the specifications of the harvester model used to maximize power extraction. This ratio is set through the configuration pins **SRCx_CFG[2:0]** (see Section 6.3) or through the I²C interface register field **SRCxREGU0.CFG0** (see Section 9.3).

The MPPT module evaluates periodically with the following sequence the open-circuit voltage (**V_{OC}**) to ensure optimal power extraction at any time.

- The AEM13920 stops extracting power from the **SRCx** during **T_{MPPT,WAIT}** to allow the **SRCx** voltage to rise to **V_{OC}**.
- Once this delay elapses, the AEM13920 performs the measurement of **V_{OC}** during **T_{MPPT,MEASURE}**.
- After the measurement, the AEM13920 resumes extracting power by regulating the **SRCx** voltage to the newly determined **V_{MPP}**.
- This MPPT evaluation is repeated every **T_{MPPT,PERIOD}**.

T_{MPPT,WAIT} and **T_{MPPT,PERIOD}** are set through the configuration pins **SRCx_CFG[4:3]** (see Section 6.3) or by configuring the **SRCxREGU1** register (see Section 9.3) while **T_{MPPT,MEASURE}** is constant for any configuration (see Table 8).

The total time during which the AEM13920 does not extract power correspond to the sum of **T_{MPPT,WAIT}** and **T_{MPPT,MEASURE}** (as shown in Figure 7).

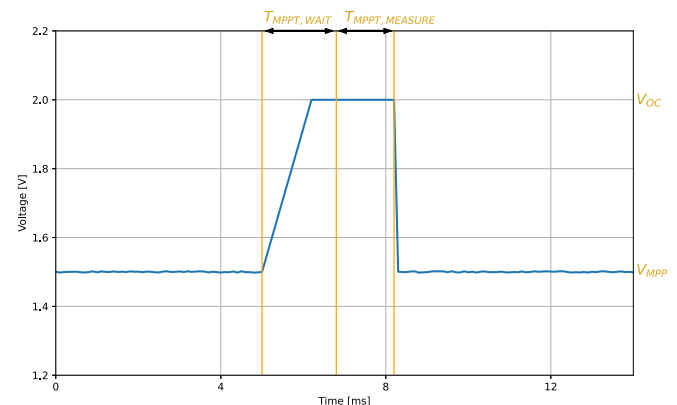


Figure 7: MPPT evaluation behavior (with **T_{MPPT,WAIT}** of 1.8 ms)

5.2.2.2. ZMPP

Some harvesters provide better performance when connected to a circuit with constant input resistance. The AEM13920 boost converter #1 MPPT can be set as constant input resistance by enabling the ZMPP feature. This can be done by setting all **SRC1_CFG[2:0]** pins HIGH (see Section 6.3) or by setting the field **SRC1REGU0.CFG0** to 0x07 (see Section 9.3).

In ZMPP mode, the AEM13920 regulates the input resistance of **SRC1** to match the resistance R_{ZMPP} connected to the **ZMPP** pin. Operation is similar to that of the V_{MPP} / V_{OC} mechanism described in Section 5.2.2.1:

- Every $T_{MPPT,PERIOD}$, the AEM13920 disconnects **SRC1** from **BUFSRC1**, and connects the **ZMPP** pin to GND, making R_{ZMPP} the source load resistance.
- After a $T_{MPPT,WAIT}$ delay, the AEM13920 measures the voltage on **SRC1**, loaded by R_{ZMPP} . The measured voltage is the new **SRC1** input regulation voltage.

When in ZMPP mode, $T_{MPPT,WAIT}$ is determined as it is done in open-circuit voltage ratio mode: either by **SRC1_CFG[4:3]** (see Section 6.3) or by the SRC1REGU1.CFG1 I²C register field (see Section 9.3).

Please note that the ZMPP feature is only available on boost converter 1, or when a harvester is connected on both boost converters simultaneously, as described in Section 5.2.5.

5.2.3. Source Constant Voltage Regulation

This section describes the AEM13920 behavior when the source regulation mode is set to constant voltage. Switching to this mode is done by setting the **SRCx_MODE** to LOW or by setting SRCxREGU0.MODE I²C register field bit to LOW (see Sections 6.2 and 9.3).

During **START STATE**, **OVDIS STATE** and **SUPPLY STATE**, the voltage on **SRCx** is regulated to a fixed voltage configured through the **SRCx_CFG[4:0]** pins (see Section 6.2) or through the SRCxREGUx registers (see Section 9.3).

In constant voltage regulation mode, the AEM13920 behaves as follows:

- If the open-circuit voltage V_{OC} of the harvester is lower than $V_{SRCx,REG}$, the AEM13920 does not extract power from the source.
- If V_{OC} is higher than $V_{SRCx,REG}$, the AEM13920 regulates V_{SRCx} to $V_{SRCx,REG}$ and thus, extracts power from the source.
- If $V_{SRCx,REG}$ is configured by I²C below V_{SRCLOW} thanks to SRCxREGUx and SRCLOW registers, the AEM13920 enters **SLEEP STATE** (see Section 5.9.6 and Section 9.4).

5.2.4. Automatic High-Power Mode

When the AEM13920 detects that the energy available on **SRCx** is high enough, the boost converter automatically switches to high-power mode to double the current extraction compared to low-power mode.

Preventing the AEM13920 to switch to high-power mode may allow to use an inductor with half peak current rating for L_{BOOSTx} (see Section 9.6). On the other hand, allowing the AEM13920 to switch to high-power mode increases the maximum current that the AEM13920 can harvest from **SRCx** to **STO**.

Automatic high-power mode is enabled by default and can be disabled by setting the BSTxCFG.ENHP register bit to 0 through the I²C interface.

5.2.5. Using Both Boost Converters in Parallel

It is possible to use the two boost converters in parallel to double the current that can be extracted from a single harvester.

To do so, the user must configure the AEM13920 as follows:

- Connect the harvester simultaneously on **SRC1** and **SRC2**.
- Configure both boost converters input with identical regulation settings (MPPT with the same ratio and timings or constant voltage with the same target voltage).
- C_{SRC1} , C_{SRC2} , L_{BOOST1} and L_{BOOST2} must all be populated and both boost timings must be configured accordingly.

Please note that ZMPP can be configured to work with interconnected sources. In this case, the AEM13920 must be configured as shown in Figure 8:

- **SRC1** and **SRC2** as MPPT (**SRC1_MODE** and **SRC2_MODE** HIGH).
- **SRC1** MPPT ratio to ZMPP.
- **SRC2** MPPT ratio to 100%.
- A single R_{ZMPP} resistor connected between **ZMPP** and **SRC1/SRC2**.

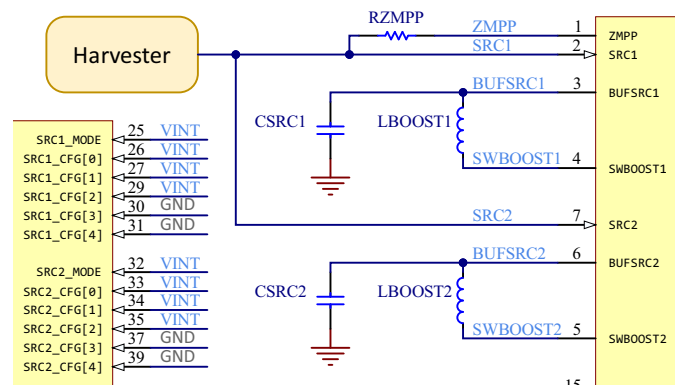


Figure 8: ZMPP connection with both boost converters used in parallel

See Section 5.2.2.2 for further information about ZMPP.

5.3. Buck Converter

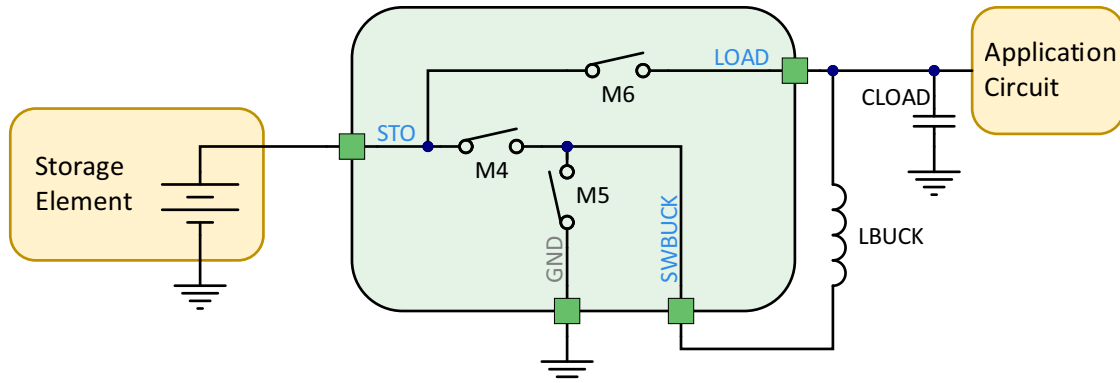


Figure 9: Simplified schematic view of the buck converter

The buck (step-down) converter transfers energy from the storage element connected to **STO** to the regulated **LOAD** output. The switching transistors of the buck converter are M4 and M5. The reactive power component of this converter is the external inductor L_{BUCK} . **LOAD** is decoupled by the capacitor C_{LOAD} , which smooths the voltage against the current pulses from the buck converter and the consumption of the external circuit connected to **LOAD**.

Setting the **LOAD** regulation voltage V_{LOAD} or disabling the buck converter is done through the **LOAD_CFG[2:0]** pins (see Section 6.6) or the I²C register BUCKCFG (see Section 9.7).

After cold start, if the buck converter is enabled and if the temperature is within the range, the buck converter starts:

- When V_{STO} is higher than V_{CHRDY} if the 5 V charger is not connected, or
- When V_{STO} is higher than V_{OVDIS} if the 5 V charger is connected.

NOTE: When using the 5 V charger, make sure that the configured $I_{5V,CC}$ is high enough to supply the circuit connected on **LOAD**. If not, once V_{STO} rises above V_{OVDIS} , **LOAD** is directly enabled, and the current drained on the storage element will directly discharge V_{STO} below V_{OVDIS} , thus, disabling the **LOAD** output.

When V_{STO} drops below V_{OVDIS} , the AEM13920 waits for $T_{CRIT,ST}$ to set the STATUS0.OVDIS register field HIGH and to schedule to disable the buck converter. The **LOAD** output remains enabled until the end of T_{CRIT} .

When the temperature gets out of the storage element discharge temperature range (TEMPCOLDDIS and TEMPHOTDIS), the AEM13920 rises the IRQFLG0.TEMPDIS and waits for $T_{CRIT,ST}$. If the temperature is still out of range after $T_{CRIT,ST}$, the AEM13920 schedules to disable the buck converter. The **LOAD** output remains enabled until the end of T_{CRIT} .

The maximum current supplied to the **LOAD** pin depends on both the value of L_{BUCK} and the buck converter timings for charging and discharging L_{BUCK} , and thus, its peak current $I_{LBUCK,PEAK}$. The buck timings can be configured thanks to the I²C register field BUCKCFG.TMULT. See Section 6.6.2 for default buck timing values and Section 9.7 for further information and typical combinations of L_{BUCK} inductor value and buck converter timings.

Using the buck converter is not mandatory. If not used, the user must do the following:

- Connect all **LOAD_CFG[2:0]** to GND (LOW) to disable the buck converter.
- Leave **SWBUCK** and **LOAD** floating.

When the difference between V_{STO} and V_{LOAD} is too small for the buck converter to keep working properly, it switches to “bang-bang” controlled converter mode:

- When V_{LOAD} is too low, the switch M6 connects **STO** directly to **LOAD**, making V_{LOAD} rise.
- When V_{LOAD} is too high, M6 disconnects **STO** and **LOAD** so that V_{LOAD} decreases.

This happens when the following condition is satisfied:

$$V_{STO} - V_{LOAD} < 0.25V$$

In that case, efficiency is lower than in buck mode.

5.4. Thermal Monitoring

The AEM13920 thermal monitoring allows for protecting the storage element from cold and hot temperatures by monitoring the ambient temperature and comparing it to the configured charge and discharge protection thresholds.

The charge minimum (cold) and maximum (hot) temperature thresholds are used to disable the boost converters in order to stop charging the storage element if the temperature is out of range.

The discharge minimum (cold) and maximum (hot) temperature thresholds are used to disable the buck converter in order to stop discharging the storage element through the **LOAD** if the temperature is out of range. Please note that in this case, **VINT** will continue to be supplied from **STO**.

To use the thermal monitoring feature, a resistor (R_{DIV}) and a NTC thermistor (R_{TH}) are required in order to form a resistive divider (see Figure 12). The **TH_REF** terminal allows for applying a reference voltage to the resistive divider while **TH_MON** is the measuring point. The temperature evaluation is done periodically every $T_{TEMP,MON}$ (see Table 8). To spare power, the divider is biased only during this evaluation.

See Section 6.7 for thermal monitoring configuration.

Thermal monitoring is optional, if not used, connect **TH_MON** to **VINT** and leave **TH_REF** floating.

5.5. Average Power Monitoring

The AEM13920 implements four different Average Power Monitoring (APM) modules:

- **APMSRC1**: power transferred to **STO** from **SRC1** (boost #1 power converter).
- **APMSRC2**: power transferred to **STO** from **SRC2** (boost #2 power converter).
- **APMLOAD**: power transferred from **STO** to **LOAD** (buck power converter).
- **APMCHG5V**: Percentage of APM window during which the **5V_IN** input has been charging the storage element (5 V charger).

Please note that all APM measures are related to **STO** (energy provided to/from the storage) as shown on Figure 10.

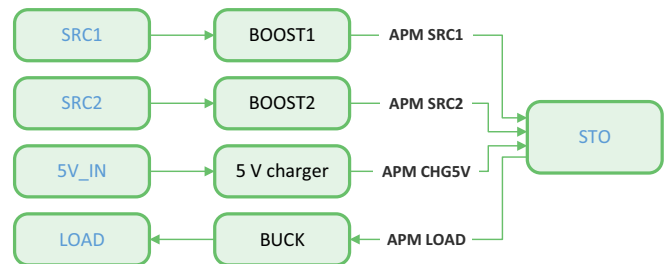


Figure 10: Average Power Monitoring in the power chain

5.5.1. SRCx and LOAD APM

The SRCx and LOAD APM modules work the same, so for these APM module explanations, the following is defined:

- **APM_IN** as the input node measured by the APM module.
- **APM_OUT** as the output node measured by the APM module.

Those APM modules are able to determine the transferred energy by counting the number of current pulses transferred from **APM_IN** to **APM_OUT** by the DCDC converter over a configurable time window, and thus, evaluates the corresponding energy.

Two modes are available:

- **Pulse Counter mode**: The APM modules count the number of DCDC pulses happening during the APM window.
- **Power Meter mode**: The APM modules integrate the energy transferred from **APM_IN** to **APM_OUT** during the APM window.

Refer to Section 9.17 for further details about how to set the modes and the window, and how to convert registers value to Joule.

If **SRC1** and **SRC2** are connected in parallel (to double the AEM13920 harvesting current capability), the user might notice that at currents low enough to be harvested by a single source, one source is extracting most of the current. This is due to a slight difference between **SRC1** and **SRC2** voltage regulation set point: the one having the higher set point voltage senses the voltage as too low to pull current, thus letting the other one pull the whole available current.

5.5.2. 5 V Charger APM

The 5 V charger APM module (APMCHG5V) provides the percentage of the APM window during which the 5 V charger has been charging the storage element.

Refer to Section 9.17.2 for further details about how to interpret the 5 V charger APM register.

5.5.3. APM IRQ

If the IRQEN1.APMDONE field has been set to 1 (see Section 9.14), a rising edge on the **IRQ** pin along with the IRQFLG1.APMDONE field indicates the end of an APM window and thus, that new APM values are available and ready to be read in the APM data register (see Section 9.17).

Please note that all four APM modules operate synchronously with the same configured APM window.

When the **IRQ** pin rises due to the IRQFLG1.APMDONE, new APM data is available for all enabled APM modules. The APM data register should be read before the end of the next APM window, after which the APM data will be overwritten.

5.5.4. APM Accumulator

In order to get the APM data without waking up the application/MCU too often, the APM accumulator can be used.

This feature enables accumulation of APM data from the enabled APM modules over multiple consecutive APM windows. Instead of triggering the **IRQ** pin after each individual APM window, the AEM13920 sums the APM data from a configured number of consecutive windows. The **IRQ** pin is set by the IRQFLG1.APMDONE register only after the specified number of windows has been completed.

Please note that when the APM accumulator is used, the **IRQ** pin will be raised only after multiple APM windows, but the APM data register is still updated after each APM window.

Once the **IRQ** pin rises due to the IRQFLG1.APMDONE, the APM data register should be read before the end of the next APM window, after which the APM data will be overwritten.

Refer to Section 9.13 for further details about how to configure the APM accumulator feature.

5.6. IRQ Pin

The **IRQ** pin allows the application circuit to be notified of various events occurring in the AEM13920 (rising edge on the **IRQ** pin). At startup, the only event that is enabled is I2CRDY, signaling that the AEM13920 has finished to coldstart and thus, that it is out from **RESET STATE**. Other events can be enabled by writing the IRQEN0 and IRQEN1 registers (see Section 9.14).

When the **IRQ** pin shows a rising edge, the event that triggered it can be determined by reading the IRQFLG0 and IRQFLG1 registers (see Section 9.15). The **IRQ** pin is reset when the corresponding IRQFLGx register is read.

5.7. 5 V Charger

The AEM13920 is equipped with a 5 V charger (**5V_IN**) that can be used for fast charging of the storage element (**STO**) when the following conditions are met:

- $V_{5V_IN} \geq 3.60 \text{ V}$.
- $V_{5V_IN} \geq V_{STO} + 200 \text{ mV}$.
- The temperature is within the configured operating range.
- The AEM13920 is either in **START STATE**, **SUPPLY STATE**, or **OVDIS STATE**.

When the 5 V charger is connected, both boost converters are automatically disabled.

The 5 V charger operates in CC mode (see Section 5.7.1) or in CC/CV mode (see Section 5.7.2).

Using the 5 V charger is not mandatory. When not used, leave both **5V_IN** and **5V_IMAX** pins floating.

5.7.1. CC Mode

By default, the 5 V charger operates in constant current (CC) mode only. In this mode, the storage element is charged from the **5V_IN** input with the configured maximum charging current ($I_{5V,CC}$). As V_{STO} approaches V_{OVCH} , the AEM13920 gradually reduces the charging duty cycle to zero, progressively lowering the effective charging current through pulse-charging operation.

The maximum charging current is configurable in a range from 13.5 mA to 135 mA thanks to R_{5V_IMAX} resistor connected to the **5V_IMAX** pin (see Section 6.8 for further details about R_{5V_IMAX} configuration).

NOTE: When the storage element is over-discharged ($V_{STO} < V_{OVDIS}$), the charging current is limited to $I_{5V,OVDIS,CC}$ (see Table 7).

5.7.2. CC/CV Mode

If enabled, the optional constant voltage (CV) mode allows the user to define a configurable end-of-charge voltage ($V_{5V,STOP}$) for the storage element. The 5 V charger operates in constant current (CC) mode during the main charging phase and maintains the configured maximum charging current until the storage element voltage approaches $V_{5V,STOP}$. The AEM13920 then switches to CV mode and reduces the charge current smoothly down to zero as V_{STO} reaches $V_{5V,STOP}$.

If $V_{5V,STOP}$ is configured above V_{OVCH} , the CV mode has no effect and the charger behaves as if the feature was disabled. For proper operation, $V_{5V,STOP}$ must be set below V_{OVCH} .

See Section 6.8 for further details about how to configure the 5 V charger and Section 9.8 for CV mode enabling and $V_{5V,STOP}$ configuration.



5.8. Shipping Mode

The shipping mode feature allows to force the AEM13920 in **RESET STATE** (see Figure 11 and Section 5.9.1), thus, disabling all AEM13920 functionalities including both boost converters, the buck converter, the 5 V charger, and the I²C interface. Only **VINT** is charged if energy is available from **SRCx**. The storage element is no longer charged or discharged.

See Section 6.9 for shipping mode configuration.

5.9. State Machine Description

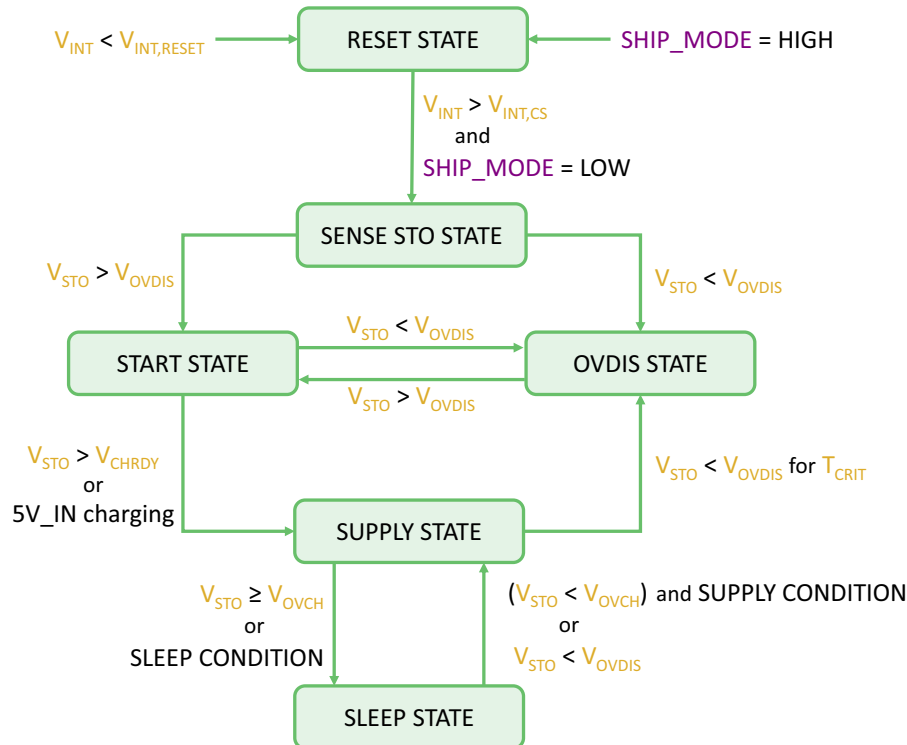


Figure 11: State machine

5.9.1. RESET STATE

The AEM13920 enters **RESET STATE** if one of the following is true:

- V_{INT} is below $V_{INT,RESET}$ (see Table 9).
- Shipping mode is enabled ($SHIP_MODE$ is HIGH).

In **RESET STATE**, the AEM13920 behaves as follows:

- Both boost converters and the buck converter are disabled.
- The AEM13920 internal supply (V_{INT}) energy is provided by $SRCx$ or $5V_IN$.
- Only $I_{Q,RESET}$ is drawn from the storage element connected to STO .
- ST_STO is LOW.

The AEM13920 stays in **RESET STATE** until the power available on either $SRCx$ or on $5V_IN$ meets the cold-start requirements long enough to make V_{INT} reach $V_{INT,CS}$ (see Table 7 and Table 9). Then:

- If shipping mode is disabled ($SHIP_MODE$ is LOW), the AEM13920 reads the value on all configuration pins and switches to **SENSE STO STATE**.
- If shipping mode is enabled ($SHIP_MODE$ is HIGH), the AEM13920 stays in **RESET STATE** until shipping mode is disabled by setting $SHIP_MODE$ LOW. Please note that $SHIP_MODE$ is read every $T_{GPIO,MON}$.

Please note that, from any state, the AEM13920 will switch to **RESET STATE** if V_{INT} drops below $V_{INT,RESET}$.

5.9.2. SENSE STO STATE

In **SENSE STO STATE**, the AEM13920 behaves as follows:

- A first measure of V_{STO} is performed by the AEM13920.
- The levels of SDA and SCL pins are evaluated to decide whether the I²C should be enabled (see Section 6.1.2)
- None of the DCDC converters are running.
- ST_STO is LOW.

The **SENSE STO STATE** lasts for about 2 ms.



From **SENSE STO STATE**, the AEM13920 switches to:

- **START STATE** if $V_{STO} > V_{OVDIS}$.
- **OVDIS STATE** if $V_{STO} < V_{OVDIS}$.

From this point, the I²C interface is available (see Section 6.1.2).

5.9.3. START STATE

When in **SENSE STO STATE**, the AEM13920 switches to **START STATE** if V_{STO} is above V_{OVDIS} .

In **START STATE**, the AEM13920 behaves as follows:

- The storage element connected to **STO** is charged by the boost converters until V_{STO} reaches V_{CHRDY} .
- **VINT** internal supply energy is provided by the storage element regardless of the power available on **SRCx** and **5V_IN**.
- The buck converter (**LOAD**) is disabled.
- **ST_STO** is LOW.

From **START STATE**, the AEM13920 switches to:

- **SUPPLY STATE** if $V_{STO} > V_{CHRDY}$ or if the 5 V charger is charging.
- **OVDIS STATE** if $V_{STO} < V_{OVDIS}$.

5.9.4. SUPPLY STATE

When in **START STATE**, the AEM13920 switches to **SUPPLY STATE** if V_{STO} is above V_{CHRDY} or if the 5 V charger is charging.

In **SUPPLY STATE**, the AEM13920 behaves the same as when in **START STATE**, but with the following differences:

- The **LOAD** is supplied by the buck converter if all of the following conditions are true:
 - The buck converter has been enabled by the user (see Section 6.6.1).
 - The temperature is within the configured storage element discharge temperature range (**TEMPCOLDDIS** and **TEMPHOTDIS**, see Section 9.10).
- **ST_STO** is HIGH.

From **SUPPLY STATE**, the AEM13920:

- Switches to **SLEEP STATE** if one of the following conditions is met:
 - $V_{STO} \geq V_{OVCH}$.
 - **SLEEP CONDITION** (see Section 5.9.6).
- Schedules to switch to **OVDIS STATE** if $V_{STO} < V_{OVDIS}$ for $T_{CRIT,ST}$, and waits for the end of T_{CRIT} to actually switch to **OVDIS STATE**. If V_{STO} recovers above V_{OVDIS} after $T_{CRIT,ST}$ but before the end of T_{CRIT} , the AEM13920 will still enter **OVDIS STATE** at the end of T_{CRIT} .

5.9.5. OVDIS STATE

The AEM13920 switches to **OVDIS STATE** if:

- V_{STO} is below V_{OVDIS} when in **SENSE STO STATE** or **START STATE**.
- V_{STO} remains below V_{OVDIS} for more than $T_{CRIT,ST}$ when in **SUPPLY STATE** (waiting for the end of T_{CRIT} to actually switch).

In **OVDIS STATE**, the AEM13920 behaves as follows:

- The storage element connected to **STO** is charged by the boost converters and/or by the 5 V charger, until V_{STO} rises above V_{OVDIS} .
- **VINT** internal supply energy is provided by **SRCx** or **5V_IN**.
- The buck converter (**LOAD**) is disabled.
- **ST_STO** is LOW.

From **OVDIS STATE**, the AEM13920 switches to:

- **START STATE** if V_{STO} rises above V_{OVDIS} .
- **RESET STATE** if not enough power is available on **SRCx** and on **5V_IN** to maintain V_{INT} above $V_{INT,RESET}$. In this case, the **STO** pin is set to high impedance, so that virtually no current is drawn from the storage element connected to **STO** ($I_{Q,RESET}$ as defined in Section 3.4).

5.9.6. SLEEP STATE

SLEEP STATE allows for reducing the AEM13920 internal circuit consumption when none of the **SRCx** provides enough power or when the storage element is fully charged. Thus, storage element discharging is kept minimal. **SLEEP STATE** is also reached when charging is not allowed (temperature outside range, boost converters disabled, V_{SRCx} below the V_{SRCLOW} threshold).



The following conditions are defined:

- SLEEP CONDITION is true if one of the following conditions is true:
 - Temperature outside of the range (see Section 6.7).
 - All boost converters are disabled through I²C (see Section 9.6).
 - Voltage on both SRCx is below the V_{SRCLOW} threshold (see Section 9.4).
- SUPPLY CONDITION is true if all the following conditions are true:
 - Temperature within the range (see Section 6.7).
 - At least one boost converter is enabled (see Section 9.6).
 - Target voltage on one of SRCx (i.e., V_{MPP}) is above the V_{SRCLOW} threshold (see Section 9.4).

In SLEEP STATE, the AEM13920 behaves as follows:

- The storage element connected to STO is not charged by SRCx, allowing for reducing the quiescent current on VINT and thus, on STO.
- If V_{STO} is below V_{OVCH}, the storage element connected to STO can be charged from the 5 V charger by connecting a power source on 5V_IN.
- VINT internal supply energy is provided by the storage element regardless of the power available on SRCx and 5V_IN.
- The buck converter (LOAD) is enabled (if the buck converter enabling conditions are true, see Section 5.9.4).
- ST_STO is HIGH.

From SLEEP STATE, the AEM13920 switches back to SUPPLY STATE if one of the following conditions is met:

- SUPPLY CONDITION and V_{STO} < V_{OVCH}.
- V_{STO} < V_{OVDIS}.

6. System Configuration

6.1. Configuration Pins and I²C

6.1.1. Configuration Pins

After a cold start, the AEM13920 reads the configuration pins. Those are then read periodically every $T_{GPIO,MON}$. The configuration pins can be changed on-the-fly. The floating configuration pins are read as HIGH, except for the `SHIP_MODE` pin, which is read as LOW.

NOTE: all the read-only registers (with addresses from 0x15 to 0x28 and from 0xE0 to 0xE4) can be read and contain valid data even if the AEM13920 is configured through the GPIO. The only exception is the APM-related registers that are no longer updated when using the GPIO configuration.

6.1.2. Configuration by I²C

To configure the AEM13920 through the I²C interface after a cold start, the user must wait for the `IRQ` pin to rise, showing that the AEM13920 is out of `RESET STATE` and is ready to communicate with I²C. Please note that the `IRQ` pin is always low during `RESET STATE`. See Section 5.6 for further informations about the `IRQ` pin.

Registers are stored in a volatile memory, so their value is lost when `VINT` drops below the reset voltage $V_{INT,RESET}$, making the AEM13920 switch to `RESET STATE`. `VDDIO` is only for I²C communication bus reference voltage, so register values are kept whether `VDDIO` is supplied or not once registers are written.

Once the above procedure is done, the user can write to the desired registers and validate the configuration by setting the `CTRL.UPDATE` register field. All configuration pins are then ignored (except the `SHIP_MODE` pin) and all configurations are set by the register values. All registers have a default value that can be found in Table 18.

NOTE: It is important to note that if both `SDA` and `SCL` pins are read LOW (`GND`) during `SENSE STO STATE`, the I²C interface will be disabled. In this case, the `IRQ` pin will remain LOW, even if the AEM13920 is out of `RESET STATE`. To enable the I²C after this point, the master must first send an I²C message (`START + DATA`) to the AEM13920, such as the address of the AEM13920. Once the I²C interface is enabled, the `IRQ` pin is set HIGH along with the `IRQFLG0.I2CRDY` field to indicate that the I²C interface is ready.



6.2. Source Constant Voltage Regulation

The following configurations apply when **SRCx_MODE** is LOW, so that the boost converter is in constant voltage mode. The user can set the regulation voltage with **SRCx_CFG[4:0]** (see Table 11), or through the SRCxREGUx registers (see Section 9.3.3).

Configuration pins					Voltage [V]
SRCx_CFG[4:0]					V _{SRCx,REG}
L	L	L	L	L	0.14
L	L	L	L	H	0.30
L	L	L	H	L	0.36
L	L	L	H	H	0.42
L	L	H	L	L	0.48
L	L	H	L	H	0.51
L	L	H	H	L	0.53
L	L	H	H	H	0.54
L	H	L	L	L	0.56
L	H	L	L	H	0.57
L	H	L	H	L	0.60
L	H	L	H	H	0.66
L	H	H	L	L	0.72
L	H	H	L	H	0.74
L	H	H	H	L	0.75
L	H	H	H	H	0.77

Configuration pins					Voltage [V]
SRCx_CFG[4:0]					V _{SRCx,REG}
H	L	L	L	L	0.78
H	L	L	L	H	0.81
H	L	L	H	L	0.87
H	L	L	H	H	0.93
H	L	H	L	L	0.99
H	L	H	L	H	1.10
H	L	H	H	L	1.20
H	L	H	H	H	1.31
H	H	L	L	L	1.40
H	H	L	L	H	1.50
H	H	L	H	L	1.61
H	H	L	H	H	1.70
H	H	H	L	L	1.79
H	H	H	L	H	1.90
H	H	H	H	L	1.99
H	H	H	H	H	2.10

Table 11: Configuration of the source constant voltage regulation with SRCx_CFG[4:0] pins

6.3. Maximum Power Point Tracking

The following configurations apply when **SRCx_MODE** is HIGH, so that the boost converter is in MPPT ratio mode. When configuring the MPPT module, the user can set the MPPT ratio and timings with **SRCx_CFG[4:0]** (see Tables 12 and 13), or through the SRCxREGUx registers (see Section 9.3).

Configuration pins			MPPT Ratio
SRCx_CFG[2:0]			R_{MPPT}
L	L	L	35 %
L	L	H	50 %
L	H	L	65 %
L	H	H	70 %
H	L	L	75 %
H	L	H	80 %
H	H	L	85 %
H	H	H	ZMPP (SRC1) / 100 % (SRC2)

Table 12: MPPT ratio configuration with SRCx_CFG[2:0] pins

Configuration pins		MPPT wait time [ms]	MPPT Period [ms]
SRCx_CFG[4:3]		T_{MPPT, WAIT} ¹	T_{MPPT, PERIOD}
L	L	1.8	116 ²
L	H	7.3	465
H	L	29	1862
H	H	233	14895

Table 13: MPPT timing configuration with SRCx_CFG[4:3] pins

1. The total time spent in open-circuit is the sum of **T_{MPPT, WAIT}** (configurable, see table above) and **T_{MPPT, MEASURE}** (fixed, see Table 8).
2. If **T_{MPPT, PERIOD}** is set to 116 ms for any of the two **SRCx**, the APM WINDOW will automatically be set to 116 ms by the AEM13920 for the two boost converters, the buck converter, and the 5 V charger APM modules.

6.4. Storage Element Protection Thresholds

The storage element protection thresholds and the temperature thresholds are configurable to match various storage element types through the **STO_CFG[2:0]** pins (see Tables 14 and 15), or through the following I²C registers:

- VOVDIS, VCHRDY, and VOVCH: to configure the storage element protection voltage thresholds **VOVDIS**, **VCHRDY**, and **VOVCH** (see Section 9.5).
- TEMPCOLDCH, TEMPHOTCH, TEMPCOLDDIS, and TEMPHOTDIS: to configure the storage element protection temperature thresholds (see Sections 9.9 and 9.10).

For the configuration of **VOVDIS**, **VCHRDY**, and **VOVCH** via I²C, ensure that the following condition is respected:

$$V_{OVDIS} < V_{CHRDY} < V_{OVCH}$$

*NOTE: It is recommended to provide a minimum margin of 100 mV between **VOVDIS**, **VCHRDY**, and **VOVCH** to avoid unintended state switching due to small storage element voltage variations.*

Configuration pins			Overdischarge voltage [V]	Charge ready voltage [V]	Overcharge voltage [V]	Storage element type
STO_CFG[2:0]			VOVDIS	VCHRDY	VOVCH	
L	L	L	2.51	2.61	3.79	Lithium-ion Super Capacitor (LiC)
L	L	H	2.51	2.61	3.49	Lithium-ion Super Capacitor 85 °C (LiC)
L	H	L	3.00	3.21	4.13	Lithium-ion
L	H	H	3.00	3.21	3.90	Lithium-ion (long life)
H	L	L	3.51	3.60	3.90	Lithium-ion (super long life)
H	L	H	3.00	3.30	4.13	Lithium Polymer (LiPo)
H	H	L	2.81	3.11	3.62	Lithium Iron Phosphate (LiFePO4)
H	H	H	2.61	2.70	3.90	Tadiran HLC1020

Table 14: Storage element thresholds configuration with **STO_CFG[2:0]** pins

Configuration pins			Minimum charging temperature [°C]	Maximum charging temperature [°C]	Minimum discharging temperature [°C]	Maximum discharging temperature [°C]	Storage element type
STO_CFG[2:0]			TEMPCOLDCH	TEMPHOTCH	TEMPCOLDDIS	TEMPHOTDIS	
L	L	L	-15	60	-15	60	Lithium-ion Super Capacitor (LiC)
L	L	H	-25	85	-25	85	Lithium-ion Super Capacitor 85 °C (LiC)
L	H	L	0	45	0	45	Lithium-ion
L	H	H	0	45	0	45	Lithium-ion (long life)
H	L	L	0	45	0	45	Lithium-ion (super long life)
H	L	H	0	45	0	45	Lithium Polymer (LiPo), NiMH
H	H	L	0	45	0	45	Lithium Iron Phosphate (LiFePO4)
H	H	H	-40	85	-40	85	Tadiran HLC1020

Table 15: Default temperature thresholds depending on **STO_CFG[2:0]** configuration with the recommended RDIV and RTH

DISCLAIMER: Storage element protection thresholds and temperature thresholds provided for each storage element type in the table above are indicative to support a wide range of storage element variants. They are provided as is to the best knowledge of e-peas's application laboratory. They should not replace the actual values provided in the storage element manufacturer's specifications and datasheet.

6.5. Boost Converter Timings

The boost converter timing multiplier (T_{MULT}) default value when the I²C is not used is x3.

A different boost timing multiplier value can be configured through the BSTxCFG registers (see Section 9.6), thus, configuring the peak current of the boost converter inductor (available by I²C register only).

Please refer to Table 37 for the different minimum and recommended L_{BOOSTx} values for each timing multiplier value.

6.6. Buck Converter

6.6.1. Load Voltage

The regulated voltage on **LOAD** output can be configured with the **LOAD_CFG[2:0]** pins (see Table 16), or through the BUCKCFG register (see Section 9.7).

Configuration pins			LOAD voltage [V]
LOAD_CFG[2:0]			V_{LOAD}
L	L	L	Buck disabled
L	L	H	0.6
L	H	L	0.9
L	H	H	1.2
H	L	L	1.5
H	L	H	1.8
H	H	L	2.2
H	H	H	2.5 ¹

Table 16: Configuration of LOAD voltage with **LOAD_CFG[2:0]**

1. This configuration is only available if $V_{OVDIS} \geq 2.5$ V.

NOTE: the configuration of **LOAD_CFG[2:0]** can only be changed when the AEM13920 is in **RESET STATE** (see Section 5.9.1 for more information).

6.6.2. Buck Converter Timings

The buck converter timing multiplier (T_{MULT}) default value when the I²C is not used is x2.

A different buck timing multiplier value can be configured through the BUCKCFG register (see Section 9.7), thus configuring the peak current of the buck converter inductor (available by I²C register only).

Please refer to Table 39 for the different L_{BUCK} values for each timing multiplier value.

6.7. Thermal Monitoring

Once the thermal monitoring has been enabled by connecting R_{DIV} and R_{TH} as shown in Figure 12, the thermal protection can be used.

The typical values of R_{DIV} and R_{TH} , found in Table 10, allow for having the default temperature thresholds depending on **STO_CFG[2:0]** configuration as shown in Table 15.

To use different temperature protection thresholds for storage element charge (see Section 9.9) and for storage element discharge (see Section 9.10), the TEMPCOLDCH, TEMPHOTCH, TEMPCOLDDIS, and TEMPHOTDIS registers can be used along with the following equations:

$$THRESH(T) = \frac{256 \cdot R_{TH}(T)}{R_{TH}(T) + R_{DIV}}$$

$$R_{TH}(T) = R_0 \cdot e^{B \cdot \left(\frac{1}{T} - \frac{1}{T_0} \right)}$$

$$T = \frac{B}{\ln\left(\frac{R_{TH}(T)}{R_0}\right) + \frac{B}{T_0}}$$

- THRESH is the unsigned 8-bit value to be written in the registers to set the temperature threshold to the temperature T [K].
- R_0 [Ω] is the resistance of the NTC thermistor at ambient temperature $T_0 = 298.15$ K (25 °C).
- $R_{TH}(T)$ [Ω] is the resistance of the thermistor at temperature T [K].
- T_0 [K] = 298.15 K (25 °C)
- T [K] is the current ambient temperature of the circuit.
- B is the characteristic constant of the thermistor, allowing to determine the resistance of the thermistor for a given temperature.

Please note that the thermistor must be of the NTC (Negative Temperature Coefficient) type.

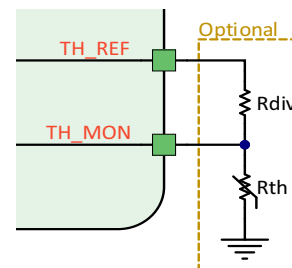


Figure 12: **TH_REF** and **TH_MON** connections

If thermal protection is not needed, it can be disabled through TEMPPROTECT register (see Section 9.11). In that case:

- If R_{DIV} and R_{TH} are connected, the temperature will continue to be measured periodically every $T_{TEMP,MON}$ but the protection thresholds will be ignored. The temperature data can be read through the TEMP register.
- If TH_MON is connected to $VINT$ and TH_REF left floating, the thermal monitoring will be disabled.

6.8. 5 V Charger

As explained in the Section 5.7, the 5 V charger implements a constant current (CC) mode and a constant voltage (CV) mode.

When in constant current mode, the maximum charging current $I_{5V,CC}$ can be set by connecting a resistor R_{5V_IMAX} between $5V_IMAX$ and GND:

$$I_{5V,CC} = \frac{50}{R_{5V_IMAX}}$$

R_{5V_IMAX} must be chosen so that $I_{5V,CC}$ complies with the range defined in Table 7. Example values can be found in Table 17:

Resistor [Ω]	Maximum Charging Current [mA]
R_{5V_IMAX}	$I_{5V,CC}$
370	135.0
680	73.5
1500	33.3
3700	13.5

Table 17: Typical resistor values for setting 5 V charger max. current

If $V_{STO} < V_{OVDIS}$, the charging current is limited to $I_{5V,OVDIS,CC}$.

To use the CV mode (disabled by default), the user must enable CV operation mode and configure the 5 V charger stop voltage $V_{5V,STOP}$ by I²C (see Section 9.8).

Please note that, regardless of the operating mode, the rise time of the voltage applied on the $5V_IN$ pin must not be too short. Thus, it is recommended to add an RC circuit in series with the $5V_IN$ pin which matches the following, with R_{5V} in series and C_{5V} between $5V_IN$ and GND:

$$R_{5V} \cdot C_{5V} > T_{5V,RISE}$$

- $T_{5V,RISE}$ is the minimum rise time from 0 V to 5 V on the $5V_IN$ pin (see Table 8). Comparing this to the RC constant adds a margin as the RC constant defines 63 % of the final voltage.
- R_{5V} must be determined so that, for the configured $I_{5V,CC}$, the voltage on the $5V_IN$ pin is:
 - above 3.60 V.
 - above $V_{STO} + 200$ mV.
- C_{5V} is determined from the value of R_{5V} using the equation above. A low charging current allows for high R_{5V} value and thus, for a low C_{5V} value.

6.9. Shipping Mode

The shipping mode, described in Section 5.8, is configured as follows:

- Shipping mode enabled by connecting the $SHIP_MODE$ pin to STO .
- Shipping mode disabled by connecting the $SHIP_MODE$ pin to GND or by leaving it floating.

7. I²C Serial Interface Protocol

The AEM13920 uses I²C communication for configuration as well as to provide information about system status and measurement data. Communication requires a serial data line (**SDA**) and a serial clock line (**SCL**). A device sending data is defined as a transmitter and a device receiving data as a receiver. The device that controls the communication is called a master and the device it controls is defined as the slave.

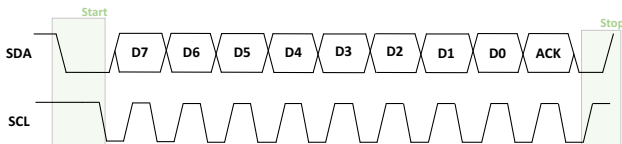


Figure 13: I²C transmission frame

The master is in charge of generating the clock, managing bus accesses and generating the start and stop bits. The AEM13920 is a slave that will receive configuration data or send the informations requested by the master.

The AEM13920 supports I²C Standard-mode (100 kHz maximum clock rate), Fast-mode (400 kHz maximum clock rate), and Fast-mode Plus (1 MHz maximum clock rate) device. Data is sent with the most significant bit first.

Here are some typical I²C interface states:

- When the communication is idle, both transmission lines are pulled-up (**SDA** and **SCL** are open drain outputs);
- Start bit (S): to initiate the transmission, the master switches the **SDA** line low while keeping **SCL** high. This is called the start bit;
- Stop bit (P): to end the transmission, the master switches the **SDA** line from low to high while keeping **SCL** high. This is called a stop bit;
- Repeated Start bit (Sr): it is used as a back-to-back start and stop bit. It is similar to a start condition, but when the bus is not on idle;
- ACK: to acknowledge a transmission, the device receiving the data (master in case of a read mode transmission, slave in case of a write mode transmission) switches **SDA** low;
- NACK: when the device receiving data keeps **SDA** high after the transmission of a byte. When reading a byte, this can mean that the master is done reading bytes from the slave.

To initiate the communication, the master sends a byte with the following informations:

- Bits [7:1] is the slave address, which is 0x41 for the AEM13920.

- Bit [0] is the communication mode: 1 for 'read mode' (used when the master reads informations from the slave) and 0 for 'write mode' (when the master writes informations to the slave);
- Slave replies with an ACK to acknowledge that the address has been successfully transmitted.

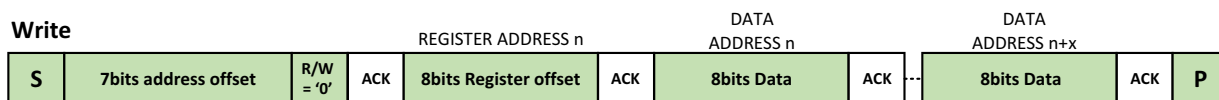
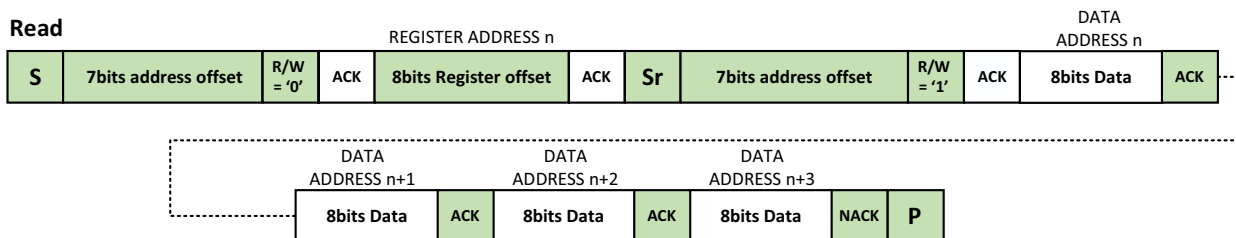
Here is the procedure for the master to write a slave register:

- Master sends the address of the slave in 'write' mode;
- Slave sends an ACK;
- Master sends the address of the register to be written. For example, for the TEMPCOLDCH register, the master sends the value 0x0B;
- Slave sends an ACK;
- Master sends the data to write to the register;
- Slave sends an ACK;
- If the master wants to write register at the next address (TEMPHOTCH in our example), it sends next value to write, without having to specify the address again. This can be done several times in a row for writing several consecutive registers;
- Else the master sends a stop bit (P).

Here is the procedure for the master to read a slave register:

- Master sends the address of the slave in 'write' mode;
- Slave sends an ACK;
- Master sends the address of the register to be read. For example, for the APM0LOAD register, the master sends the value 0x1F;
- Slave sends an ACK;
- Master sends a repeated start bit (Sr);
- Master sends the address of the slave in 'read' mode;
- Slave sends an ACK;
- Master provides the clock on **SCL** to allow the slave to shift the data of the read register on **SDA**;
- If the master wants to read register at the next address (APM1LOAD in our example), it sends an ACK and provides the clock for the slave to shift its following 8 bits of data. This can be done several times in a row for reading several registers;
- If the master wants to end the transmission, it sends a NACK to notify the slave that the transmission is over, and then sends a stop bit (P).

Both communications are described in Figure 14. Refer to Table 18 for all register addresses.

Write

Read


S = START

ACK = ACKNOWLEDGE

P = STOP

R/W = READ/WRITE

NACK = NOT ACKNOWLEDGE

Sr = RESTART

Figure 14: Read and write transmission

8. Register Map

Please note that the AEM13920 device address is 0x41.

Address	Name	Bit	Field Name	Access	Reset	Description
0x00	VERSION	[7:0]	VERSION	R	-	AEM13920 version number.
0x01	SRC1REGU0	[0:0]	MODE	R/W	0x01	SRC1 regulation mode.
		[3:1]	CFG0	R/W	0x00	
0x02	SRC1REGU1	[2:0]	CFG1	R/W	0x00	SRC1 regulation mechanism configuration.
		[5:3]	CFG2	R/W	0x00	
0x03	SRC2REGU0	[0:0]	MODE	R/W	0x01	SRC2 regulation mode.
		[3:1]	CFG0	R/W	0x00	
0x04	SRC2REGU1	[2:0]	CFG1	R/W	0x00	SRC2 regulation mechanism configuration.
		[5:3]	CFG2	R/W	0x00	
0x05	VOVDIS	[5:0]	THRESH	R/W	0x06	Storage element overdischarge threshold.
0x06	VCHRDY	[6:0]	THRESH	R/W	0x05	Storage element charge ready threshold.
0x07	VOVCH	[6:0]	THRESH	R/W	0x3A	Storage element overcharge threshold.
0x08	BST1CFG	[0:0]	EN	R/W	0x01	Boost SRC1 enable.
		[1:1]	HPEN	R/W	0x01	Boost SRC1 high-power mode enable.
		[4:2]	TMULT	R/W	0x01	Boost SRC1 current configuration.
0x09	BST2CFG	[0:0]	EN	R/W	0x01	Boost SRC2 enable.
		[1:1]	HPEN	R/W	0x01	Boost SRC2 high-power mode enable.
		[4:2]	TMULT	R/W	0x01	Boost SRC2 current configuration.
0x0A	BUCKCFG	[2:0]	VLOAD	R/W	0x00	Buck output voltage configuration.
		[5:3]	TMULT	R/W	0x03	Buck current configuration.
0x0B	TEMPCOLDCH	[7:0]	THRESH	R/W	0xD1	Cold temperature threshold for storage element charging.
0x0C	TEMPHOTCH	[7:0]	THRESH	R/W	0x18	Hot temperature threshold for storage element charging.
0x0D	TEMPCOLDDIS	[7:0]	THRESH	R/W	0xD1	Cold temperature threshold for storage element discharging.
0x0E	TEMPHOTDIS	[7:0]	THRESH	R/W	0x18	Hot temperature threshold for storage element discharging.
0x0F	TEMPPROTECT	[0:0]	EN	R/W	0x01	Thermal protection enable.
0x10	SRCLOW	[2:0]	SRC1THRESH	R/W	0x00	V _{SRCLOW} threshold for SRC1.
		[5:3]	SRC2THRESH	R/W	0x00	V _{SRCLOW} threshold for SRC2.
0x11	APM	[0:0]	SRC1EN	R/W	0x00	APM SRC1 enable.
		[1:1]	SRC2EN	R/W	0x00	APM SRC2 enable.
		[2:2]	LOADEN	R/W	0x00	APM LOAD enable.
		[3:3]	MODE	R/W	0x00	APM mode.
		[4:4]	WINDOW	R/W	0x00	APM window.
		[5:5]	CHG5VEN	R/W	0x00	APM 5 V charger enable.

Table 18: Register map (part 1)



Address	Name	Bit	Field Name	Access	Reset	Description
0x12	IRQEN0	[0:0]	I2CRDY	R/W	0x01	IRQ serial interface ready enable.
		[1:1]	VOVDIS	R/W	0x00	IRQ VOVDIS enable.
		[2:2]	VCHRDY	R/W	0x00	IRQ VCHRDY enable.
		[3:3]	VOVCH	R/W	0x00	IRQ VOVCH enable.
		[4:4]	SRCLOW	R/W	0x00	IRQ source low threshold (SRCx) enable.
		[5:5]	TEMPCH	R/W	0x00	IRQ temperature (charge) enable.
		[6:6]	TEMPDIS	R/W	0x00	IRQ temperature (discharge) enable.
		[7:7]	CHG5VCONN	R/W	0x00	IRQ 5 V charger connected enable.
0x13	IRQEN1	[0:0]	SRC1MPPTSTART	R/W	0x00	IRQ MPPT start (SRC1) enable.
		[1:1]	SRC1MPPTDONE	R/W	0x00	IRQ MPPT done (SRC1) enable.
		[2:2]	SRC2MPPTSTART	R/W	0x00	IRQ MPPT start (SRC2) enable.
		[3:3]	SRC2MPPTDONE	R/W	0x00	IRQ MPPT done (SRC2) enable.
		[4:4]	STODONE	R/W	0x00	IRQ STO measurement done enable.
		[5:5]	TEMPDONE	R/W	0x00	IRQ temperature measurement done enable.
		[6:6]	APMDONE	R/W	0x00	IRQ APM done enable.
		[7:7]	APMERR	R/W	0x00	IRQ APM error enable.
0x14	CTRL	[0:0]	UPDATE	R/W	0x00	Load I ² C registers configuration.
		[1:1]	-	R	-	Reserved.
		[2:2]	SYNCBUSY	R	0x00	Synchronization busy flag.
0x15	IRQFLG0	[0:0]	I2CRDY	R	0x00	IRQ serial interface ready flag.
		[1:1]	VOVDIS	R	0x00	IRQ VOVDIS flag.
		[2:2]	VCHRDY	R	0x00	IRQ VCHRDY flag.
		[3:3]	VOVCH	R	0x00	IRQ VOVCH flag.
		[4:4]	SRCLOW	R	0x00	IRQ source low threshold (SRCx) flag.
		[5:5]	TEMPCH	R	0x00	IRQ temperature (charge) flag.
		[6:6]	TEMPDIS	R	0x00	IRQ temperature (discharge) flag.
		[7:7]	CHG5VCONN	R	0x00	IRQ 5 V charger connected flag.
0x16	IRQFLG1	[0:0]	SRC1MPPTSTART	R	0x00	IRQ MPPT start (SRC1) flag.
		[1:1]	SRC1MPPTDONE	R	0x00	IRQ MPPT done (SRC1) flag.
		[2:2]	SRC2MPPTSTART	R	0x00	IRQ MPPT start (SRC2) flag.
		[3:3]	SRC2MPPTDONE	R	0x00	IRQ MPPT done (SRC2) flag.
		[4:4]	STODONE	R	0x00	IRQ STO measurement done flag.
		[5:5]	TEMPDONE	R	0x00	IRQ temperature measurement done flag.
		[6:6]	APMDONE	R	0x00	IRQ APM done flag.
		[7:7]	APMERR	R	0x00	IRQ APM error flag.
0x17	STATUS0	[0:0]	OVDIS	R	0x00	Status overdischarge.
		[1:1]	CHRDY	R	0x00	Status charge ready.
		[2:2]	OVCH	R	0x00	Status overcharge.
		[3:3]	SRC1LOW	R	0x00	Status source low threshold (SRC1).
		[4:4]	SRC2LOW	R	0x00	Status source low threshold (SRC2).
		[5:5]	CHG5VCONN	R	0x00	Status 5 V charger connected.

Table 18: Register map (part 2)



Address	Name	Bit	Field Name	Access	Reset	Description
0x18	STATUS1	[0:0]	TEMPCOLDCH	R	0x00	Status cold temperature (charge).
		[1:1]	TEMPHOTCH	R	0x00	Status hot temperature (charge).
		[2:2]	TEMPCOLDDIS	R	0x00	Status cold temperature (discharge).
		[3:3]	TEMPHOTDIS	R	0x00	Status hot temperature (discharge).
0x19	APM0SRC1	[7:0]	DATA	R	0x00	APM data 0 (SRC1).
0x1A	APM1SRC1	[7:0]	DATA	R	0x00	APM data 1 (SRC1).
0x1B	APM2SRC1	[7:0]	DATA	R	0x00	APM data 2 (SRC1).
0x1C	APM0SRC2	[7:0]	DATA	R	0x00	APM data 0 (SRC2).
0x1D	APM1SRC2	[7:0]	DATA	R	0x00	APM data 1 (SRC2).
0x1E	APM2SRC2	[7:0]	DATA	R	0x00	APM data 2 (SRC2).
0x1F	APM0LOAD	[7:0]	DATA	R	0x00	APM data 0 (LOAD).
0x20	APM1LOAD	[7:0]	DATA	R	0x00	APM data 1 (LOAD).
0x21	APM2LOAD	[7:0]	DATA	R	0x00	APM data 2 (LOAD).
0x22	APMERR	[0:0]	SRC1OV	R	0x00	APM counter overflow SRC1 .
		[1:1]	SRC1NVLD	R	0x00	APM counter corrupted SRC1 .
		[2:2]	SRC2OV	R	0x00	APM counter overflow SRC2 .
		[3:3]	SRC2NVLD	R	0x00	APM counter corrupted SRC2 .
		[4:4]	LOADOV	R	0x00	APM counter overflow LOAD .
		[5:5]	LOADNVLD	R	0x00	APM counter corrupted LOAD .
		[6:6]	CHG5VLIM	R	0x00	5 V charger current limited due to overdischarged storage element.
0x23	TEMP	[7:0]	DATA	R	0x00	Temperature monitoring value.
0x24	STO	[7:0]	DATA	R	0x00	STO monitoring value.
0x25	SRC1	[7:0]	DATA	R	0x00	SRC1 monitoring value.
0x26	SRC2	[7:0]	DATA	R	0x00	SRC2 monitoring value.
0x27	APM0CHG5V	[7:0]	DATA	R	0x00	APM data 0 (CHG5V).
0x28	APM1CHG5V	[7:0]	DATA	R	0x00	APM data 1 (CHG5V).
0x29	CHG5V	[0:0]	EN	R/W	0x01	5 V charger enable.
		[1:1]	CVEN	R/W	0x00	Constant voltage (CV) mode enable.
		[6:2]	THRESH	R/W	0x00	5 V charger stop voltage threshold.
0x2A	APMACC	[7:0]	CFG	R/W	0x00	Number of APM window accumulations.
...	RSVD	-	-	R	-	Reserved.
0xE0	PN0	[7:0]	DATA	R	0x30	Part number 0 data.
0xE1	PN1	[7:0]	DATA	R	0x32	Part number 1 data.
0xE2	PN2	[7:0]	DATA	R	0x39	Part number 2 data.
0xE3	PN3	[7:0]	DATA	R	0x33	Part number 3 data.
0xE4	PN4	[7:0]	DATA	R	0x31	Part number 4 data.

Table 18: Register map (part 3)

9. Registers Configuration

9.1. I²C Control (CTRL)

The CTRL register allows for updating the AEM13920 configurations and checking registers synchronization status.

CTRL register	0x14	R/W		
Bit [7:3]	Bit [2]	Bit [1]	Bit [0]	
RESERVED	SYNCBUSY	RESERVED	UPDATE	
0x00	0x00	0x00	0x00	

Table 19: CTRL register

Bit [2]: system registers synchronization (CTRL.SYNCBUSY)

This field indicates whether the synchronization from the I²C registers to the system registers is ongoing or not.

- 0: CTRL register not synchronizing.
- 1: CTRL register synchronizing.

Bit [0]: system registers update (CTRL.UPDATE)

This field is used to control the source of the AEM13920 configurations (GPIO or I²C) and to update the configurations with the current I²C register values or GPIO states.

- 0: GPIO
 - W: load configurations from the GPIO.
 - R: configurations from the GPIO is currently used if read as 0.
- 1: I²C
 - W: load configurations from the I²C registers.
 - R: configurations from the I²C registers is currently used if read as 1.

NOTE: if the AEM13920 is already configured through the I²C registers, writing any register does not have any effect until 1 is written to the CTRL.UPDATE field, leading to the AEM13920 to read the new register values and apply them.

NOTE: when using I²C register configuration, the user can switch back to GPIO configuration by writing 0 to the CTRL.UPDATE field. In that case, the settings previously written to the IRQEN registers are still valid even when using GPIO configuration, as well as the data in IRQFLG register.

9.2. Version Register (VERSION)

The VERSION register holds the version of the AEM13920.

VERSION register	0x00	R
Bit [7:0]		
VERSION		
-		

Table 20: VERSION register

Bit [7:0]: version number (VERSION.VERSION)



9.3. Source Regulation (SRCxREGUx)

The SRCxREGUx registers allow the configuration of the SRCx regulation mechanism.

Table 21 shows the use of SRCxREGUx registers according to the source regulation mode.

- **SRCxREGU0.MODE = 0**: constant voltage regulation mode, as described in Section 9.3.3.
 - LVL [7:0] - defines the constant regulation voltage.
- **SRCxREGU0.MODE = 1**: MPPT ratio regulation mode, as described in Section 9.3.4.
 - MPPT_RATIO [2:0] - defines the MPPT ratio.
 - MPPT_WAIT [2:0] - defines the MPPT wait time.
 - MPPT_PERIOD [2:0] - defines the MPPT period.

	SRCxREGU1								SRCxREGU0							
	Bit [7]	Bit [6]	Bit [5]	Bit [4]	Bit [3]	Bit [2]	Bit [1]	Bit [0]	Bit [7]	Bit [6]	Bit [5]	Bit [4]	Bit [3]	Bit [2]	Bit [1]	Bit [0]
Register Field			CFG2				CFG1						CFG0		MODE	
Mode 0: Constant Voltage			LVL[7:6]				LVL[5:3]						LVL [2:0]		MODE	
Mode 1: MPPT			MPPT_PERIOD [2:0]				MPPT_WAIT [2:0]						MPPT_RATIO [2:0]		MODE	

Table 21: Summary of SRCxREGUx register fields

9.3.1. SRCxREGU0

The SRCxREGU0 is the first set of registers for configuring the SRCx regulation mechanism.

SRC1REGU0 register SRC2REGU0 register	0x01 0x03	R/W R/W
Bit [7:4]	Bit [3:1]	Bit [0]
RESERVED	CFG0	MODE
0x00	0x00	0x01

Table 22: SRCxREGU0 register

Bit [3:1]: SRCx configuration 0 (SRCxREGU0.CFG0)

This fields is used to configure the following SRCx regulation mechanism parameters:

- If **MODE = 0**: SRCxREGU0.CFG0 configures the SRCx regulation constant voltage, along with the other SRCxREGU1.CFGx register fields (see Section 9.3.3).
- If **MODE = 1**: SRCxREGU0.CFG0 configures the MPPT ratio (see Table 25).

Bit [0]: SRCx regulation mode (SRCxREGU0.MODE)

This fields is used to configure the SRCx regulation mode:

- 0: constant voltage regulation mode.
- 1: MPPT ratio (V_{MPP}/V_{OC} or ZMPP) regulation mode.

9.3.2. SRCxREGU1

The SRCxREGU1 is the second set of registers for configuring the SRCx regulation mechanism.

	SRC1REGU1 register			0x02		R/W
	SRC2REGU1 register			0x04		R/W
	Bit [7]	Bit [6]	Bit [5]	Bit [4]	Bit [3]	Bit [2:0]
Mode 0: Constant Voltage	RESERVED			CFG2		CFG1
	0x00			0x00		0x00
Mode 1: MPPT	RESERVED		CFG2			CFG1
	0x00		0x00			0x00

Table 23: SRCxREGU1 register

Bit [5:3]: SRCx configuration 2 (SRCxREGU1.CFG2)

This fields is used to configure the following SRCx regulation mechanism parameters:

- If **MODE = 0**: SRCxREGU1.CFG2 configures the SRCx constant regulation voltage, along with the other SRCxREGUx.CFGx register fields (see Section 9.3.3).
- If **MODE = 1**: SRCxREGU1.CFG2 configures the SRCx MPPT period $T_{MPPT,PERIOD}$ (see Table 27).

NOTE: When SRCxREGU0.MODE = 0 (constant voltage mode) the SRCxREGU1.CFG2 field uses bits [4:3] of the SRCxREGU1 register, instead of bits [5:3] when MODE = 1 as shown in Table 23.

Bit [2:0]: SRCx configuration 1 (SRCxREGU1.CFG1)

This fields is used to configure the following SRCx regulation mechanism parameters:

- If **MODE = 0**: SRCxREGU1.CFG1 configures the SRCx constant regulation voltage, along with the other SRCxREGUx.CFGx register fields (see Section 9.3.3).
- If **MODE = 1**: SRCxREGU1.CFG1 configures the SRCx MPPT wait time $T_{MPPT,WAIT}$ (see Table 26).



9.3.3. Constant Voltage Configuration

Table 24 describes how to configure SRCxREGUx registers when the AEM13920 source regulation mode is set to constant voltage (SRCxREGU0.MODE = 0).

See Table 21 for a description of how LVL[7:0] is distributed across SRCxREGUx registers.

LVL [7:0]	V _{SRCx,REG} [V]	LVL [7:0]	V _{SRCx,REG} [V]	LVL [7:0]	V _{SRCx,REG} [V]	LVL [7:0]	V _{SRCx,REG} [V]	LVL [7:0]	V _{SRCx,REG} [V]	LVL [7:0]	V _{SRCx,REG} [V]
0x00	Source Low ¹	0x2A	0.375	0x4A	0.855	0x6A	1.335	0x8A	1.970	0xAA	3.182
...		0x2B	0.390	0x4B	0.870	0x6B	1.350	0x8B	1.993	0xAB	3.227
0x0C		0x2C	0.405	0x4C	0.885	0x6C	1.365	0x8C	2.015	0xAC	3.273
0x0D	0.120	0x2D	0.420	0x4D	0.900	0x6D	1.380	0x8D	2.037	0xAD	3.318
0x0E	0.128	0x2E	0.435	0x4E	0.915	0x6E	1.395	0x8E	2.060	0xAE	3.364
0x0F	0.135	0x2F	0.450	0x4F	0.930	0x6F	1.410	0x8F	2.082	0xAF	3.409
0x10	0.143	0x30	0.465	0x50	0.945	0x70	1.425	0x90	2.104	0xB0	3.455
0x11	0.150	0x31	0.480	0x51	0.960	0x71	1.440	0x91	2.127	0xB1	3.500
0x12	0.158	0x32	0.495	0x52	0.975	0x72	1.455	0x92	2.149	0xB2	3.545
0x13	0.165	0x33	0.510	0x53	0.990	0x73	1.470	0x93	2.172	0xB3	3.591
0x14	0.173	0x34	0.525	0x54	1.005	0x74	1.478	0x94	2.194	0xB4	3.636
0x15	0.180	0x35	0.540	0x55	1.020	0x75	1.500	0x95	2.227	0xB5	3.682
0x16	0.188	0x36	0.555	0x56	1.035	0x76	1.522	0x96	2.273	0xB6	3.727
0x17	0.195	0x37	0.570	0x57	1.050	0x77	1.545	0x97	2.318	0xB7	3.773
0x18	0.203	0x38	0.585	0x58	1.065	0x78	1.567	0x98	2.364	0xB8	3.818
0x19	0.210	0x39	0.600	0x59	1.080	0x79	1.590	0x99	2.409	0xB9	3.864
0x1A	0.218	0x3A	0.615	0x5A	1.095	0x7A	1.612	0x9A	2.455	0xBA	3.909
0x1B	0.225	0x3B	0.630	0x5B	1.110	0x7B	1.634	0x9B	2.500	0xBB	3.955
0x1C	0.233	0x3C	0.645	0x5C	1.125	0x7C	1.657	0x9C	2.545	0xBC	4.000
0x1D	0.240	0x3D	0.660	0x5D	1.140	0x7D	1.679	0x9D	2.591	0xBD	4.045
0x1E	0.248	0x3E	0.675	0x5E	1.155	0x7E	1.701	0x9E	2.636	0xBE	4.091
0x1F	0.255	0x3F	0.690	0x5F	1.170	0x7F	1.724	0x9F	2.682	0xBF	4.136
0x20	0.263	0x40	0.705	0x60	1.185	0x80	1.746	0xA0	2.727	0xC0	4.182
0x21	0.270	0x41	0.720	0x61	1.200	0x81	1.769	0xA1	2.773	0xC1	4.227
0x22	0.278	0x42	0.735	0x62	1.215	0x82	1.791	0xA2	2.818	0xC2	4.273
0x23	0.285	0x43	0.750	0x63	1.230	0x83	1.813	0xA3	2.864	0xC3	4.318
0x24	0.293	0x44	0.765	0x64	1.245	0x84	1.836	0xA4	2.909	0xC4	4.364
0x25	0.300	0x45	0.780	0x65	1.260	0x85	1.858	0xA5	2.955	0xC5	4.409
0x26	0.315	0x46	0.795	0x66	1.275	0x86	1.881	0xA6	3.000	0xC6	4.455
0x27	0.330	0x47	0.810	0x67	1.290	0x87	1.903	0xA7	3.045	...	
0x28	0.345	0x48	0.825	0x68	1.305	0x88	1.925	0xA8	3.091	...	
0x29	0.360	0x49	0.840	0x69	1.320	0x89	1.948	0xA9	3.136	0xFF	

Table 24: SRCx constant voltage values configured by SRCxREGUx (SRCxREGU0.MODE = 0)

1. Setting SRCxREGUx.LVL lower than 0x0D causes the AEM13920 to consider the SRCx voltage to be lower than the default V_{SRCLOW}, which may cause the AEM13920 to enter **SLEEP STATE** if the SLEEP condition is met (see Section 5.9.6).



9.3.4. MPPT Configuration

This section describes how to configure the MPPT module through the SRCxREGUx registers when the AEM13920 source regulation mode is set to MPPT ratio (SRCxREGU0.MODE = 1).

See Table 21 for the distribution of MPPT_RATIO [2:0], MPPT_WAIT [2:0], MPPT_PERIOD [2:0] values across SRCxREGUx registers.

- Table 25 shows the configuration of SRCxREGU0.CFG0 register field to set the MPPT ratio.
- Table 26 shows the configuration of SRCxREGU1.CFG1 register field to set the MPPT wait time before open-circuit voltage measure.
- Table 27 shows the configuration of SRCxREGU1.CFG2 register field to set the MPPT period.

SRCxREGU0.CFG0 MPPT_RATIO [2:0]			R _{MPPT}
0	0	0	35 %
0	0	1	50 %
0	1	0	65 %
0	1	1	70 %
1	0	0	75 %
1	0	1	80 %
1	1	0	85 %
1	1	1	SRC1: ZMPP SRC2: 100 %

Table 25: SRCx MPPT ratio/ZMPP configured by SRCxREGUx (SRCxREGU0.MODE = 1)

SRCxREGU1.CFG1 MPPT_WAIT [2:0]			T _{MPPT,WAIT} ¹ [ms]
0	0	0	1.8
0	0	1	3.6
0	1	0	7.3
0	1	1	15
1	0	0	29
1	0	1	116
1	1	0	233
1	1	1	465

Table 26: SRCx MPPT wait time configured by SRCxREGUx (SRCxREGU0.MODE = 1)

1. The total time spent in open-circuit is the sum of $T_{MPPT,WAIT}$ (configurable, see table above) and $T_{MPPT,MEASURE}$ (fixed, see Table 8).

SRCxREGU1.CFG2 MPPT_PERIOD [2:0]			T _{MPPT,PERIOD} [ms]
0	0	0	116 ¹
0	0	1	233
0	1	0	465
0	1	1	931
1	0	0	1862
1	0	1	3724
1	1	0	7447
1	1	1	14895

Table 27: SRCx MPPT period configured by SRCxREGUx (SRCxREGU0.MODE = 1)

1. If $T_{MPPT,PERIOD}$ is set to 116 ms for any of the two SRCx, the APM WINDOW will automatically be set to 116 ms by the AEM13920 for all the APM modules.

9.4. Source Low Threshold (SRCLOW)

The SRCLOW register allows the configuration of the SRCx voltage threshold (V_{SRCLOW}) below which the AEM13920 switches to SLEEP STATE. When V_{MPP} is higher than V_{SRCLOW} , the source is considered available.

NOTE: The source low threshold is mainly useful in MPPT ratio mode. When using the constant voltage mode, leave the SRCLOW register at its default value unless forcing the reset state is desired.

SRCLOW register		0x10	R/W
Bit [7:6]	Bit [5:3]	Bit [2:0]	
RESERVED	SRC2THRESH	SRC1THRESH	
0x00	0x00	0x00	

Table 28: SRCLOW register

Bit [5:3]: source 2 low threshold (SRCLOW.SRC2THRESH)

This field allows the configuration of the V_{SRCLOW} threshold of SRC2.

Bit [2:0]: source 1 low threshold (SRCLOW.SRC1THRESH)

This field allows the configuration of the V_{SRCLOW} threshold of SRC1.

Table 29 shows the threshold voltages V_{SRCLOW} according to the configuration of SRCLOW.SRCxTHRESH fields.

SRCLOW.SRCxTHRESH	Source low voltage threshold V_{SRCLOW} [V]
0x00	0.113
0x01	0.203
0x02	0.255
0x03	0.300
0x04	0.360
0x05	0.405
0x06	0.510
0x07	0.600

Table 29: V_{SRCLOW} thresholds as configured by SRCLOW register

9.5. Storage Element Threshold Voltages (VOVDIS / VCHRDY / VOVCH)

The storage element protection thresholds, described in Section 6.4, can be set independently by the registers VOVDIS, VCHRDY and VOVCH.

9.5.1. Overdischarge Voltage (VOVDIS)

The VOVDIS register allows the configuration of the storage element voltage below which the storage element is considered to be fully depleted, and must not be discharged any further (V_{OVDIS}).

VOVDIS register		0x05	R/W
Bit [7:6]		Bit [5:0]	
RESERVED		THRESH	
0x00		0x06	

Table 30: VOVDIS register

Bit [5:0]: storage element overdischarge threshold (VOVDIS.THRESH)

This field allows the configuration of V_{OVDIS} based on Table 31 values or based on the following formula:

$$V_{OVDIS} = 2.400 + THRESH \cdot 0.01875$$

NOTE: Respect $V_{OVDIS} < V_{CHRDY} < V_{OVCH}$ and keep a 100 mV minimum margin between them to avoid false state transitions.

VOVDIS. THRESH	V_{OVDIS} [V]	VOVDIS. THRESH	V_{OVDIS} [V]	VOVDIS. THRESH	V_{OVDIS} [V]	VOVDIS. THRESH	V_{OVDIS} [V]
0x00	2.400	0x10	2.700	0x20	3.000	0x30	3.300
0x01	2.419	0x11	2.719	0x21	3.019	0x31	3.319
0x02	2.438	0x12	2.738	0x22	3.038	0x32	3.338
0x03	2.456	0x13	2.756	0x23	3.056	0x33	3.356
0x04	2.475	0x14	2.775	0x24	3.075	0x34	3.375
0x05	2.494	0x15	2.794	0x25	3.094	0x35	3.394
0x06	2.513	0x16	2.813	0x26	3.113	0x36	3.413
0x07	2.531	0x17	2.831	0x27	3.131	0x37	3.431
0x08	2.550	0x18	2.850	0x28	3.150	0x38	3.450
0x09	2.569	0x19	2.869	0x29	3.169	0x39	3.469
0x0A	2.588	0x1A	2.888	0x2A	3.188	0x3A	3.488
0x0B	2.606	0x1B	2.906	0x2B	3.206	0x3B	3.506
0x0C	2.625	0x1C	2.925	0x2C	3.225	0x3C	3.525
0x0D	2.644	0x1D	2.944	0x2D	3.244	0x3D	3.544
0x0E	2.663	0x1E	2.963	0x2E	3.263	0x3E	3.563
0x0F	2.681	0x1F	2.981	0x2F	3.281	0x3F	3.581

Table 31: Storage element V_{OVDIS} configuration by VOVDIS register

9.5.2. Charge Ready Voltage (VCHRDY)

The VCHRDY register allows the configuration of the storage element voltage required to switch to **SUPPLY STATE** (V_{CHRDY}).

VCHRDY register		0x06	R/W
Bit [7]		Bit [6:0]	
RESERVED		THRESH	
0x00		0x05	

Table 32: VCHRDY register

Bit [6:0]: storage element charge ready threshold (VCHRDY.THRESH)

This field allows the configuration of V_{CHRDY} based on Table 33 values or based on the following formula:

$$V_{CHRDY} = 2.456 + THRESH \cdot 0.01875$$

NOTE: Respect $V_{OVDIS} < V_{CHRDY} < V_{OVCH}$ and keep a 100 mV minimum margin between them to avoid false state transitions.

VCHRDY. THRESH	V_{CHRDY} [V]	VCHRDY. THRESH	V_{CHRDY} [V]	VCHRDY. THRESH	V_{CHRDY} [V]	VCHRDY. THRESH	V_{CHRDY} [V]
0x00	2.456	0x19	2.925	0x32	3.394	0x4B	3.862
0x01	2.475	0x1A	2.944	0x33	3.413	0x4C	3.881
0x02	2.494	0x1B	2.963	0x34	3.431	0x4D	3.900
0x03	2.513	0x1C	2.981	0x35	3.450	0x4E	3.918
0x04	2.531	0x1D	3.000	0x36	3.469	0x4F	3.937
0x05	2.550	0x1E	3.019	0x37	3.488	0x50	3.956
0x06	2.569	0x1F	3.038	0x38	3.506	0x51	3.975
0x07	2.588	0x20	3.056	0x39	3.525	0x52	3.993
0x08	2.606	0x21	3.075	0x3A	3.544	0x53	4.012
0x09	2.625	0x22	3.094	0x3B	3.563	0x54	4.031
0x0A	2.644	0x23	3.113	0x3C	3.581	0x55	4.050
0x0B	2.663	0x24	3.131	0x3D	3.600	0x56	4.068
0x0C	2.681	0x25	3.150	0x3E	3.619	0x57	4.087
0x0D	2.700	0x26	3.169	0x3F	3.638	0x58	4.106
0x0E	2.719	0x27	3.188	0x40	3.656	0x59	4.125
0x0F	2.738	0x28	3.206	0x41	3.675	0x5A	4.143
0x10	2.756	0x29	3.225	0x42	3.693	0x5B	4.162
0x11	2.775	0x2A	3.244	0x43	3.712	0x5C	4.181
0x12	2.794	0x2B	3.263	0x44	3.731	0x5D	4.200
0x13	2.813	0x2C	3.281	0x45	3.750	0x5E	4.218
0x14	2.831	0x2D	3.300	0x46	3.768	0x5F	4.237
0x15	2.850	0x2E	3.319	0x47	3.787	0x60	4.256
0x16	2.869	0x2F	3.338	0x48	3.806	0x61	4.275
0x17	2.888	0x30	3.356	0x49	3.825	0x62	4.293
0x18	2.906	0x31	3.375	0x4A	3.843	0x63	4.312

Table 33: Storage element V_{CHRDY} configuration by VCHRDY register



9.5.3. Overcharge Voltage (VOVCH)

The VOVCH register allows the configuration of the storage element maximum voltage before disabling its charging (V_{OVCH}).

VOVCH register		0x07	R/W
Bit [7]		Bit [6:0]	
RESERVED		THRESH	
0x00		0x3A	

Table 34: VOVCH register

Bit [6:0]: storage element overcharge threshold (VOVCH.THRESH)

This field allows the configuration of V_{OVCH} based on Table 35 values or based on the following:

- $0x00 \leq \text{VOVCH.THRESH} \leq 0x65$:

$$V_{OVCH} = 2.700 + \text{THRESH} \cdot 0.01875$$
- $\text{VOVCH.THRESH} \geq 0x66$:

$$V_{OVCH} = 4.594V$$

NOTE: Respect $V_{OVDIS} < V_{CHRDY} < V_{OVCH}$ and keep a 100 mV minimum margin between them to avoid false state transitions.

VOVCH.THRESH	V_{OVCH} [V]	VOVCH.THRESH	V_{OVCH} [V]	VOVCH.THRESH	V_{OVCH} [V]	VOVCH.THRESH	V_{OVCH} [V]
0x00	2.700	0x1B	3.206	0x36	3.713	0x51	4.219
0x01	2.719	0x1C	3.225	0x37	3.731	0x52	4.238
0x02	2.738	0x1D	3.244	0x38	3.750	0x53	4.256
0x03	2.756	0x1E	3.263	0x39	3.769	0x54	4.275
0x04	2.775	0x1F	3.281	0x3A	3.788	0x55	4.294
0x05	2.794	0x20	3.300	0x3B	3.806	0x56	4.313
0x06	2.813	0x21	3.319	0x3C	3.825	0x57	4.331
0x07	2.831	0x22	3.338	0x3D	3.844	0x58	4.350
0x08	2.850	0x23	3.356	0x3E	3.863	0x59	4.369
0x09	2.869	0x24	3.375	0x3F	3.881	0x5A	4.388
0x0A	2.888	0x25	3.394	0x40	3.900	0x5B	4.406
0x0B	2.906	0x26	3.413	0x41	3.919	0x5C	4.425
0x0C	2.925	0x27	3.431	0x42	3.938	0x5D	4.444
0x0D	2.944	0x28	3.450	0x43	3.956	0x5E	4.463
0x0E	2.963	0x29	3.469	0x44	3.975	0x5F	4.481
0x0F	2.981	0x2A	3.488	0x45	3.994	0x60	4.500
0x10	3.000	0x2B	3.506	0x46	4.013	0x61	4.519
0x11	3.019	0x2C	3.525	0x47	4.031	0x62	4.538
0x12	3.038	0x2D	3.544	0x48	4.050	0x63	4.556
0x13	3.056	0x2E	3.563	0x49	4.069	0x64	4.575
0x14	3.075	0x2F	3.581	0x4A	4.088	0x65	4.594
0x15	3.094	0x30	3.600	0x4B	4.106	...	
0x16	3.113	0x31	3.619	0x4C	4.125	0x7F	
0x17	3.131	0x32	3.638	0x4D	4.144		
0x18	3.150	0x33	3.656	0x4E	4.163		
0x19	3.169	0x34	3.675	0x4F	4.181		
0x1A	3.188	0x35	3.694	0x50	4.200		

Table 35: Storage element V_{OVCH} configuration by VOVCH register

9.6. Boost Converters (BSTxCFG)

The BSTxCFG register allows the configuration of the boost converters.

BST1CFG register		0x08	R/W
BST2CFG register		0x09	R/W
Bit [7:5]	Bit [4:2]	Bit [1]	Bit [0]
RESERVED	TMULT	HPEN	EN
0x00	0x01 ¹	0x01	0x01

Table 36: BSTxCFG registers

1. The boost converter timing multiplier default value is different depending on whether the AEM13920 is configured by I²C or not. See Section 6.5 for the default value when I²C is not used.

Bit [4:2]: boost converter timing multiplier (BSTxCFG.TMULT)

This field allows the modification of the peak current of each boost inductor by increasing/decreasing the on/off timings of the corresponding boost converter. The higher the timing multiplier, the higher the boost inductor peak current, and thus the higher the average source current pulled from SRCx to STO.

The peak current in the inductor also depends on the value of the inductor (see Table 37).

Bit [1]: boost converter high-power mode enable (BSTxCFG.ENHP)

This field allows enabling or disabling the automatic high-power mode.

- 0: disable the high-power mode.
- 1: enable the automatic high-power mode. The AEM13920 will automatically enter high-power mode if needed, allowing for more power to be harvested from SRCx (see Section 5.2.4).

Bit [0]: boost converter enable (BSTxCFG.EN)

This field allows enabling or disabling each boost converter.

- 0: disable the boost converter.
- 1: enable the boost converter.

BSTxCFG.TMULT	Timing Multiplier	Minimum ¹ L _{BOOSTx} [μH]	Recommended ² L _{BOOSTx} [μH]
0x00	x1	3.3	10
0x01	x2	6.6	15
0x02	x3	9.9	33
0x03	x4	13.2	47
0x04	x6	19.8	68
0x05	x8	26.4	100
0x06	x12	39.6	120
0x07	x16	52.8	180

Table 37: Timing multiplier configured by TMULT field with corresponding boost inductor value

1. Never install an inductor with an inductance (effective value including tolerance, derating, etc.) lower than those values for each setting of the timing multiplier. This would cause permanent damage to the AEM13920.
2. Those values provide the best efficiency/current capability trade-off according to the tests carried out in the e-peas laboratory.



9.7. Buck Converter (BUCKCFG)

The BUCK register allows the configuration of the buck converter, which output is the [LOAD](#) pin.

NOTE: the configuration of the BUCKCFG.VLOAD field must not be set to "OFF" (0x00) if the buck converter was previously enabled, whether it was through the [LOAD_CFG\[2:0\]](#) pins or using the BUCKCFG register.

BUCKCFG register		0x0A	R/W
Bit [7:6]	Bit [5:3]	Bit [2:0]	
RESERVED	TMULT	VLOAD	
0x00	0x03 ¹	0x00	

Table 38: BUCKCFG register

1. The buck converter timing multiplier default value is different depending on whether the AEM13920 is configured by I²C or not. See Section 6.6.2 for the default value when I²C is not used.

Bit [5:3]: buck converter timing multiplier (BUCKCFG.TMULT)

This field allows the modification of the peak current of the buck inductor by increasing/decreasing the on/off timings of the buck converter. The higher the timing multiplier, the higher the buck inductor peak current, and thus the higher the average current pulled from [STO](#) to [LOAD](#).

The peak current in the inductor depends also on the value of the inductor (see Table 39).

Bit [2:0]: buck converter output regulation voltage (BUCKCFG.VLOAD)

This field allows the configuration of the buck converter regulation output voltage (V_{LOAD}). The available configurations can be found in Table 40. To switch off the buck converter, set BUCK.VLOAD to 0x00.

BUCKCFG.TMULT	Timing Multiplier	Minimum ¹ L_{BUCK} [μH]	Recommended ² L_{BUCK} [μH]
0x00	x1	1.7	6.8
0x01	x2	3.3	10
0x02	x3	5.0	15
0x03	x4	6.6	22
0x04	x6	9.9	33
0x05	x8	13.2	47
0x06	x12	19.8	68
0x07	x16	26.4	100

Table 39: Buck inductor values according to buck timing

1. Never install an inductor with an inductance (real value including tolerance, derating, etc.) lower than those values for each setting of the timing multiplier. This would cause permanent damage to the AEM13920.
2. Those values provide the best efficiency/current capability trade-off according to the tests carried out in the e-peas laboratory.

BUCKCFG.VLOAD	V_{LOAD} [V]
0x00	OFF
0x01	0.6
0x02	0.9
0x03	1.2
0x04	1.5
0x05	1.8
0x06	2.2
0x07	2.5 ¹

Table 40: V_{LOAD} settings by BUCKCFG.VLOAD register

1. This configuration is only available if $V_{OVDIS} \geq 2.5$ V.

9.8. 5 V Charger (CHG5V)

The CHG5V register allows the configuration of the 5 V charger (as explained in Section 5.7).

CHG5V register		0x29	R/W	
Bit [7]	Bit [6:2]	Bit [1]	Bit [0]	
RESERVED	THRESH	CVEN	EN	
0x00	0x00	0x00	0x01	

Table 41: CHG5V register

Bit [6:2]: 5 V charger stop voltage threshold (CHG5V.THRESH)

This field allows the configuration of the 5 V charger stop voltage threshold ($V_{5V,STOP}$), which is the storage element voltage at which the 5 V charger will stop to charge the storage element when the constant voltage (CV) mode is enabled (see Table 42).

Bit [1]: 5 V charger constant voltage mode enable (CHG5V.CVEN)

This field allows enabling or disabling of the constant voltage (CV) operation mode (see Section 5.7).

- 0: disable the 5 V charger CV mode.
- 1: enable the 5 V charger CV mode.

Bit [0]: 5 V charger enable (CHG5V.EN)

This field allows enabling or disabling of the 5 V charger.

- 0: disable the 5 V charger.
- 1: enable the 5 V charger.

CHG5V. THRESH	$V_{5V,STOP}$ [V]	CHG5V. THRESH	$V_{5V,STOP}$ [V]	CHG5V. THRESH	$V_{5V,STOP}$ [V]	CHG5V. THRESH	$V_{5V,STOP}$ [V]
0x00	2.65	0x08	3.35	0x10	3.75	0x18	4.15
0x01	2.75	0x09	3.40	0x11	3.80	0x19	4.20
0x02	2.85	0x0A	3.45	0x12	3.85	0x1A	4.25
0x03	2.95	0x0B	3.50	0x13	3.90	0x1B	4.30
0x04	3.05	0x0C	3.55	0x14	3.95	0x1C	4.40
0x05	3.15	0x0D	3.60	0x15	4.00	0x1D	4.50
0x06	3.25	0x0E	3.65	0x16	4.05	0x1E	4.60
0x07	3.30	0x0F	3.70	0x17	4.10	0x1F	4.70

Table 42: $V_{5V,STOP}$ configuration by CHG5V register

9.9. STO Charge Temperature Protection (TEMPCOLDCH and TEMPHOTCH)

Those fields are used when the thermal protection monitoring (TEMPPROTECT) is enabled (see Section 9.11) to configure the minimum (cold) and maximum (hot) temperature protection thresholds for charging the storage element connected to [STO](#).

If the temperature is out of the charge temperature range, the boost converters will be disabled to stop charging the storage element.

THRESH value is determined as follows from the desired temperature T:

- Determine the resistance of the thermo resistor R_{TH} at the desired temperature.
- Calculate THRESH using the following formula:

$$THRESH(T) = \frac{256 \cdot R_{TH}(T)}{R_{TH}(T) + R_{DIV}}$$

See Section 6.7 for further information about thermal monitoring configuration.

9.9.1. TEMPCOLDCH

The TEMPCOLDCH register allows the configuration of the minimum temperature for storage element charging.

TEMPCOLDCH register	0x0B	R/W
Bit [7:0]		
THRESH		
0xD1		

Table 43: TEMPCOLDCH register

Bit [7:0]: storage element charge minimum temperature threshold (TEMPCOLDCH.THRESH)

This field allows the configuration of the minimum temperature (cold) threshold for storage element charging.

9.9.2. TEMPHOTCH

The TEMPHOTCH register allows the configuration of the maximum temperature for storage element charging.

TEMPHOTCH register	0x0C	R/W
Bit [7:0]		
THRESH		
0x18		

Table 44: TEMPHOTCH register

Bit [7:0]: storage element charge maximum temperature threshold (TEMPHOTCH.THRESH)

This field allows the configuration of the maximum temperature (hot) threshold for storage element charging.

9.10. STO Discharge Temperature Protection (TEMPCOLDDIS and TEMPHOTDIS)

Those fields are used when the thermal protection monitoring (TEMPPROTECT) is enabled (see Section 9.11) to configure the minimum (cold) and maximum (hot) temperature protection thresholds for discharging the storage element connected to [STO](#).

If the temperature is out of the discharge temperature range, the buck converter will be disabled to stop discharging the storage element through the [LOAD](#). In this case, [VINT](#) will continue to be supplied from [STO](#).

See Section 6.7 for further information about thermal monitoring configuration.

9.10.1. TEMPCOLDDIS

The TEMPCOLDDIS register allows the configuration of the minimum temperature (cold) for storage element discharging.

TEMPCOLDDIS register	0x0D	R/W
Bit [7:0]		
THRESH		
0xD1		

Table 45: TEMPCOLDDIS register

Bit [7:0]: storage element discharge minimum temperature threshold (TEMPCOLDDIS.THRESH)

This field allows the configuration of the minimum temperature (cold) threshold for storage element discharging.

9.10.2. TEMPHOTDIS

The TEMPHOTDIS register allows the configuration of the maximum temperature (hot) for storage element discharging.

TEMPHOTDIS register	0x0E	R/W
Bit [7:0]		
THRESH		
0x18		

Table 46: TEMPHOTDIS register

Bit [7:0]: storage element discharge maximum temperature threshold (TEMPHOTDIS.THRESH)

This field allows the configuration of the maximum temperature (hot) threshold for storage element discharging.

9.11. Thermal Protection Monitoring (TEMPPROTECT)

The TEMPPROTECT register allows enabling or disabling the thermal protection based on the protection thresholds described in Sections 9.9 and 9.10.

TEMPPROTECT register	0x0F	R/W
Bit [7:1]		Bit [0]
RESERVED		EN
0x00		0x01

Table 47: TEMPPROTECT register

Bit [0]: thermal protection monitoring enable (TEMPPROTECT.EN)

This field allows enabling or disabling the thermal protection monitoring:

- 0: disable the thermal protection.
- 1: enable the thermal protection.

9.12. Average Power Monitoring (APM)

The APM register allows the configuration of the Average Power Monitoring feature.

The configuration of this register affects the APM readings of both boost converters, as well as the readings of the buck converter and the 5 V charger.

For APM data interpretation, refer to the Section 9.17.

APM register				0x11			R/W
Bit [7:6]	Bit [5]	Bit [4]	Bit [3]	Bit [2]	Bit [1]	Bit [0]	
RESERVED	CHG5VEN	WINDOW	MODE	LOADEN	SRC2EN	SRC1EN	
0x00	0x00	0x00	0x00	0x00	0x00	0x00	

Table 48: APM register

Bit [5]: 5 V charger APM enable (APM.CHG5VEN)

This field allows enabling or disabling the APM for CHG5V.

- 0: disable the 5 V charger APM.
- 1: enable the 5 V charger APM.

Bit [4]: APM window (APM.WINDOW)

This field allows the configuration of the APM window.

- 0: select a window of 233 ms.
- 1: select a window of 116 ms.

NOTE: if either SRC1 or SRC2 is configured with an MPPT period of 116 ms, the APM window will automatically switch to 116 ms for both boost converters, for the buck converter, and for the 5 V charger, regardless of whether the corresponding converter is enabled or not. This is the case for the SRCxREGU1.CFG2 default value (MPPT default period is 116 ms).

Bit [3]: APM mode (APM.MODE)

This field allows the configuration of the SRCx and LOAD APM mode.

- 0: set the SRCx and LOAD APM in counter mode.
- 1: set the SRCx and LOAD APM in power meter mode.

Bit [2]: load output APM enable (APM.LOADEN)

This field allows enabling or disabling the APM for LOAD.

- 0: disable the load output APM.
- 1: enable the load output APM.

Bit [1]: source 2 APM enable (APM.SRC2EN)

This field allows enabling or disabling the APM for SRC2.

- 0: disable the source 2 APM.
- 1: enable the source 2 APM.

Bit [0]: source 1 APM enable (APM.SRC1EN)

This field allows enabling or disabling the APM for SRC1.

- 0: disable the source 1 APM.
- 1: enable the source 1 APM.

9.13. APM Accumulator (APMACC)

The APMACC register allows the configuration of the APM accumulator feature.

APMACC register		0x2A	R/W
Bit [7:0]			
CFG			
0x00			

Table 49: APMACC register

Bit [7:0]: APM accumulator configuration (APMACC.CFG)

This field allows the configuration of the number of APM windows during which the APM data is accumulated before rising the **IRQ** APMDONE flag (see Section 5.5.4). The APM accumulator can be used with both APM modes (pulse counter mode and power meter mode).

APMACC.CFG must be configured to the wanted number of accumulated windows minus 1 (e.g., set APMACC.CFG to 2 to accumulate APM data over 3 APM windows).

To disable the APM accumulator, set APMACC.CFG to 0 (the **IRQ** pin will be raised after every APM window).

9.14. IRQ Enable (IRQENx)

9.14.1. IRQEN0

The IRQEN0 register allows the configuration of which events trigger the **IRQ** pin rising edge (see also the IRQEN1 register).

IRQEN0 register						0x12		R/W
Bit [7]	Bit [6]	Bit [5]	Bit [4]	Bit [3]	Bit [2]	Bit [1]	Bit [0]	
CHG5VCONN	TEMPDIS	TEMPCH	SRCLW	VOVCH	VCHRDY	VOVDIS	I2CRDY	
0x00	0x00	0x00	0x00	0x00	0x00	0x00	0x01	

Table 50: IRQEN0 register

Bit [7]: 5 V charger connected (IRQEN0.CHG5VCONN)

This field allows enabling or disabling the generation of a rising edge on the **IRQ** pin with IRQFLG0.CHG5VCONN when the 5 V charger gets connected or disconnected (see Table 7 for V_{SV_IN} requirements).

- 0: disable the IRQ flag for 5 V charger connection.
- 1: enable the IRQ flag for 5 V charger connection.

Bit [6]: storage element discharge temperature (IRQEN0.TEMPDIS)

This field allows enabling or disabling the generation of a rising edge on the **IRQ** pin with IRQFLG0.TEMPDIS when the temperature crosses the minimum or maximum temperature allowed for storage element discharging (selected through the TEMPCOLDDIS and TEMPHOTDIS registers).

- 0: disable the IRQ flag for discharge temperature range.
- 1: enable the IRQ flag for discharge temperature range.

Bit [5]: storage element charge temperature (IRQEN0.TEMPCH)

This field allows enabling or disabling the generation of a rising edge on the **IRQ** pin with IRQFLG0.TEMPCH when the temperature crosses the minimum or maximum temperature allowed for storage element charging (selected through the TEMPCOLDCH and TEMPHOTCH registers).

- 0: disable the IRQ flag for charge temperature range.
- 1: enable the IRQ flag for charger temperature range.

Bit [4]: source low voltage (IRQEN0.SRCLW)

This field allows enabling or disabling the generation of a rising edge on the **IRQ** pin with IRQFLG0.SRCLW on any transition between a state where, at least one source voltage is above its V_{SRCLW} threshold (selected through the SRCLW.SRCxTHRESH fields), and a state where both source voltages are below their respective thresholds.

- 0: disabled the IRQ flag for source low threshold.
- 1: enable the IRQ flag for source low threshold.

Bit [3]: storage element overcharge (IRQEN0.VOVCH)

This field allows enabling or disabling the generation of a rising edge on the **IRQ** pin with IRQFLG0.VOVCH when the storage element voltage crosses V_{OVCH} (selected through the VOVCH register).

- 0: disable the IRQ flag for V_{OVCH} crossing.
- 1: enable the IRQ flag for V_{OVCH} crossing.

Bit [2]: storage element charge ready (IRQEN0.VCHRDY)

This field allows enabling or disabling the generation of a rising edge on the **IRQ** pin with IRQFLG0.VCHRDY when the storage element voltage crosses V_{CHRDY} (selected through the VCHRDY register).

- 0: disable the IRQ flag for V_{CHRDY} crossing.
- 1: enable the IRQ flag for V_{CHRDY} crossing.

Bit [1]: storage element overdischarge (IRQEN0.VOVDIS)

This field allows enabling or disabling the generation of a rising edge on the **IRQ** pin with IRQFLG0.VOVDIS after $T_{CRIT,ST}$ when the storage element voltage drops below V_{OVDIS} (selected through the VOVDIS register) or directly when the storage element voltage rises above V_{OVDIS} .

- 0: disable the IRQ flag for V_{OVDIS} crossing.
- 1: enable the IRQ flag for V_{OVDIS} crossing.

Bit [0]: I2C interface ready (IRQEN0.I2CRDY)

This field allows enabling or disabling the generation of a rising edge on the **IRQ** pin with IRQFLG0.I2CRDY when the serial interface (I²C) is ready to communicate.

- 0: disable the IRQ flag for I²C interface readiness.
- 1: enable the IRQ flag for I²C interface readiness.

9.14.2. IRQEN1

The IRQEN1 register allows the configuration of which events trigger the **IRQ** pin rising edge (see also the IRQEN0 register).

IRQEN1 register				0x13		R/W	
Bit [7]	Bit [6]	Bit [5]	Bit [4]	Bit [3]	Bit [2]	Bit [1]	Bit [0]
APMERR	APMDONE	TEMPDONE	STODONE	SRC2MPPTDONE	SRC2MPPTSTART	SRC1MPPTDONE	SRC1MPPTSTART
0x00	0x00	0x00	0x00	0x00	0x00	0x00	0x00

Table 51: IRQEN1 register

Bit [7]: APM error (IRQEN1.APMERR)

This field allows enabling or disabling the generation of a rising edge on the **IRQ** pin with IRQFLG1.APMERR when an APM error occurs.

- 0: disable the IRQ flag for APM errors.
- 1: enable the IRQ flag for APM errors.

Bit [6]: APM done (IRQEN1.APMDONE)

This field allows enabling or disabling the generation of a rising edge on the **IRQ** pin with IRQFLG1.APMDONE when APM data is available.

- 0: disable the IRQ flag for new available APM data.
- 1: enable the IRQ flag for new available APM data.

Bit [5]: temperature measurement done (IRQEN1.TEMPDONE)

This field allows enabling or disabling the generation of a rising edge on the **IRQ** pin with IRQFLG1.TEMPDONE when the temperature measurement is done.

- 0: disable the IRQ flag for temperature measurement completion.
- 1: enable the IRQ flag for temperature measurement completion.

Bit [4]: storage element voltage measurement done (IRQEN1.STODONE)

This field allows enabling or disabling the generation of a rising edge on the **IRQ** pin with IRQFLG1.STODONE when the storage element voltage measurement is done.

- 0: disable the IRQ flag for STO voltage measurement completion.
- 1: enable the IRQ flag for STO voltage measurement completion.

Bit [3]: source 2 MPPT done (IRQEN1.SRC2MPPTDONE)

This field allows enabling or disabling the generation of a rising edge on the **IRQ** pin with IRQFLG1.SRC2MPPTDONE when the source 2 MPP evaluation is done.

- 0: disable the IRQ flag for source 2 MPP evaluation completion.
- 1: enable the IRQ flag for source 2 MPP evaluation completion.

Bit [2]: source 2 MPPT start (IRQEN1.SRC2MPPTSTART)

This field allows enabling or disabling the generation of a rising edge on the **IRQ** pin with IRQFLG1.SRC2MPPTSTART when the source 2 MPP evaluation starts.

- 0: disable the IRQ flag for source 2 MPP evaluation start.
- 1: enable the IRQ flag for source 2 MPP evaluation start.

Bit [1]: source 1 MPPT done (IRQEN1.SRC1MPPTDONE)

This field allows enabling or disabling the generation of a rising edge on the **IRQ** pin with IRQFLG1.SRC1MPPTDONE when the source 1 MPP evaluation is done.

- 0: disable the IRQ flag for source 1 MPP evaluation completion.
- 1: enable the IRQ flag for source 1 MPP evaluation completion.

Bit [0]: source 1 MPPT start (IRQEN1.SRC1MPPTSTART)

This field allows enabling or disabling the generation of a rising edge on the **IRQ** pin with IRQFLG1.SRC1MPPTSTART when the source 1 MPP evaluation starts.

- 0: disable the IRQ flag for source 1 MPP evaluation start.
- 1: enable the IRQ flag for source 1 MPP evaluation start.

9.15. IRQ Flags (IRQFLGx)

The IRQFLGx registers allow the user to get a status about specific AEM13920 events. When the event happens, the **IRQ** pin switches HIGH and the register field bit that corresponds to the event will switch to 1, provided that the event flag has been enabled in the corresponding IRQENx register. The bit will stay to 1 and the **IRQ** pin will stay HIGH until the IRQFLGx register is read.

9.15.1. IRQFLG0

The IRQFLG0 register is the **IRQ** pin event flags register 0.

IRQFLG0 register				0x15 R			
Bit [7]	Bit [6]	Bit [5]	Bit [4]	Bit [3]	Bit [2]	Bit [1]	Bit [0]
CHG5VCONN	TEMPDIS	TEMPCH	SRCLW	VOVCH	VCHRDY	VOVDIS	I2CRDY
0x00	0x00	0x00	0x00	0x00	0x00	0x00	0x00

Table 52: IRQFLG0 register

Bit [7]: 5 V charger connected (IRQFLG0.CHG5VCONN)

This **IRQ** pin event flag is set when the 5 V charger gets connected or disconnected (see Table 7 for V_{5V_IN} requirements), if the corresponding event source has been previously enabled through the IRQEN0.CHG5VCONN field.

- 0: the 5 V charger was not connected or disconnected.
- 1: the 5 V charger was connected or disconnected.

Bit [6]: storage element discharge temperature (IRQFLG0.TEMPDIS)

This **IRQ** pin event flag is set when the temperature crosses the minimum or maximum temperature allowed for storage element discharging (selected through the TEMPCOLDDIS and TEMPHOTDIS registers), if the corresponding event source has been previously enabled through the IRQEN0.TEMPDIS field.

- 0: the temperature did not cross the minimum or maximum temperature allowed for discharging.
- 1: the temperature crossed the minimum or maximum temperature allowed for discharging.

Please note that if the AEM13920 is configured through the I²C registers to enable IRQ flags, the behavior of the **IRQ** pin as well as the IRQFLGx register stays the same even if the AEM13920 has switched back to GPIO configuration by writing 0 to CTRL.UPDATE.

Bit [5]: storage element charge temperature (IRQFLG0.TEMPCH)

This **IRQ** pin event flag is set when the temperature crosses the minimum or maximum temperature allowed for storage element charging (selected through the TEMPCOLDCH and TEMPHOTCH registers), if the corresponding event source has been previously enabled through the IRQEN0.TEMPCH field.

- 0: the temperature did not cross the minimum or maximum temperature allowed for charging.
- 1: the temperature crossed the minimum or maximum temperature allowed for charging.

Bit [4]: source low voltage (IRQFLG0.SRCLW)

This **IRQ** pin event flag is set on any transition between a state where, at least one source voltage is above its V_{SRCLW} threshold (selected through the SRCLW.SRCxTHRESH fields), and a state where both source voltages are below their respective thresholds, if the corresponding event source has been previously enabled through the IRQEN0.SRCLW field.

- 0: both source voltages did not fall below V_{SRCLW} .
- 1: both source voltages fell below V_{SRCLW} .

Bit [3]: storage element overcharge (IRQFLG0.VOVCH)

This **IRQ** pin event flag is set when the storage element voltage crosses V_{OVCH} (selected through the VOVCH register), if the corresponding event source has been previously enabled through the IRQEN0.VOVCH field.

- 0: the storage element voltage did not cross V_{OVCH} .
- 1: the storage element voltage crossed V_{OVCH} .

Bit [2]: storage element charge ready (IRQFLG0.VCHRDY)

This **IRQ** pin event flag is set when the storage element voltage crosses V_{CHRDY} (selected through the VCHRDY register), if the corresponding event source has been previously enabled through the IRQEN0.VCHRDY field.

- 0: the storage element voltage did not cross V_{CHRDY} .
- 1: the storage element voltage crossed V_{CHRDY} .

Bit [1]: storage element overdischarge (IRQFLG0.VOVDIS)

This **IRQ** pin event flag is set after $T_{CRIT,ST}$ when the storage element voltage drops below V_{OVDIS} (selected through the VOVDIS register) or directly when the storage element rises above V_{OVDIS} , if the corresponding event source has been previously enabled through the IRQEN0.VOVDIS field.

- 0: the storage element voltage did not cross V_{OVDIS} .
- 1: the storage element voltage crossed V_{OVDIS} .

9.15.2. IRQFLG1

The IRQFLG1 register is the **IRQ** pin event flags register 1.

IRQFLG1 register						0x16		R
Bit [7]	Bit [6]	Bit [5]	Bit [4]	Bit [3]	Bit [2]	Bit [1]	Bit [0]	
APMERR	APMDONE	TEMPDONE	STODONE	SRC2MPPTDONE	SRC2MPPTSTART	SRC1MPPTDONE	SRC1MPPTSTART	
0x00	0x00	0x00	0x00	0x00	0x00	0x00	0x00	

Table 53: IRQFLG1 register

Bit [7]: APM error (IRQFLG1.APMERR)

This **IRQ** pin event flag is set when an APM error occurs, if the corresponding event source has been previously enabled through the IRQEN1.APMERR field.

- 0: no APM error occurred.
- 1: an APM error occurred.

Bit [6]: APM done (IRQFLG1.APMDONE)

This **IRQ** pin event flag is set when APM data is available, if the corresponding event source has been previously enabled through the IRQEN1.APMDONE field.

- 0: no new APM data available.
- 1: new APM data available.

Bit [5]: temperature measurement done (IRQFLG1.TEMPDONE)

This **IRQ** pin event flag is set when the temperature measurement is done, if the corresponding event source has been previously enabled through the IRQEN1.TEMPDONE field.

- 0: temperature measurement not completed.
- 1: temperature measurement completed.

Bit [0]: I2C interface ready (IRQFLG0.I2CRDY)

This **IRQ** pin event flag is set when the serial interface (I²C) is ready to communicate, if the corresponding event source has been previously enabled through the IRQEN0.I2CRDY field.

- 0: I²C interface not yet ready to communicate.
- 1: I²C interface became ready to communicate.

Bit [4]: storage element voltage measurement done (IRQFLG1.STODONE)

This **IRQ** pin event flag is set when the storage element voltage measurement is done, if the corresponding event source has been previously enabled through the IRQEN1.STODONE field.

- 0: storage element voltage measurement not completed.
- 1: storage element voltage measurement completed.

Bit [3]: source 2 MPPT done (IRQFLG1.SRC2MPPTDONE)

This **IRQ** pin event flag is set when the source 2 MPP evaluation is done, if the corresponding event source has been previously enabled through the IRQEN1.SRC2MPPTDONE field.

- 0: source 2 MPP evaluation not completed.
- 1: source 2 MPP evaluation completed.

Bit [2]: source 2 MPPT start (IRQFLG1.SRC2MPPTSTART)

This **IRQ** pin event flag is set when the source 2 MPP evaluation starts, if the corresponding event source has been previously enabled through the IRQEN1.SRC2MPPTSTART field.

- 0: source 2 MPP evaluation not started.
- 1: source 2 MPP evaluation started.

Bit [1]: source 1 MPPT done (IRQFLG1.SRC1MPPTDONE)

This **IRQ** pin event flag is set when the source 1 MPP evaluation is done, if the corresponding event source has been previously enabled through the IRQEN1.SRC1MPPTDONE field.

- 0: source 1 MPP evaluation not completed.
- 1: source 1 MPP evaluation completed.



Bit [0]: source 1 MPPT start (IRQFLG1.SRC1MPPTSTART)

This **IRQ** pin event flag is set when the source 1 MPP evaluation starts, if the corresponding event source has been previously enabled through the IRQEN1.SRC1MPPTSTART field.

- 0: source 1 MPP evaluation not started.
- 1: source 1 MPP evaluation started.



9.16. Status (STATUSx)

9.16.1. STATUS0

The STATUS0 register is the AEM13920 status register 0.

STATUS0 register				0x17			R
Bit [7:6]	Bit [5]	Bit [4]	Bit [3]	Bit [2]	Bit [1]	Bit [0]	
RESERVED	CHG5VCONN	SRC2LOW	SRC1LOW	OVCH	CHRDY	OVDIS	
0x00	0x00	0x00	0x00	0x00	0x00	0x00	

Table 54: STATUS0 register

Bit [5]: 5 V charger connected (STATUS0.CHG5VCONN)

This status indicates whether the 5 V charger is considered connected or not (see Table 7 for V_{5V_IN} requirements).

- 0: the 5 V charger is disconnected.
- 1: the 5 V charger is connected.

Bit [4]: source 2 low voltage (STATUS0.SRC2LOW)

This status indicates whether the source 2 is higher than its V_{SRCLOW} or not.

- 0: the source 2 voltage is higher than its V_{SRCLOW} .
- 1: the source 2 voltage is equal or lower than its V_{SRCLOW} .

Bit [3]: source 1 low voltage (STATUS0.SRC1LOW)

This status indicates whether the source 1 voltage is higher than its V_{SRCLOW} or not.

- 0: the source 1 is higher than its V_{SRCLOW} .
- 1: the source 1 is equal or lower than its V_{SRCLOW} .

Bit [2]: storage element overcharge (STATUS0.OVCH)

This status indicates whether the storage element voltage is higher than V_{OVCH} or not.

- 0: the storage element voltage is lower than the overcharge threshold.
- 1: the storage element voltage is equal or higher than the overcharge threshold.

Bit [1]: storage element charge ready (STATUS0.CHRDY)

This status indicates whether the storage element voltage is higher than V_{CHRDY} or not.

- 0: the storage element voltage is lower than the charge ready threshold.
- 1: the storage element voltage is equal or higher than the charge ready threshold.

Bit [0]: storage element overdischarge (STATUS0.OVDIS)

This status indicates whether the storage element voltage has been lower than V_{OVDIS} for more than $T_{CRIT,ST}$ or not.

- 0: the storage element voltage has not been equal or lower than the overdischarge threshold for more than $T_{CRIT,ST}$.
- 1: the storage element voltage has been equal or lower than the overdischarge threshold for more than $T_{CRIT,ST}$.

9.16.2. STATUS1

The STATUS1 register is the AEM13920 status register 1.

STATUS1 register	0x18 R			
Bit [7:4]	Bit [3]	Bit [2]	Bit [1]	Bit [0]
RESERVED	TEMPHOTDIS	TEMPCOLDNIS	TEMPHOTCH	TEMPCOLDCH
0x00	0x00	0x00	0x00	0x00

Table 55: STATUS1 register

Bit [3]: storage element discharge maximum temperature (STATUS1.TEMPHOTDIS)

This status indicates whether the temperature is higher than the hot threshold (for storage element discharging) or not.

- 0: the temperature is below the hot threshold.
- 1: the temperature is equal or above the hot threshold.

Bit [2]: storage element discharge minimum temperature (STATUS1.TEMPCOLDNIS)

This status indicates whether the temperature is higher than the cold threshold (for storage element discharging) or not.

- 0: the temperature is above the cold threshold.
- 1: the temperature is equal or below the cold threshold.

Bit [1]: storage element charge maximum temperature (STATUS1.TEMPHOTCH)

This status indicates whether the temperature is higher than the hot threshold (for storage element charging) or not.

- 0: the temperature is below the hot threshold.
- 1: the temperature is equal or above the hot threshold.

Bit [0]: storage element charge minimum temperature (STATUS1.TEMPCOLDCH)

This status indicates whether the temperature is higher than the cold threshold (for storage element charging) or not.

- 0: the temperature is above the cold threshold.
- 1: the temperature is equal or below the cold threshold.

9.17. APM Data Summary (APMxSRCx, APMxCHG5V and APMxLOAD)

9.17.1. APMxSRCx and APMxLOAD Summary

The APM data registers for both boost converters and for the buck converter all share the same field structure, depending of the configured APM mode, as described in Table 56.

Pulse Counter Mode: in this mode, the value in the APM data registers is the number of pulses drawn by the DCDC converter during the APM window (see Section 5.5). The APM data in pulse counter mode depends on the APM accumulator configuration:

- If the APM accumulator is not used (APMACC.CFG set to 0), the APM data can be accessed directly in the COUNTER fields (see Table 56).
- If the APM accumulator is used, the APM data can be computed by left bit-shifting (OFFSET bits) the value in the COUNTER fields (see Table 56).

Power Meter mode: in this mode, the value in the APM data registers (POWER fields) can be computed by left bit-shifting by the value in the OFFSET field to obtain the measured APM power (P_{APM}) and energy (E_{APM}) provided to/from the storage element during the previous APM window or sum of accumulated APM windows (see Section 5.5):

$$P_{APM} = \frac{(POWER \ll OFFSET) \cdot \alpha}{APMACC.CFG + 1}$$

$$E_{APM} = P_{APM} \cdot \text{APM window}$$

Where:

P_{APM} is the average power (W) provided to/from the storage element during the previous APM window.

α is the correction factor used to convert the APM values from the APMxSRCx and APMxLOAD registers into power. This alpha factor takes into account the configured APM window and T_{MULT} , the variability between devices, the board layout, and the effective value of L_{DCDC} , as well as several other parameters that affect the APM conversion. The alpha factor can be determined by calibrating the complete system to take all these parameters into account. Please contact e-peas support for more details about alpha calibration.

(APMACC.CFG + 1) is the number of accumulated windows. Equal to 1 if the APM accumulator is not used.

E_{APM} is the energy (J) provided to/from the storage element during the previous APM window.

APM window is the duration (s) used to calculate E_{APM} . It must match the actual APM window duration, which may slightly deviate from the nominal datasheet values (116/233 ms). If the APM accumulator is used, this APM window represent the sum of the accumulated APM windows duration.

	APM2SRCx APM2LOAD								APM1SRCx APM1LOAD								APM0SRCx APM0LOAD							
	Bit [7]	Bit [6]	Bit [5]	Bit [4]	Bit [3]	Bit [2]	Bit [1]	Bit [0]	Bit [7]	Bit [6]	Bit [5]	Bit [4]	Bit [3]	Bit [2]	Bit [1]	Bit [0]	Bit [7]	Bit [6]	Bit [5]	Bit [4]	Bit [3]	Bit [2]	Bit [1]	Bit [0]
Register fields	APM2SRCx.DATA[7:0] APM2LOAD.DATA[7:0]								APM1SRCx.DATA[7:0] APM1LOAD.DATA[7:0]								APM0SRCx.DATA[7:0] APM0LOAD.DATA[7:0]							
Global APM Field	APM DATA [23:0]																							
Pulse Counter Mode (no accumulator)		COUNTER [22:16]							COUNTER [15:8]								COUNTER [7:0]							
Pulse Counter Mode (with accumulator)	OFFSET [4:0]				COUNTER [18:16]				COUNTER [15:8]								COUNTER [7:0]							
Power Meter Mode (with or without accumulator)	OFFSET [4:0]				POWER [18:16]				POWER [15:8]								POWER [7:0]							

Table 56: Summary of APMxSRCx and APMxLOAD register fields



9.17.2. APMxCHG5V Summary

The APM data registers for the 5 V charger are exclusively used in counter mode to provide the percentage of time during which the 5 V charger has been charging the storage element over the APM window, as described in Table 57.

The charging duty cycle (D_{CHG}) over the previous window, or accumulated windows, is obtained thanks to the following formulas:

$$D_{CHG} = \frac{COUNTER}{MAX\ COUNTER - 1}$$

Where MAX COUNTER depends on the configured APM window and APM accumulator:

- For APM window set to 116 ms:
MAX COUNTER = $64 \cdot (APMACC.CFG + 1)$
- For APM window set to 233 ms:
MAX COUNTER = $128 \cdot (APMACC.CFG + 1)$

Once the charging duty cycle has been defined, the average power and the energy transferred to the storage element from the **5V_IN** pin during the previous APM window can be computed with the following formulas:

$$P_{APM} = I_{5V,CC} \cdot V_{STO} \cdot D_{CHG}$$

$$E_{APM} = P_{APM} \cdot APM\ window$$

Where:

P_{APM} is the average power (W) provided to the storage element during the previous APM window.

$I_{5V,CC}$ is the 5 V charger current (A) in constant current operation. For higher accuracy, the actual charging current delivered to the storage element must be used, which may slightly deviate from the configured $I_{5V,CC}$.

V_{STO} is the average storage element voltage (V) over the previous APM window.

D_{CHG} is the charging duty cycle of the 5 V charger.

E_{APM} is the energy (J) provided to the storage element during the previous APM window.

APM window is the duration (s) used to calculate E_{APM} . It must match the actual APM window duration, which may slightly deviate from the nominal datasheet values (116/233 ms). If the APM accumulator is used, this APM window represent the sum of the accumulated APM windows.

NOTE: When V_{STO} is below V_{OVDIS} , the 5 V charger current $I_{5V,CC}$ must be replaced by $I_{5V,OVDIS,CC}$.

NOTE: When the 5 V charger constant voltage (CV) mode is enabled, the 5 V charger current will decrease from the configured $I_{5V,CC}$ as V_{STO} approaches the configured $V_{5V,STOP}$ (see Section 5.7), thus, making the above formulas incorrect.

	APM1CHG5V								APM0CHG5V							
	Bit [7]	Bit [6]	Bit [5]	Bit [4]	Bit [3]	Bit [2]	Bit [1]	Bit [0]	Bit [7]	Bit [6]	Bit [5]	Bit [4]	Bit [3]	Bit [2]	Bit [1]	Bit [0]
Pulse Counter Mode	COUNTER [15:8]								COUNTER [7:0]							

Table 57: Summary of APMxCHG5V register fields

9.18. SRCx APM Data (APMxSRCx)

9.18.1. APM0SRCx

The APM0SRCx is the first set of registers providing the SRCx APM data.

APM0SRC1 register	0x19	R
APM0SRC2 register	0x1C	R
Bit [7:0]		
DATA		
0x00		

Table 58: APM0SRCx registers

Bit [7:0]: source APM data (APM0SRCx.DATA)

This field contains the bits [7:0] of the SRCx APM data (see Table 56 for detailed fields according to APM mode).

9.18.2. APM1SRCx

The APM1SRCx is the second set of registers providing the SRCx APM data.

APM1SRC1 register	0x1A	R
APM1SRC2 register	0x1D	R
Bit [7:0]		
DATA		
0x00		

Table 59: APM1SRCx registers

Bit [7:0]: source APM data (APM1SRCx.DATA)

This register contains the bits [15:8] of the SRCx APM data (see Table 56 for detailed fields according to APM mode).

9.18.3. APM2SRCx

The APM2SRCx is the third set of registers providing the SRCx APM data.

APM2SRC1 register	0x1B	R
APM2SRC2 register	0x1E	R
Bit [7:0]		
DATA		
0x00		

Table 60: APM2SRCx registers

Bit [7:0]: source APM data (APM2SRCx.DATA)

This register contains the bits [23:16] of the SRCx APM data (see Table 56 for detailed fields according to APM mode).

9.19. LOAD APM Data (APMxLOAD)

9.19.1. APM0LOAD

The APM0LOAD is the first register providing the LOAD APM data.

APM0LOAD register	0x1F	R
Bit [7:0]		
DATA		
0x00		

Table 61: APM0LOAD register

9.19.2. APM1LOAD

The APM1LOAD is the second register providing the LOAD APM data.

APM1LOAD register	0x20	R
Bit [7:0]		
DATA		
0x00		

Table 62: APM1LOAD register

9.19.3. APM2LOAD

The APM2LOAD is the third register providing the LOAD APM data.

APM2LOAD register	0x21	R
Bit [7:0]		
DATA		
0x00		

Table 63: APM2LOAD register

Bit [7:0]: LOAD APM data (APM0LOAD.DATA)

This register contains the bits [7:0] of the LOAD APM data (see Table 56 for detailed fields according to APM mode).

Bit [7:0]: LOAD APM data (APM1LOAD.DATA)

This register contains the bits [15:8] of the LOAD APM data (see Table 56 for detailed fields according to APM mode).

Bit [7:0]: LOAD APM data (APM2LOAD.DATA)

This register contains the bits [23:16] of the LOAD APM data (see Table 56 for detailed fields according to APM mode).

9.20. CHG5V APM Data (APMxCHG5V)

9.20.1. APM0CHG5V

The APM0CHG5V is the first register providing the 5 V charger APM data.

APM0CHG5V register	0x27	R
Bit [7:0]		
DATA		
0x00		

Table 64: APM0CHG5V register

9.20.2. APM1CHG5V

The APM1CHG5V is the second register providing the 5 V charger APM data.

APM1CHG5V register	0x28	R
Bit [7:0]		
DATA		
0x00		

Table 65: APM1CHG5V register

Bit [7:0]: 5 V charger APM data (APM0CHG5V.DATA)

This register contains the bits [7:0] of the CHG5V APM data (see Table 57 for detailed fields).

Bit [7:0]: 5 V charger APM data (APM1CHG5V.DATA)

This register contains the bits [15:8] of the CHG5V APM data (see Table 57 for detailed fields).

9.21. APM Error (APMERR)

The APMERR register provides the APM errors status.

NOTE: When the APM accumulator is used (see Section 5.5.4), if an APM error occurs, the corresponding APMERR field will remain high during the whole accumulation window. The IRQFLG1.APMERR will be raised after the end of the total accumulation window.

APMERR register				0x22		R	
Bit [7]	Bit [6]	Bit [5]	Bit [4]	Bit [3]	Bit [2]	Bit [1]	Bit [0]
RESERVED	CHG5VLIM	LOADNVLD	LOADOV	SRC2NVLD	SRC2OV	SRC1NVLD	SRC1OV
0x00	0x00	0x00	0x00	0x00	0x00	0x00	0x00

Table 66: APMERR register

Bit [6]: 5 V charger current limited (APMERR.CHG5VLIM)

This field warns the user that the 5 V charging current $I_{5V,CC}$ is limited to $I_{5V,OVDIS,CC}$ because V_{STO} is below V_{OVDIS} .

- 0: $I_{5V,CC}$ is not limited due to V_{OVDIS} .
- 1: $I_{5V,CC}$ is limited to $I_{5V,OVDIS,CC}$.

Bit [5]: LOAD APM data invalid (APMERR.LOADNVLD)

This field indicates whether the LOAD APM data is corrupted or not, which occurs when the buck converter runs in “bang-bang” controlled converter mode (see Section 5.3), when:

$$V_{STO} - V_{LOAD} < 0.25V$$

This field is updated both in pulse counter mode and in power meter mode.

- 0: the LOAD APM data is valid.
- 1: the LOAD APM data is corrupted.

Bit [4]: LOAD APM counter overflow (APMERR.LOADOV)

This field indicates whether a LOAD APM counter overflow error occurred or not.

This field is updated both in pulse counter mode and in power meter mode.

- 0: no LOAD APM counter overflow error occurred.
- 1: a LOAD APM counter overflow error occurred

Bit [3]: source 2 APM data invalid (APMERR.SRC2NVLD)

This field indicates whether the SRC2 APM data is corrupted or not. This occurs when $V_{SRC2} > V_{STO}$ and the boost converter runs in diode conduction mode.

This field is updated only if the APM is configured in power meter mode.

- 0: the SRC2 APM data is valid.
- 1: the SRC2 APM data is corrupted.

Bit [2]: source 2 APM counter overflow (APMERR.SRC2OV)

This field indicates whether a SRC2 APM counter overflow error occurred or not.

This field is updated both in pulse counter mode and in power meter mode.

- 0: no SRC2 APM counter overflow error occurred.
- 1: a SRC2 APM counter overflow error occurred.

Bit [1]: source 1 APM data invalid (APMERR.SRC1NVLD)

This field indicates whether the SRC1 APM data is corrupted or not. This occurs when $V_{SRC1} > V_{STO}$ and the boost converter runs in diode conduction mode.

This field is updated only if the APM is configured in power meter mode.

- 0: the SRC1 APM data is valid.
- 1: the SRC1 APM data is corrupted.

Bit [0]: source 1 APM counter overflow (APMERR.SRC1OV)

This field indicates whether a SRC1 APM counter overflow error occurred or not.

This field is updated both in pulse counter mode and in power meter mode.

- 0: no SRC1 APM counter overflow error occurred.
- 1: a SRC1 APM counter overflow error occurred.

9.22. Temperature Monitoring Data (TEMP)

The TEMP register allows providing the temperature monitoring data as long as R_{DIV} and R_{TH} are connected (see Figure 12).

TEMP register	0x23	R
Bit [7:0]		
DATA		
0x00		

Table 67: TEMP register

Bit [7:0]: temperature monitoring data (TEMP.DATA)

This field gives the code that results from the temperature monitoring.

9.23. Storage Element Voltage Data (STO)

The STO register allows providing the storage element monitoring data.

STO register	0x24	R
Bit [7:0]		
DATA		
0x00		

Table 68: STO register

R_{TH} can be determined using the following equation:

$$R_{TH} = \frac{R_{DIV} \cdot DATA}{256 - DATA}$$

Thus the temperature T in Kelvin can be obtained with the following formula.

$$T = \frac{B}{\ln\left(\frac{R_{TH}}{R_0}\right) + \frac{B}{T_0}}$$

See Section 5.4 for further information.

Bit [7:0]: storage element monitoring data (STO.DATA)

This field contains the code that results from the storage element monitoring.

V_{STO} can be determined using the following formula:

$$V_{STO} = \frac{4.8 \cdot DATA}{256}$$

9.24. Sources Voltage Data (SRCx)

The SRCx registers are the source data registers for SRC1 and SRC2 voltages.

SRC1 register	0x25	R
SRC2 register	0x26	R
Bit [7:0]		
DATA		
0x00		

Table 69: SRCx register

Bit [7:0]: source voltage data (SRCx.DATA)

This field contains the code that results from the source voltage measurement for the MPPT regulation. Maximum value is 0xB9.

To convert data from the register to Volts, use either the formulas from Table 70 or the values from Table 71 used as a lookup table. Please note the value in register SRCx.DATA are always within the ranges given in those two tables.

SRCx.DATA Range	Formula [V]
0x00 - 0x06	0.113
0x07 - 0x12	$0.09 + (2 \cdot \text{DATA} - 9) \cdot 0.0075$
0x13 - 0x39	$0.3 + (2 \cdot \text{DATA} - 37) \cdot 0.015$
0x68 - 0x79	$\frac{0.3 + (2 \cdot \text{DATA} - 165) \cdot 0.015}{0.67}$
0x9F - 0xB9	$\frac{0.3 + (2 \cdot \text{DATA} - 293) \cdot 0.015}{0.33}$

Table 70: Source voltage V_{SRCx} from SRCx.DATA register value (formula)

SRCx.DATA	V_{SRCx} [V]	SRCx.DATA	V_{SRCx} [V]	SRCx.DATA	V_{SRCx} [V]	SRCx.DATA	V_{SRCx} [V]
0x00	0.113	0x1D	0.615	0x36	1.365	0xA2	2.318
...	...	0x1E	0.645	0x37	1.395	0xA3	2.409
0x06	0.113	0x1F	0.675	0x38	1.425	0xA4	2.500
0x07	0.128	0x20	0.705	0x39	1.455	0xA5	2.591
0x08	0.143	0x21	0.735	0x68	1.410	0xA6	2.682
0x09	0.158	0x22	0.765	0x69	1.455	0xA7	2.773
0x0A	0.173	0x23	0.795	0x6A	1.500	0xA8	2.864
0x0B	0.188	0x24	0.825	0x6B	1.545	0xA9	2.955
0x0C	0.203	0x25	0.855	0x6C	1.590	0xAA	3.045
0x0D	0.218	0x26	0.885	0x6D	1.634	0xAB	3.136
0x0E	0.233	0x27	0.915	0x6E	1.679	0xAC	3.227
0x0F	0.248	0x28	0.945	0x6F	1.724	0xAD	3.318
0x10	0.263	0x29	0.975	0x70	1.769	0xAE	3.409
0x11	0.278	0x2A	1.005	0x71	1.813	0xAF	3.500
0x12	0.293	0x2B	1.035	0x72	1.858	0xB0	3.591
0x13	0.315	0x2C	1.065	0x73	1.903	0xB1	3.682
0x14	0.345	0x2D	1.095	0x74	1.948	0xB2	3.773
0x15	0.375	0x2E	1.125	0x75	1.993	0xB3	3.864
0x16	0.405	0x2F	1.155	0x76	2.037	0xB4	3.955
0x17	0.435	0x30	1.185	0x77	2.082	0xB5	4.045
0x18	0.465	0x31	1.215	0x78	2.127	0xB6	4.136
0x19	0.495	0x32	1.245	0x79	2.172	0xB7	4.227
0x1A	0.525	0x33	1.275	0x9F	2.045	0xB8	4.318
0x1B	0.555	0x34	1.305	0xA0	2.136	0xB9	4.409
0x1C	0.585	0x35	1.335	0xA1	2.227		

Table 71: Source voltage V_{SRCx} from SRCx.DATA register value (lookup table)

9.25. Part Number (PNx)

PN0 register	0xE0	R
Bit [7:0]		
DATA		
0x30		

Table 72: PN0 register

PN1 register	0xE1	R
Bit [7:0]		
DATA		
0x32		

Table 73: PN1 register

PN2 register	0xE2	R
Bit [7:0]		
DATA		
0x39		

Table 74: PN2 register

PN3 register	0xE3	R
Bit [7:0]		
DATA		
0x33		

Table 75: PN3 register

PN4 register	0xE4	R
Bit [7:0]		
DATA		
0x31		

Table 76: PN4 register

10. Typical Application Circuits

10.1. Example Circuit 1

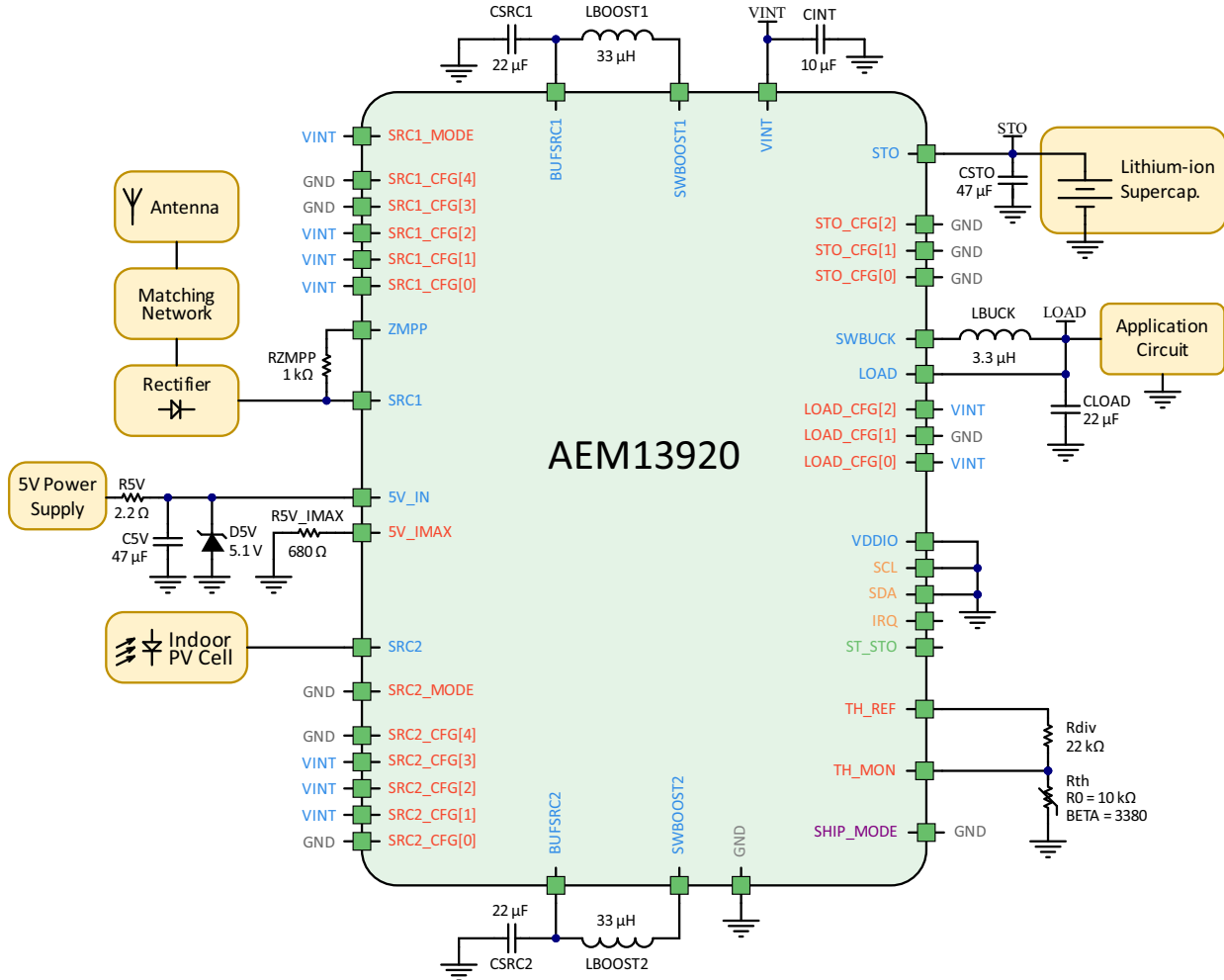


Figure 15: Typical application circuit 1

Figure 15 shows a typical application circuit of the AEM13920.

Configuration of SRC1

The first energy source is a RF harvester (antenna, matching network and RF rectifier), which has been optimized to provide maximum power when loaded with 1 kΩ. The fastest MPPT wait/period are set to ensure maximum reactivity, as RF signal level is likely to change quickly.

- SRC1_MODE = H (MPPT mode).
- SRC1_CFG[2:0] = HHH (ZMPP mode).
- SRC1_CFG[4:3] = LL.
 - $T_{MPPT, WAIT} = 1.8 \text{ ms.}$
 - $T_{MPPT, PERIOD} = 116 \text{ ms.}$

- $R_{ZMPP} = 1 \text{ k}\Omega.$
- $L_{BOOST1} = 33 \text{ }\mu\text{H}$ for best efficiency/current capability trade-off with default boost timings when I²C is not used (see Sections 6.5 and 9.6).

Configuration of SRC2

The second energy source is an indoor PV cell which has a constant 0.75 V MPP voltage. SRC2 is configured as follows:

- SRC2_MODE = L (constant voltage mode).
- SRC2_CFG[4:0] = LHHHL (0.75 V regulation).
- $L_{BOOST2} = 33 \text{ }\mu\text{H}$ for best efficiency/current capability trade-off with default boost timings when I²C is not used (see Sections 6.5 and 9.6).

Configuration of **STO**

The storage element is a Lithium-ion supercapacitor, so storage element threshold voltages are set as follows:

- **STO_CFG[2:0]** = LLL:
- $V_{OVDIS} = 2.51 \text{ V}$.
- $V_{CHRDY} = 2.61 \text{ V}$.
- $V_{OVCH} = 3.79 \text{ V}$.

Configuration of **LOAD**

The application circuit is supplied with 1.8 V with current peaks up to 100 mA. The buck converter is configured as follows:

- **LOAD_CFG[2:0]** = HLH (1.8 V).
- $L_{BUCK} = 3.3 \mu\text{H}$ for high current capability with default buck timings when I²C is not used (see Sections 6.6.2 and 9.7).

Configuration of **5V_IN**

The maximum allowed current to charge the storage element is 75 mA. Closest standard series resistor is 680 Ω , which leads to a 73.5 mA maximum current.

- $R_{5V_IMAX} = 680 \Omega$.
- $I_{5V,CC} = 73.5 \text{ mA}$.

The RC filter, which role is to slow down the rise time of the 5 V source, can be determined with the following steps.

R_{5V} is calculated so that the voltage drop across it ensures a V_{5V_IN} higher than the maximum value between $V_{OVCH} + 200 \text{ mV}$ and 3.60 V:

$$V_{OVCH} + 0.2 \text{ V} > 3.60 \text{ V}$$

$$I_{5V,CC} \cdot R_{5V} < 5\text{V} - V_{OVCH} - 0.2\text{V}$$

$$R_{5V} < \frac{5\text{V} - V_{OVCH} - 0.2\text{V}}{I_{5V,CC}} \Leftrightarrow R_{5V} < \frac{5\text{V} - 3.79\text{V} - 0.2\text{V}}{73.5 \times 10^{-3}}$$

$$R_{5V} < 13.74\Omega$$

C_{5V} is calculated so that the **5V_IN** voltage rise time remains below $T_{5V,RISE}$:

$$R_{5V} \cdot C_{5V} > T_{5V,RISE}$$

$$R_{5V} \cdot C_{5V} > 50\mu\text{s}$$

To meet these two conditions, the following component values have been selected:

- $R_{5V} = 2.2 \Omega$
- $C_{5V} = 47 \mu\text{F}$

The 5 V source is expected to have ripple and/or over voltages up to 5.5 V, so a 5.1 V zener diode D_{5V} is added to prevent those to damage the AEM13920.

The minimum required power rating of D_{5V} is computed as follows, from its maximum reverse current I_{D5V} , its voltage V_{D5V} and the resistor R_{5V} :

$$P_{D5V} \geq I_{D5V} \cdot V_{D5V} \Leftrightarrow P_{D5V} \geq \frac{5.5\text{V} - 5.1\text{V}}{R_{5V}} \cdot 5.1\text{V}$$

$$P_{D5V} \geq \frac{5.5\text{V} - 5.1\text{V}}{2.2} \cdot 5.1\text{V} \Leftrightarrow P_{D5V} \geq 927\text{mW}$$

R_{5V} dissipated power $P_{R5V,idle}$ when the 5 V charger does not pull any current to charge the storage element is determined as follows:

$$P_{R5V,idle} = \frac{(5.5\text{V} - 5.1\text{V})^2}{R_{5V}} \Leftrightarrow P_{R5V,idle} = \frac{(5.5\text{V} - 5.1\text{V})^2}{2.2}$$

$$P_{R5V,idle} = 73\text{mW}$$

Furthermore, R_{5V} dissipated power $P_{R5V,CC}$ at $I_{5V,CC}$ current (73.5 mA) is determined as follows:

$$P_{R5V,CC} = R_{5V} \cdot I_{5V,CC}^2 = 2.2\Omega \cdot (73.5\text{mA})^2 = 12\text{mW}$$

The minimum required power rating of R_{5V} is the maximum of $P_{R5V,idle}$ and $P_{R5V,CC}$, thus, 73 mW.

I²C configuration

I²C is not used:

- **VDDIO**, **SDA**, and **SCL** are tied to GND.
- **IRQ** is left floating.

Temperature monitoring

Temperature monitoring is used to protect the storage element from being charged and discharged when temperature is outside the range:

- R_{TH} :
 - $R_0 = 10 \text{ k}\Omega$.
 - $BETA = 3380$.
- $R_{DIV} = 22 \text{ k}\Omega$.
- **STO_CFG[2:0]** = LLL:
 - $TEMPCOLDCH = -15 \text{ }^\circ\text{C}$.
 - $TEMPHOTCH = 60 \text{ }^\circ\text{C}$.
 - $TEMPCOLDNIS = -15 \text{ }^\circ\text{C}$.
 - $TEMPHOTNIS = 60 \text{ }^\circ\text{C}$.

Shipping mode

Shipping mode is not used.

- **SHIP_MODE** is connected to GND.

10.2. Example Circuit 2

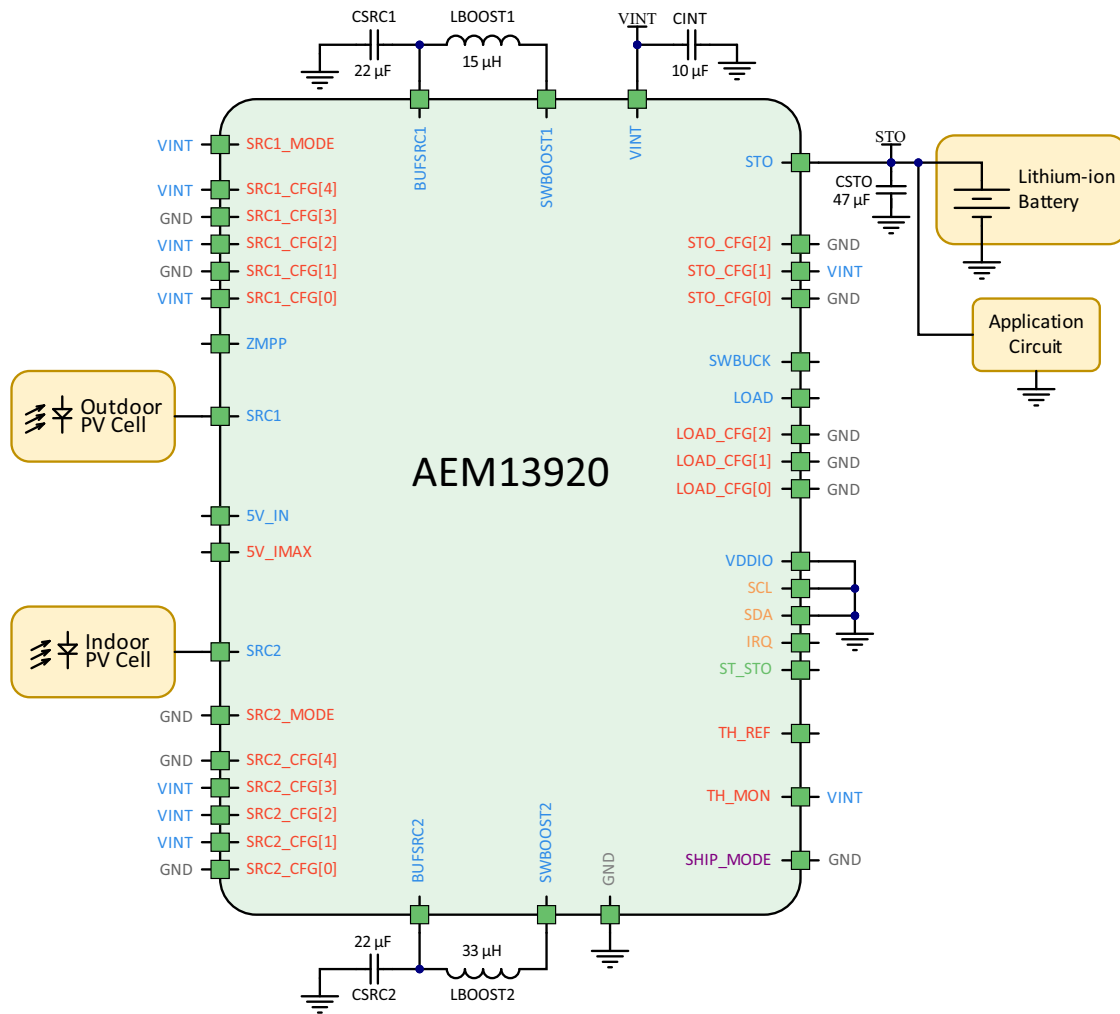


Figure 16: Typical application circuit 2

Figure 16 shows a typical application circuit of the AEM13920.

Configuration of SRC1

The first energy source is an outdoor PV cell. The MPPT is used with a 80 % ratio. A medium MPPT wait/period are set to ensure that the PV cell has enough time to reach its open circuit voltage during the V_{OC} evaluation while still keeping a good reactivity to lighting changes.

- **SRC1_MODE** = H (MPPT mode).
- **SRC1_CFG[2:0]** = HLH (80 %).
- **SRC1_CFG[4:3]** = HL.
 - $T_{MPPT, WAIT}$ = 29 ms.
 - $T_{MPPT, PERIOD}$ = 1862 ms.
- **LBOOST1** = 15 μ H for high current capability with default boost timing when I²C is not used (see Sections 6.5 and 9.6).

Configuration of SRC2

The second energy source is an indoor PV cell which has a constant 0.75 V maximum power point voltage. **SRC2** is configured as follows:

- **SRC2_MODE** = L (constant voltage mode).
- **SRC2_CFG[4:0]** = LHHHL (0.75 V regulation).
- **LBOOST2** = 33 μ H for best efficiency/current capability trade-off with default boost timing when I²C is not used (see Sections 6.5 and 9.6).

Configuration of STO

The storage element is a Lithium-ion battery, so storage element threshold voltages are set as follows:

- **STO_CFG[2:0]** = LHL.
- **V_{OVDIS}** = 3.00 V.
- **V_{CHRDY}** = 3.21 V.
- **V_{OVCH}** = 4.13 V.

Configuration of LOAD

The application circuit is supplied from the storage element, so the **LOAD** output is not used:

- **LOAD_CFG[2:0]** = LLL: buck converter is disabled.
- **SWBUCK** and **LOAD** are left floating.

Configuration of 5V_IN

The 5 V charger is not used:

- **5V_IN** and **5V_IMAX** are left floating.

I²C configuration

I²C is not used:

- **VDDIO**, **SDA**, and **SCL** are tied to GND.
- **IRQ** is left floating.

Temperature monitoring

Temperature monitoring is not used:

- **TH_MON** is connected to **VINT**.
- **TH_REF** is left floating.

Shipping mode

Shipping mode is not used.

- **SHIP_MODE** is connected to GND.

10.3. Example Circuit 3

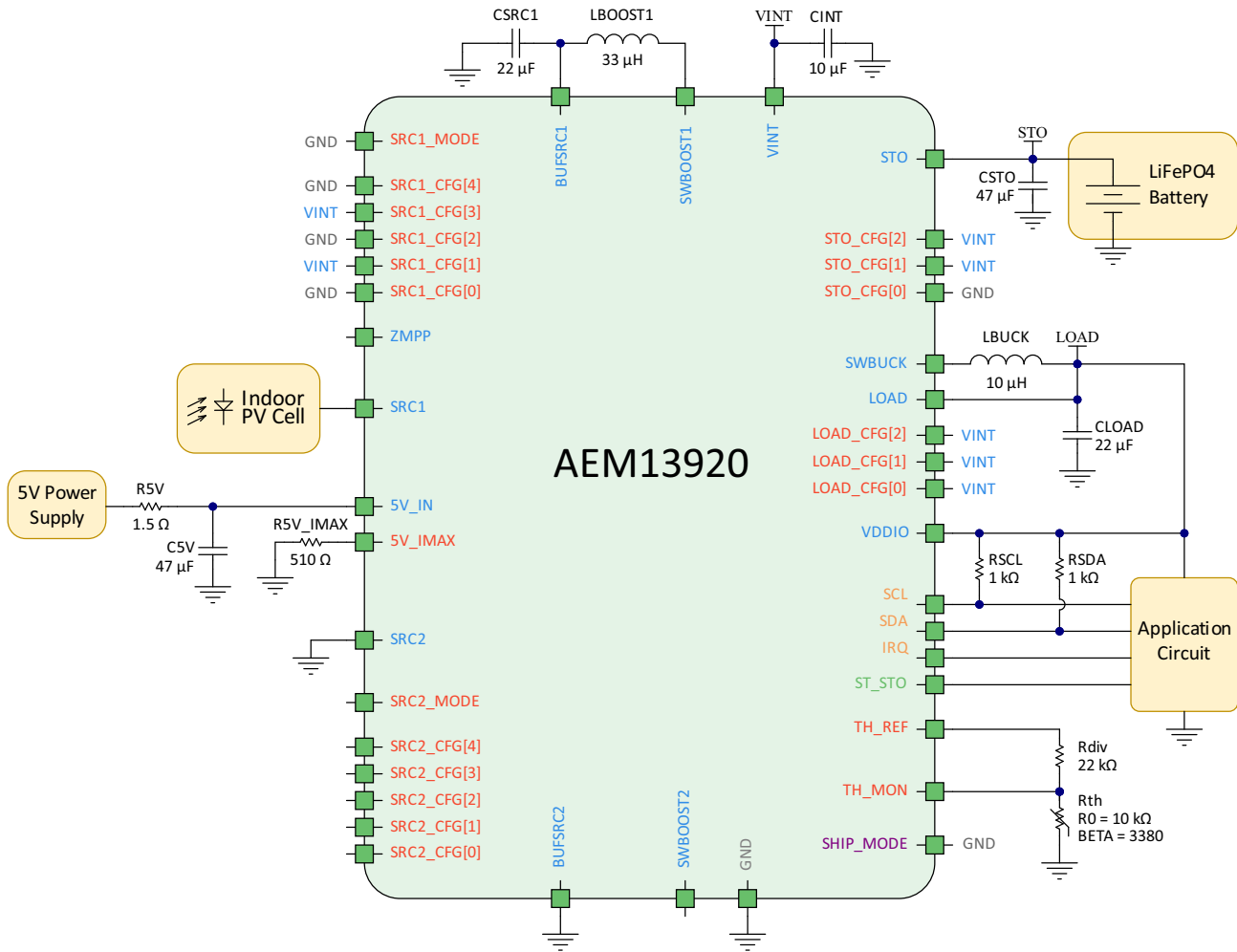


Figure 17: Typical application circuit 3

Figure 17 shows a typical application circuit of the AEM13920.

Configuration of SRC1

The energy source is an indoor PV cell which has a constant 0.60 V maximum power point voltage. SRC1 is configured as follows:

- SRC1_MODE = L (constant voltage mode).
- SRC1_CFG[4:0] = LHLHL (0.60 V regulation).
- LBOOST1 = 33 μH for best efficiency/current capability trade-off with the configured boost timings multiplier.
- T_{MULT} = x3 (configured through I²C register, see Table 77).

Configuration of SRC2

SRC2 boost converter is not used.

- SRC2, BUFSRC2 are connected to GND
- SWBOOST2 is left floating.
- SRC2_MODE and SRC2_CFG[4:0] are left floating.

Configuration of STO

The storage element is a LiFePO₄ battery, so storage element threshold voltages are set as follows:

- STO_CFG[2:0] = HHL.
- V_{OVDIS} = 2.81 V.
- V_{CHRDY} = 3.11 V.
- V_{OVCH} = 3.62 V.

Configuration of **LOAD**

The application circuit is supplied with 2.5 V with current peaks up to 20 mA. The buck converter is configured as follows:

- **LOAD_CFG[2:0]** = HHH (2.5 V)
- **L_{BUCK}** = 10 µH for best efficiency/current capability trade-off with the configured buck timings multiplier.
- **T_{MULT}** = x2 (configured through I²C register, see Table 77).

Configuration of **5V_IN**

The maximum allowed current to charge the storage element is 100 mA. Closest standard series resistor is 510 Ω, which leads to a 98 mA maximum current.

- **R_{5V_IMAX}** = 510 Ω.
- **I_{5V,CC}** = 98 mA.

The power supply connected on the **5V_IN** provides a voltage from minimum 4 V to maximum 5 V. The RC filter, which role is to slow down the rise time of the 5 V source, can be selected with the following steps to ensure the 5 V charger operates properly.

R_{5V} is calculated so that the voltage drop across it ensures a **V_{5V_IN}** higher than the maximum value between **V_{OVCH}** + 200 mV and 3.60 V, considering the minimum voltage available from the power supply:

$$V_{OVCH} + 0.2 \text{ V} > 3.60 \text{ V}$$

$$I_{5V,CC} \cdot R_{5V} < 4 \text{ V} - V_{OVCH} - 0.2 \text{ V}$$

$$R_{5V} < \frac{4 \text{ V} - V_{OVCH} - 0.2 \text{ V}}{I_{5V,CC}} \Leftrightarrow R_{5V} < \frac{4 \text{ V} - 3.62 \text{ V} - 0.2 \text{ V}}{98 \times 10^{-3}}$$

$$R_{5V} < 1.84 \Omega$$

C_{5V} is calculated so that the **5V_IN** voltage rise time remains below **T_{5V,RISE}**:

$$R_{5V} \cdot C_{5V} > T_{5V,RISE}$$

$$R_{5V} \cdot C_{5V} > 50 \mu\text{s}$$

To meet these two conditions, the following component values have been selected:

- **R_{5V}** = 1.5 Ω
- **C_{5V}** = 47 µF

R_{5V} dissipated power **P_{R5V,CC}** at **I_{5V,CC}** (98 mA) is determined as follows:

$$P_{R5V,CC} = R_{5V} \cdot I_{5V,CC}^2 = 1.5 \Omega \cdot (98 \text{ mA})^2 = 14.4 \text{ mW}$$

I²C configuration

I²C is used to configure the AEM13920:

- **VDDIO** is connected to **LOAD**, which is the node supplying the application circuit that communicates with the AEM13920 through I²C.
- **SDA** and **SCL** are pulled-up to **VDDIO** with 1 kΩ resistors and connected to the application circuit micro controller (MCU) I²C bus.
- **IRQ** and **ST_STO** are connected to the application circuit MCU GPIOs.

The configuration is sent through the I²C bus. Please note that the configuration done through pins (**SRCx_CFG[4:0]**, **STO_CFG[2:0]**, **LOAD_CFG[2:0]**, etc.) must also be written to the registers, otherwise the default register values will be applied (see Section 9 for further details about configuring the AEM13920 with I²C registers).

See Table 77 for the whole I²C register configuration (all other registers have appropriate default values).

Temperature monitoring

Temperature monitoring is used to protect the storage element from being charged and discharged outside its acceptable temperature range. The following settings are applied (see Table 77 for the corresponding register values):

- **R_{TH}**:
 - **R0** = 10 kΩ.
 - **BETA** = 3380.
- **R_{DIV}** = 22 kΩ.
- Charging is allowed between 0°C and +45°C.
- Discharging is allowed between -20°C and +65°C.
- Thermal protection monitoring is enabled by default (see Section 9.11) so it is not mandatory to write the **TEMPPROTECT** register.



Average Power Monitoring (APM)

To set up the APM, the user must do the following:

- Enable the APM event in register IRQEN1 so that a rising edge is generated on the **IRQ** pin to notify the application MCU when a new APM data is ready. This is done by setting the IRQEN1.APMDONE field to 1.
- Configure the APM register:
 - APM.WINDOW = 0: APM window of 233 ms for longest integration.
 - APM.MODE = 1: power meter mode.
 - APM.LOADEN = 1: enable power monitoring of energy from **STO** to **LOAD**.
 - APM.SRC1EN = 1: enable APM monitoring from **SRC1** to **STO**.
 - APM.SRC2EN = 0: disable APM monitoring from **SRC2** to **STO**.
 - APM.CHG5VEN = 1: enable APM monitoring on the 5 V charger.
- Read APM data when **IRQ** raises.
- Read the IRQFLG1 register to reset the **IRQ** pin.

Shipping mode

Shipping mode is not used.

- **SHIP_MODE** is connected to **GND**.

Register Name	Value	Notes
SRC1REGU1	0x07	Constant voltage mode.
SRC1REGU0	0x02	$V_{SRCx,REG} = 0.600 \text{ V}$.
VOVDIS	0x15	$V_{OVDIS} = 2.794 \text{ V}$.
VCHRDY	0x22	$V_{CHRDY} = 3.094 \text{ V}$.
VOVCH	0x32	$V_{OVCH} = 3.638 \text{ V}$.
BST1CFG	0x11	Boost converter #1 enabled. $T_{MULT} = x3$.
BUCKCFG	0x0F	$V_{LOAD} = 2.5 \text{ V}$. $T_{MULT} = x2$.
TEMPCOLDCH	0x90	Min. 0°C for charge.
TEMPHOTCH	0x2E	Max. +45°C for charge.
TEMPCOLDNIS	0xC6	Min. -20°C for discharge.
TEMPHOTNIS	0x1B	Max. +65°C for discharge.
TEMPPROTECT	0x01	Enable thermal protection monitoring.
APM	0x1D	APM register configuration as explained above.
IRQEN1	0x40	Enable APMDONE IRQ.
CTRL	0x01	Write this register after writing the others to load I ² C register configuration.

Table 77: Summary of I²C register configuration for typical application circuit 3

11. Circuit Behavior

11.1. Start Up from SRCx

11.1.1. Configuration

- **SRC1** supplied by a 2.0 V voltage source with 100 mA compliance with 700 Ω in series:
 - $V_{OC} = 2.0$ V.
 - $V_{MPP} = 1.5$ V with $R_{MPPT} = 75\%$.
 - $I_{SRC1} = 714$ μ A.
- **SRC1_MODE** = H:
 - **SRC1** mode is MPPT.
- **SRC1_CFG[4:3]** = LH:
 - $T_{MPPT,PERIOD} = 465$ ms.
 - $T_{MPPT,WAIT} = 7.3$ ms.
- **SRC1_CFG[2:0]** = HLL:
 - $R_{MPPT} = 75\%$.
- $L_{BOOST1} = 33$ μ H.
- **STO_CFG[2:0]** = HLH:
 - $V_{OVDIS} = 3.00$ V.
 - $V_{CHRDY} = 3.30$ V.
 - $V_{OVCH} = 4.13$ V
- $C_{STO} = 1$ mF electrolytic charged at 2.8 V beforehand.
- **LOAD_CFG[2:0]** = HLH:
 - $V_{LOAD} = 1.8$ V.
- $L_{BUCK} = 10$ μ H.
- **VDDIO** = 3.3 V (external source).

11.1.2. Observations

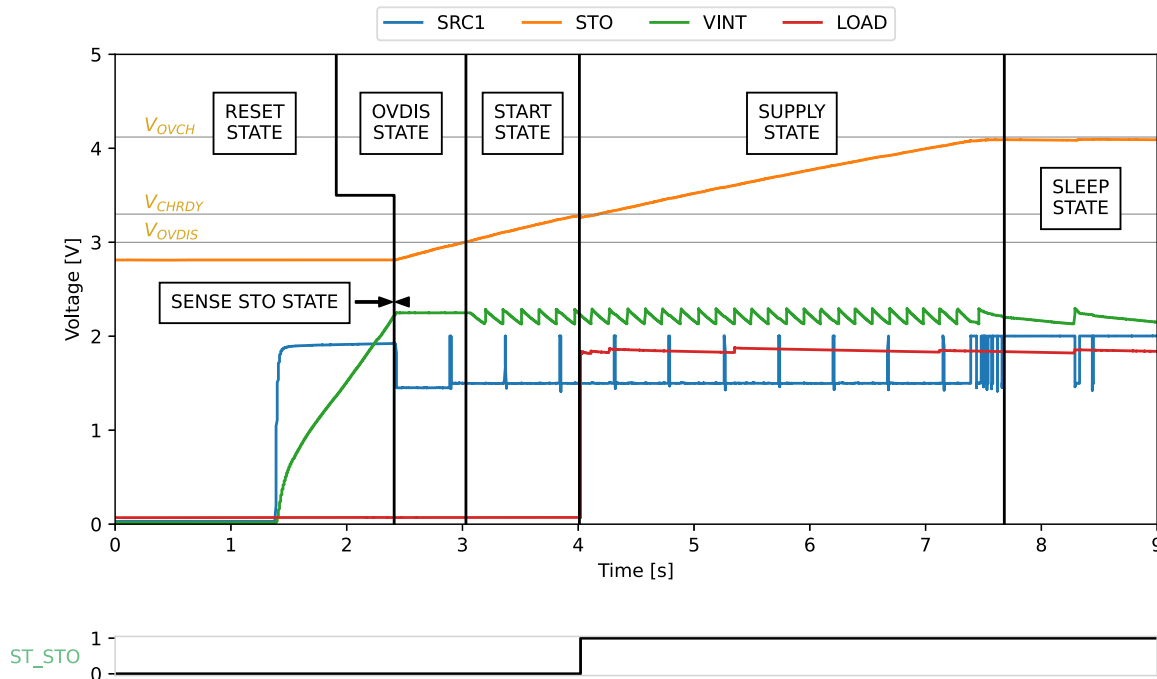


Figure 18: Behavior at start up

- The AEM13920 is initially in **RESET STATE**.
- Once the supply connected on **SRC1** is switched on, the AEM13920 coldstarts. C_{INT} is charged until **VINT** reaches $V_{INT,CS}$. The AEM13920 switches then to **SENSE STO STATE**.
- In **SENSE STO STATE**, the AEM13920 measures V_{STO} , which is slightly below V_{OVDIS} . The AEM13920 switches then to **OVDIS STATE**.
- In **OVDIS STATE**, the AEM13920 performs a first V_{OC} evaluation and charges the storage element on **STO** by harvesting the energy from **SRC1**. V_{SRC1} is regulated at 75% of V_{OC} , thanks to the MPPT module. **VINT** is supplied by **SRC1**. Once V_{STO} reaches V_{OVDIS} , the AEM13920 switches to **START STATE**.
- In **START STATE**, the AEM13920 continues to charge the storage element connected to **STO** from the harvested energy by **SRC1**. The MPPT module still ensures that V_{SRC1} is regulated at 75% of V_{OC} . The **LOAD** output remains disabled. **VINT** is supplied by **STO**. Because the 5 V charger is not used, the AEM13920 waits for V_{STO} to reach V_{CHRDY} to switch to **SUPPLY STATE**.
- In **SUPPLY STATE**, the AEM13920 keeps charging the storage element from **SRC1**. The **LOAD** output is now enabled, so that V_{LOAD} is regulated at 1.8 V and **ST_STO** pin is asserted. Once V_{STO} reaches V_{OVCH} , the AEM13920 switches to **SLEEP STATE**.
- In **SLEEP STATE**, **STO**, **LOAD** and **VINT** are fully charged. The AEM13920 stops harvesting energy from **SRC1**. Please note that around 8.3 s, the AEM13920 recharges **VINT** from **SRC1**, and around 8.5 s, the AEM13920 switches briefly to **SUPPLY STATE** to recharge **STO** from **SRC1**.



11.2. Shutdown

11.2.1. Configuration

- **SRC1** supplied by a 2.0 V voltage source with 100 mA compliance with 500 Ω in series:
 - $V_{OC} = 2.0$ V.
 - $V_{MPP} = 1.5$ V with $R_{MPPT} = 75\%$.
 - $I_{SRC1} = 1$ mA.
 - The supply is disconnected near 11.2 s.
- **SRC1_MODE** = H:
 - **SRC1** mode is MPPT.
- **SRC1_CFG[4:3]** = LH:
 - $T_{MPPT,PERIOD} = 465$ ms.
 - $T_{MPPT,WAIT} = 7.3$ ms.
- **SRC1_CFG[2:0]** = HLL:
 - $R_{MPPT} = 75\%$.
- $L_{BOOST1} = 33$ μ H.
- **STO_CFG[2:0]** = HLH:
 - $V_{OVDIS} = 3.00$ V.
 - $V_{CHRDY} = 3.30$ V.
 - $V_{OVCH} = 4.13$ V.
- $C_{STO} = 10$ mF electrolytic capacitor charged at 2.8 V beforehand.
- **LOAD_CFG[2:0]** = HLH:
 - $V_{LOAD} = 1.8$ V.
 - A 200 Ω resistor is connected between **LOAD** and **GND**, so that a 9 mA current is pulled from the **LOAD** pin.
- $L_{BUCK} = 10$ μ H.
- **VDDIO** = 3.3 V (external source).

11.2.2. Observations

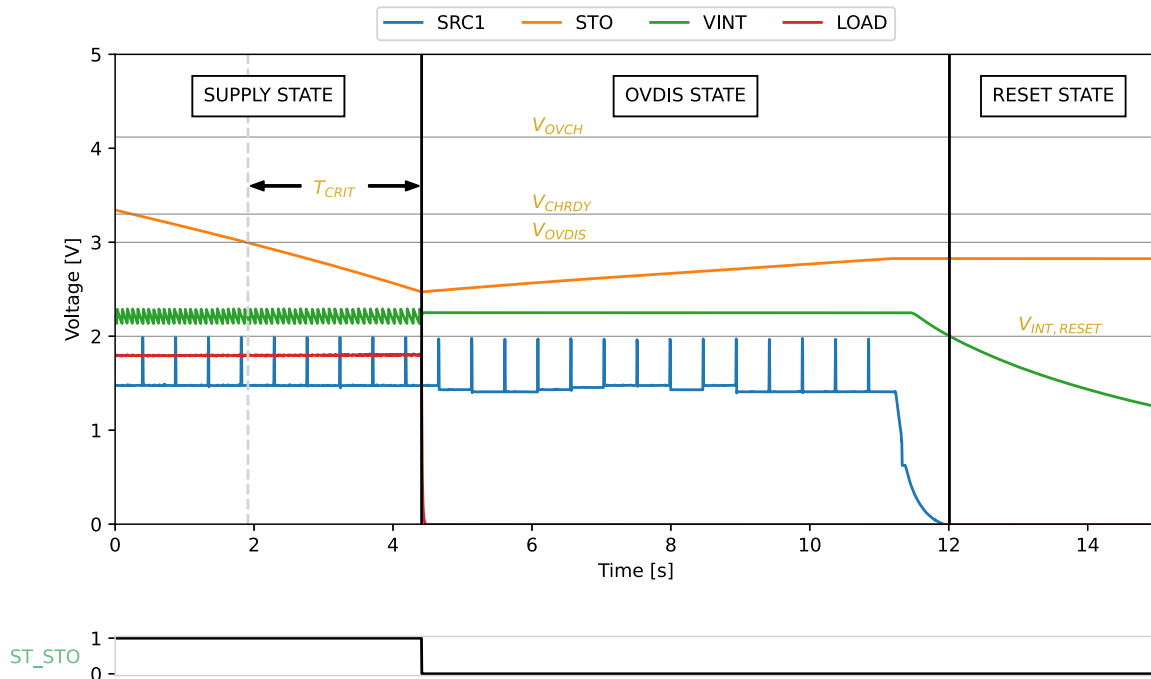


Figure 19: Behavior at shutdown

- The AEM13920 is initially in **SUPPLY STATE**. **ST_STO** pin is HIGH. The harvested power on **SRC1** is lower than the power supplied by **LOAD**, so that the storage element on **STO** is being discharged. Once V_{STO} drops below V_{OVDIS} for longer than $T_{CRIT,ST}$, the AEM13920 schedules to go in **OVDIS STATE** but stays in **SUPPLY STATE** until the end of T_{CRIT} , and then switches to **OVDIS STATE**.
- In **OVDIS STATE**, the **LOAD** output is disabled and **VINT** is supplied by **SRC1**. **ST_STO** pin is LOW. There is no more current drawn on the **LOAD** pin, and thus, on the storage element, so that the power budget is positive and the storage element is charged again.
- Near 11.2 s, the power supply connected on **SRC1** is switched off. The storage element on **STO** is no longer charged, neither is **VINT**. When V_{INT} drops below $V_{INT, RESET}$, the AEM13920 switches to **RESET STATE**.
- In **RESET STATE**, the energy present on the storage element is preserved as the **STO** pin is set to high impedance.



12. Minimum BOM

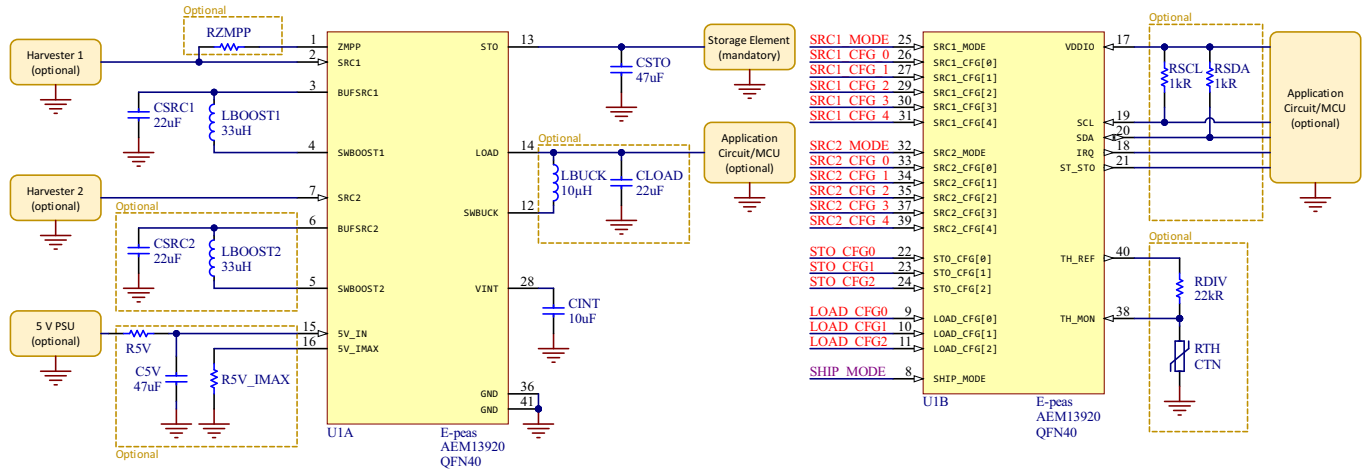


Figure 20: Schematic

Designator	Description	Quantity	Manufacturer	Part Number
Mandatory	U1	1	e-peas	order at sales@e-peas.com
	Storage Element	1	To be defined by user.	
	CINT	1	Murata	GRM155R60J106ME44D
	CSTO ¹	1	Murata	GRM188R60J476ME15D
	CSRC1 ²	1	Murata	GRM188R61A226ME15D
	LBOOST1 ²	1	Coilcraft	LPS4018-333MRB
Optional	RZMPP ²	1	To be defined by user.	
	CSRC2 ²	1	Murata	GRM188R61A226ME15D
	LBOOST2 ²	1	Coilcraft	LPS4018-333MRB
	C5V ²	1	Murata	GRM188R60J476ME15D
	R5V ²	1	To be defined by user.	
	R5V_IMAX ²	1	To be defined by user.	
	CLOAD	1	Murata	GRM188R61A226ME15D
	LBUCK	1	TDK	VLS252012CX-100M-1
	RSCL	1	Multicomp	MCWR06X1001FTL
	RSDA	1	Multicomp	MCWR06X1001FTL
	RDIV	1	Yageo	RC0402FR-0722KL
	RTH	1	Murata	NCP15XH103J03RC

Table 78: Minimum BOM

1. Recommended CSTO for optimal efficiency, particularly with high-ESR storage elements. If using a smaller value, ensure it meets the minimum requirement (see Table 10).
2. The AEM13920 must have at least one energy source to work: boost #1 (SRC1), boost #2 (SRC2) or 5 V input (5V_IN), or any combination of those. If using SRC1, CSRC1 and LBOOST1 are mandatory. If using SRC2, CSRC2 and LBOOST2 are mandatory. If using 5V_IN, CSV, R5V and R5V_IMAX are mandatory.

13. Layout

13.1. Guidelines

Figure 21 shows an example of PCB layout with AEM13920.

The following guidelines must be applied for best performances:

- Make sure that ground and power signals are routed with large tracks. If an internal ground plane is used, place vias as close as possible to the components, especially for decoupling capacitors.
- Reactive components related to the boost converters and the buck converter must be placed as close as possible to the corresponding pins (**SWBOOSTx**, **BUFSRCx**, **SWBUCK**, **LOAD** and **STO**), and be routed with large tracks/polygons.
- Keep distance between inductors to avoid magnetic coupling.
- PCB track capacitance must be reduced as much as possible on the boost converters switching nodes **SWBOOSTx**. This is done as follows:
 - Keep the connection between the **SWBOOSTx** / **SWBUCK** pins and their corresponding inductors short.
 - Remove the ground and power planes under the **SWBOOSTx** / **SWBUCK** nodes. The polygon on the opposite external layer may also be removed.
 - Increase the distance between **SWBOOSTx** / **SWBUCK** and the ground polygon on the external PCB layer where the AEM13920 is mounted.
- PCB track capacitance must be reduced as much as possible on the **TH_REF** node. Same principle as for **SWBOOSTx** may be applied.

13.2. Example

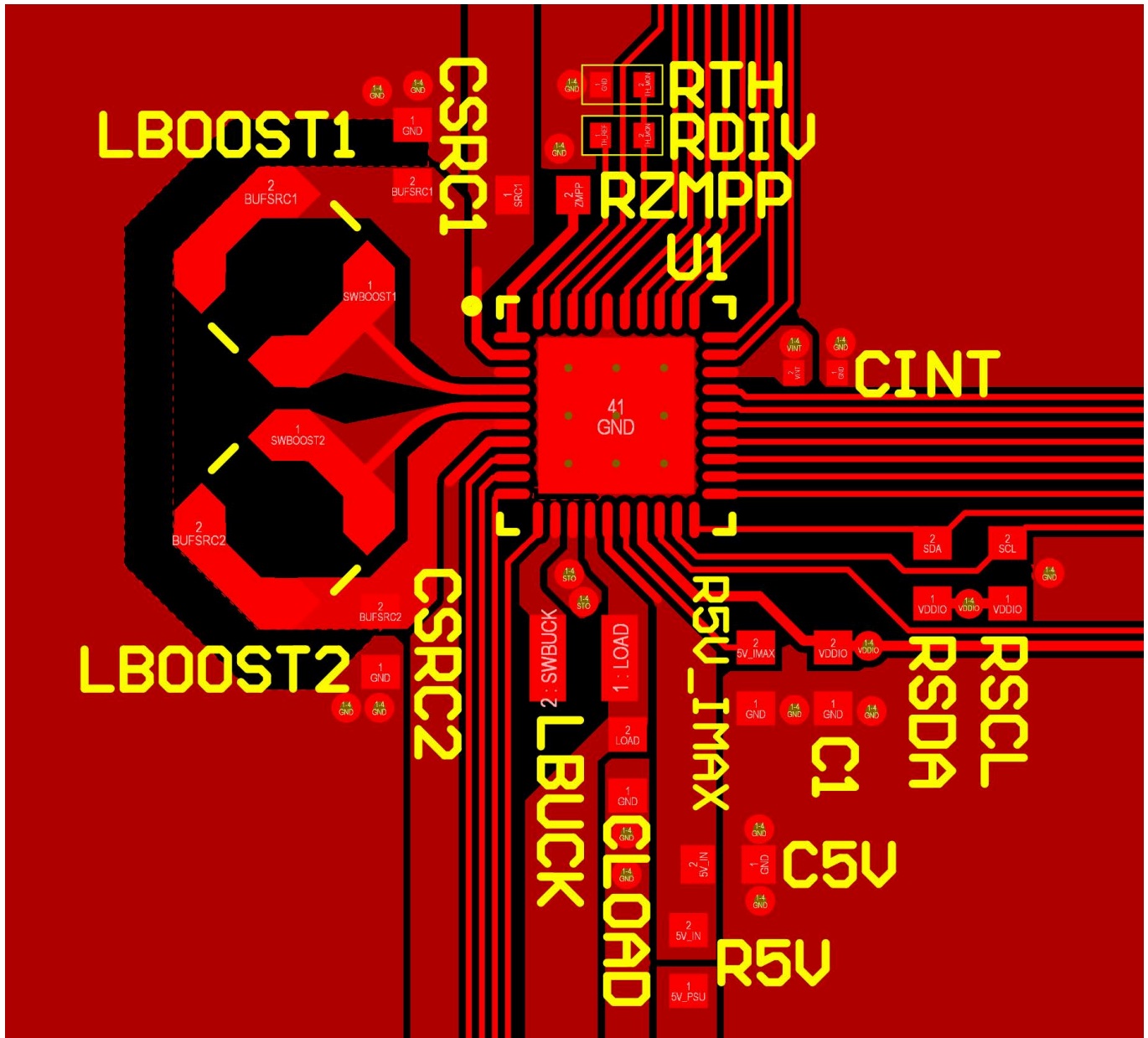


Figure 21: Layout example

14. Package Information

14.1. Moisture Sensitivity Level

Package	Moisture Sensitivity Level (MSL) ¹
QFN-40	Level 1

Table 79: Moisture sensitivity level

1. According to JEDEC 22-A113 standard.

14.2. RoHS Compliance

e-peas product complies with RoHS requirement.

e-peas defines “RoHS” to mean that semiconductor end-products are compliant with RoHS regulation for all 10 RoHS substances.

This applies to silicon, die attached adhesive, gold wire bonding, lead frames, mold compound, and lead finish (pure tin).

14.3. REACH Compliance

The component and elements used by e-peas subcontractors to manufacture e-peas PMICs and devices are REACH compliant. For more detailed information, please contact e-peas sales team.

14.4. Tape and Reel Dimensions

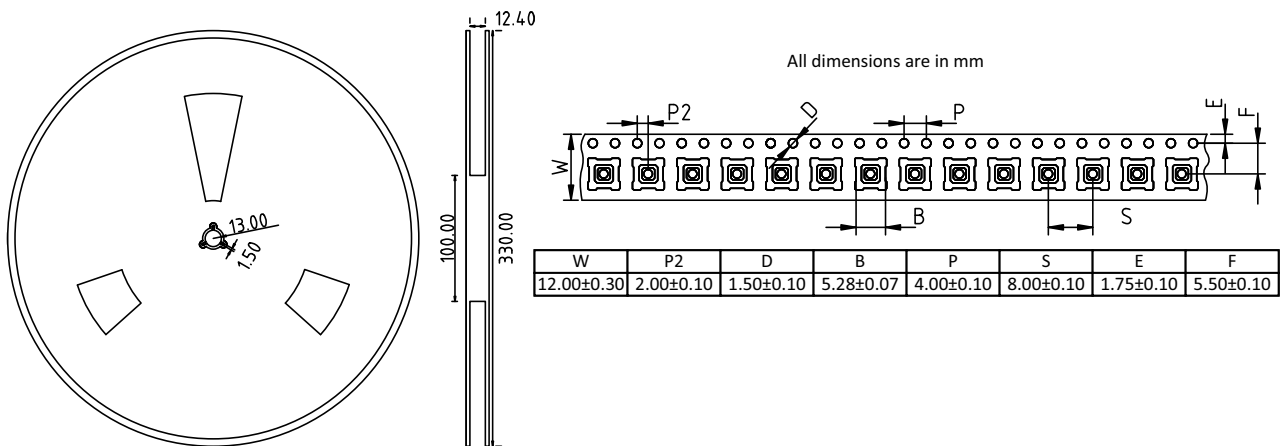


Figure 22: Tape and reel dimensions

14.5. Package Dimensions

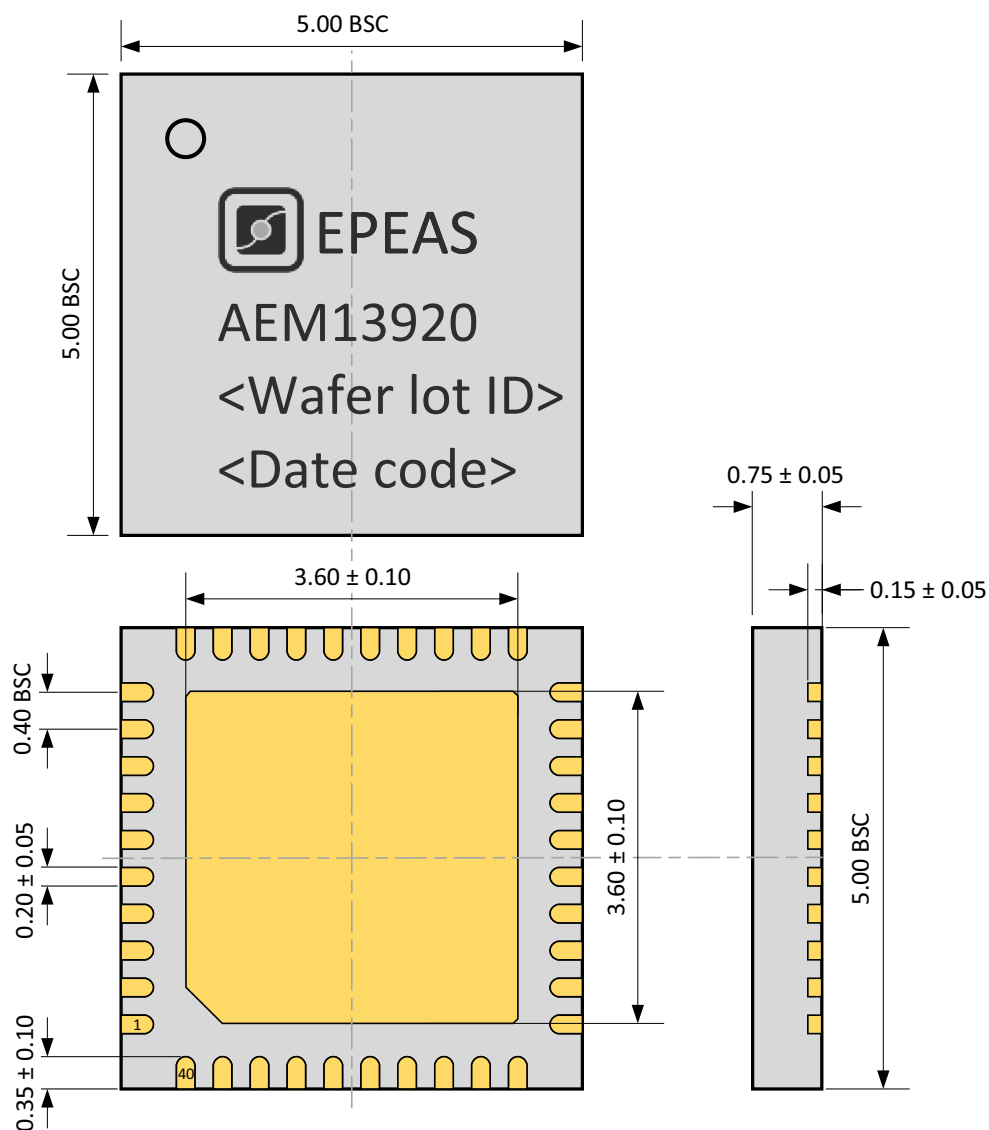


Figure 23: QFN 40-pin 5x5mm drawing (all dimensions in mm)

14.6. Board Layout

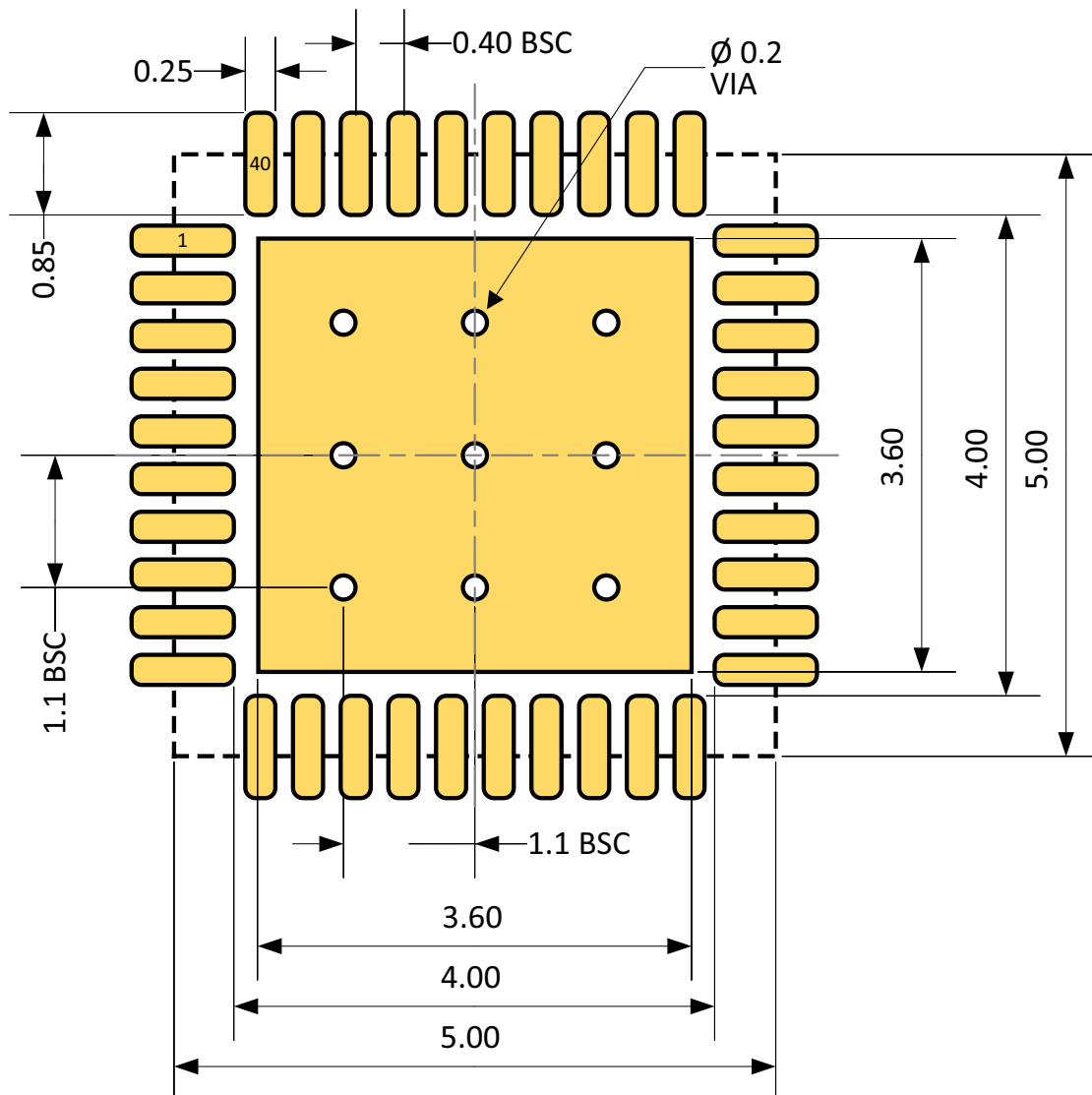


Figure 24: Recommended board layout for QFN40 package (all dimensions in mm)

15. Glossary

C_{5V}	Decoupling capacitor on the 5V_IN pin.	I_{Q,SUPPLY}	AEM13920 internal quiescent current drawn on STO in SUPPLY STATE .
C_{INT}	Decoupling capacitor on the VINT pin.	I_{Q,RESET}	AEM13920 internal quiescent current drawn on STO in RESET STATE .
C_{LOAD}	Decoupling capacitor on LOAD pin.	I_{SRCx}	Current extracted from the harvester connected on SRCx .
C_{SRCx}	Decoupling capacitor on the BUFSRCx pin.	L_{BOOSTx}	Boost converter x inductor.
C_{STO}	Decoupling capacitor on the STO pin.	L_{BUCK}	Buck converter inductor.
D_{5V}	Zener diode that ensures that the voltage on 5V_IN stays below 5.5 V at any time.	P_{APM}	Power provided to/from the storage element measured through the APM (see Section 9.17).
D_{CHG}	5 V charger APM charging duty cycle (see Section 9.17.2).	P_{R5V,CC}	Power dissipated by the R_{5V} when the 5 V charger is in constant current (CC) mode.
E_{APM}	Energy that was provided to/from the storage element during the APM window (see Section 9.17).	P_{R5V,idle}	Power dissipated by R_{5V} when no current is pulled by the 5 V charger (current only flowing in the zener protection diode).
I_{5V,CC}	Current provided to the storage element by the 5V_IN when in constant current mode.	R_{5V}	Resistor creating a RC filter with C_{5V} on 5V_IN to limit T_{5V,RISE} .
I_{5V,OVDIS,CC}	Charging current of the 5 V charger when in constant current (CC) mode and V_{STO} < V_{OVDIS} .	R_{5V,IMAX}	Resistor connected between 5V_IMAX and GND that defines the maximum current provided to the storage element by the 5 V charger (5V_IN pin).
I_{5V,CV}	Current provided to the storage element by the 5V_IN when in constant voltage mode.	R_{DIV}	Along with R_{TH} , resistor creating a resistive voltage divider connected to TH_MON , used for thermal monitoring.
I_{LBOOST,PEAK}	Peak current in L_{BOOSTx} when the boost converter is running.	R_{MPPT}	For the boost converters, ratio between the open circuit voltage V_{OC} and the voltage regulation V_{MPP} determined by the MPPT (when the boost converter is in MPPT mode).
I_{LBUCK,PEAK}	Peak current in L_{BUCK} when the buck converter is running.	R_{SCL} / R_{SDA}	Pull-up resistors used for the I ² C communication bus.
I_{Q,SHIP}	AEM13920 internal quiescent current on STO in shipping mode (SHIP_MODE is HIGH) with or without energy available on SRCx .		
I_{Q,SLEEP}	AEM13920 internal quiescent current drawn on STO in SLEEP STATE .		

R_{TH}	Along with R_{DIV} , thermistor creating a resistive voltage divider connected to TH_MON , used for thermal monitoring.	V_{CHRDY}	In START STATE , voltage required on the storage element to switch to SUPPLY STATE (see Section 6.4).
R_{ZMPP}	Resistor used for ZMPP module, connected between SRC1 and ZMPP .	V_{CHRDY,BUCK}	Minimum voltage accepted on the storage element before starting to supply the LOAD if the 5 V charger is not used.
T_{5V,RISE}	Minimum voltage rise time on the 5V_IN pin.	V_{ESD}	Electrostatic discharge voltage.
T_{CRIT}	Delay for the AEM13920 to go in OVDIS STATE and to disable the LOAD output (see Section 5.9).	V_{INT}	Voltage on the VINT pin.
T_{CRIT,ST}	Delay for the AEM13920 to notify the application about an overdischarged storage element, a temperature out of discharge temperature range, and to schedule to disable the LOAD output (see Section 5.3).	V_{INT,CS}	Minimum voltage on VINT to allow the AEM13920 to switch from RESET STATE to SENSE STO STATE .
T_{GPIO,MON}	GPIO reading rate.	V_{INT,RESET}	Minimum voltage on VINT before switching to RESET STATE (from any other state).
T_{MPPT,MEASURE}	Duration of V_{OC} measurement during MPP evaluation.	V_{LOAD}	Voltage on the LOAD pin.
T_{MPPT,PERIOD}	Time between the start of two MPP evaluations (see Table 13).	V_{MPP}	Target regulation voltage on SRCx when extracting power (when SRCx regulation mode is MPPT).
T_{MPPT,WAIT}	Time interval during which the AEM13920 stops extracting power and allows the source voltage to rise to the open-circuit voltage before V_{OC} measurement begins (see Table 13).	V_{OC}	Open circuit voltage of the harvester connected on SRCx .
T_{MULT}	Boost or buck converter inductor charging timings multiplier.	V_{OVCH}	Maximum voltage accepted on the storage element before disabling its charging (see Section 6.4).
T_{STO,MON}	Storage element voltage monitoring rate.	V_{OVDIS}	Voltage below which the storage element is considered to be fully depleted, and must not be discharged any further (see Section 6.4).
T_{TEMP,MON}	Temperature monitoring rate.	V_{OVDIS,BUCK}	Minimum storage element voltage accepted before: - Starting to supply the LOAD if the 5 V charger is used, - Stopping to supply the LOAD when the storage element voltage is too low.
V_{5V_IN}	Voltage on the 5V_IN pin.	V_{SRCLOW}	V_{SRCx} threshold below which the AEM13920 switches to SLEEP STATE , as described in Section 9.4.
V_{5V_IN,MIN}	Minimum voltage on the 5V_IN pin.	V_{SRCx}	Voltage on the SRCx pin.
V_{5V,STOP}	Voltage on STO at which the 5 V charger stops charging the storage element (see Section 5.7).		

$V_{\text{SRCx,CS}}$

Minimum **SRCx** voltage required for the AEM13920 to coldstart.

 $V_{\text{SRCx,REG}}$

Target regulation voltage of the source, depending on **SRCx_CFG[4:0]** configuration or I²C register (when **SRCx** regulation mode is constant voltage).

 V_{STO}

Voltage on the **STO** pin.

 V_{VDDIO}

Voltage on the **VDDIO** pin.

16. Revision History

Revision	Date	Description
1.0	April, 2023	Creation of the document.
1.1	August, 2023	Finished register descriptions. Added: - I²C protocol. - Efficiencies. - Behavior graphs. - Typical application circuits. - Glossary. - Quiescent currents.
1.2	September, 2023	Added: - Missing storage element voltage info on buck efficiency graph. - Boost efficiency graph: source current as graph title instead of x axis title. - Minimum BOM section. - Updated minimum cold-start power.
1.3	December, 2023	- Replaced "0/1" by "L/H" for configuration pins.
1.4	April, 2024	- Layout improvements. - Added precision about CTRL and IRQFLGx registers. - Fixed RDIV part number in "Minimum BOM" table. - Added PCB routing/layout guidelines. - Minimum BOM table: LBUCK unit "μH" instead of "μF". - Boost efficiency: source voltage noted "VSRC" instead of "VMPP". - Reordered register descriptions to regroup them by functionality. - Added cautionary statement for storage element threshold voltages. - Added C5V decoupling capacitor. - Fixed typo in I2C protocol description figure.
1.5	June, 2024	- BSTxCFG register configuration: added recommended inductor values. - Added cautionary statement along with storage element threshold voltages. - Added CSTO min. and typical value in "Recommended external components" table. - Behavior section: fixed typo in SRC1 configuration (VOVCH changed to VOC).

Table 80: Revision history (part 1)

Revision	Date	Description
1.6	November, 2024	- Fixed EVK part number on first page. - Updated application list on first page. - Updated minimum source regulation voltage to 0.120 V. - Renamed "Performance Data" section to "Typical Characteristics". - Created a "Specifications" section to place the characteristic and ratings sections in it. - Added operating temperature, storage temperature, and minimum input voltage values in "Absolute Maximum Ratings". - Added ESD ratings and thermal resistance θ_{JA} and θ_{JC} values. - In "Electrical Characteristics" table: - Updated V_{MPP} and $V_{SRCx,REG}$ maximum values. - Added I_{QOFF}. - Updated the quiescent currents. - Corrected $T_{MPPT,PERIOD}$ minimum and maximum values. - Added precisions about optional/mandatory components in "Recommended external components" table. - Added precisions about C_{SRCx} value range. - Updated minimum L_{BOOSTx} and L_{BUCK} values according to boost/buck timings. - Updated L_{BUCK} typical value to 10 μH in the "Recommended external components" table. - Aesthetic modifications on "AEM13920 state machine" figure. - Explained 5V_IN voltage minimum rise time and minimum value. - Added R5V resistor in figures, typical application circuits and 5 V charger sections. - Renamed "Source Voltage Regulation" sections to "Source Constant Voltage Regulation". - Added "Boost Converter Timings" and "Buck Converter Timings" sections in "System Configuration" to specify the timings default values. - Added explanation detailing that the I²C interface will be disabled if I²C pins are read LOW during SENSE STO STATE. - Fixed rounding errors on SRCx related voltages. - Rounded all source regulation voltages to 3 decimal places. - Renamed "sleep threshold" to "source low threshold" for disambiguation. - Changed "I²C Serial Interface Protocol", "Register Map" and "Registers Configuration" subsections to sections. - Corrected CTRL.SYNCBUSY bit to read only "R" instead of read and write in the "Register map" table. - SRCxREGUx.LVL register: lowest source regulation voltages lead to SRCx being considered as "source low". - SRCx register: fixed formula ranges as well as lookup table ranges. - Removed the ambiguous "interrupt" term about the IRQ pin. - Modified L_{BOOST_LP} to 15 μH instead of 10 μH in "Typical application circuit 2". - Added calculations for R5V, C5V and D5V to example circuits 1 and 3 sections. - Added Moisture Sensitivity Level, RoHS Compliance and REACH Compliance in "Package Information" section. - Updated package dimensions and added markings in "QFN 40-pin 5x5mm drawing" figure. - Updated board layout dimensions and added vias in "Recommended board layout for QFN40 package" figure.
1.7	December, 2024	Fixed various typos throughout the document.

Table 80: Revision history (part 2)



Revision	Date	Description
1.8	June, 2025	- Minor edits throughout the document (grammatical corrections, error fixes, and sentence improvements). - Updated timing values across the datasheet to reflect actual device behavior. No changes to device functionality or configuration. - Updated terminology for MPPT timing parameters throughout the document for improved clarity and consistency: - $T_{MPPT,SAMPLING}$ modified to $T_{MPPT,WAIT}$. - $T_{MPPT,MEASURE}$ added. - LOAD pin maximum voltage modified to 2.75 V in the absolute maximum ratings table. - In the electrical characteristics table: - Rounded $V_{SRCx,REG}$ minimum and maximum values with GPIO configuration to two decimal places. - Removed V_{STO} from $V_{SRCx,REG}$ maximum value with GPIO configuration. - Updated V_{MPP} maximum value to $0.85 \times V_{OC}$. - Updated V_{OC} maximum value to V_{STO}. - Removed $P_{SV_IN,MIN}$. - Updated V_{OVDIS}, V_{CHRDY}, and V_{OVCH} minimum and maximum values to match the configuration table values. - Removed V_{INT} minimum and maximum values. - Clarified storage element protection threshold configuration conditions when configured by I²C to ensure proper margin and avoid unstable state transition. - Renamed I_{QOFF} to $I_{Q,RESET}$. - Updated R_{ZMPP} maximum value. - Rounded up the values in the source constant voltage regulation configuration table. - Updated the storage element protection threshold values. - Added Average Power Monitoring (APM) explanations. - Improved the SRCLOW IRQ (IRQEN0.SRCLOW) explanation. - Updated the I2CRDY field of the IRQFLG0 register to 0x00. - Reworked register descriptions for improved clarity and readability. No functional changes. - Added tape and reel dimensions figure. - Glossary items listed alphabetically instead of by category.

Table 80: Revision history (part 3)

Revision	Date	Description
2.0	August, 2025	- Updated AEM13920 part number to 10AEM13920J0001. - Editorial improvements and clarifications throughout the document. - Buck conversion efficiency updated on first page. - Updated the SHIP_MODE pin enabling connection to STO in the shipping mode configuration section. - In the electrical characteristics table: - Updated V_{OC} and V_{MPP} maximum values respectively to 5 V and V_{STO}. - Updated $T_{STO,MON}$ values and conditions. - Updated T_{CRIT} description. - Updated V_{5V_IN} minimum value. - Added $T_{CRIT,ST}$, $I_{5V,OVDIS,CC}$, and $V_{5V,STOP}$. - Added $I_{Q,SLEEP}$ maximum value. - Updated C_{STO} typical value to 47 μF in the recommended external components table. - Updated the simplified schematic view of the buck converter to add M6 switch between STO and LOAD. - Improved buck converter section to explain LOAD output disabling based on overdischarged storage element of out of temperature range. - Updated conditions to go in SUPPLY STATE from START STATE if 5 V charger used. - 5 V charger behavior reworked (constant voltage mode disabled by default) and added 5 V charger configuration register CHG5V. - Added APM module for the 5 V charger (APMCHG5V and the corresponding APM enabling and data registers). - Added APM accumulator feature and the corresponding APMACC register. - Updated storage element thresholds configuration by $STO_CFG[2:0]$ and added associated temperature thresholds. - Reworked register descriptions for improved clarity and readability. - Thermal protection enabling register renamed from TMON to TEMPPROTECT. - Updated APM behavior (data measured over the full APM window) and APM.WINDOW register configuration modified to 116 ms and 233 ms. - Increased V_{CHRDY} configurable voltage range through VCHRDY register. - Updated the recommended L_{BOOSTx} and L_{BUCK} for each timings multiplier in the BSTxCFG and BUCKCFG register sections. - Updated APM data registers (APM2SRCx and APM2BUCK) on 8 bits: - COUNTER[22:16] in pulse counter mode without APM accumulator. - OFFSET[4:0] and COUNTER[18:16] in pulse counter mode with accumulator. - OFFSET[4:0] and POWER[18:16] in power meter mode. - Added part number registers (PNx). - Updated the boost timings multiplier configuration in the typical application circuit example 3. - Reworked the configurations and observations in the circuit behavior sections.
2.1	October, 2025	- Updated boost converter efficiency graphs. - VERSION register MAJOR and MINOR fields replaced by a single field named VERISON in the register map table. - Renamed the VOUT field of the BUCKCFG register to VLOAD. - Renamed the BUCKEN field of the APM register to LOADEN. - Renamed the SRCxSRCLOW field of the STATUS0 register to SRCxLOW. - Renamed the APMxBUCK registers to APMxLOAD. - Renamed the BUCKOV and BUCKNVLD fields of the APMERR register to LOADOV and LOADNVLD.

Table 80: Revision history (part 4)

Revision	Date	Description
2.2	October, 2025	- Updated buck converter efficiency graphs. - Reworked the APM data summary section and the 5 V charger APM formulas.
2.3	October, 2025	- Note about changing the configuration of LOAD_CFG[2:0] on-the-fly. - Note about switching off the buck converter through the BUCKCFG register.
2.4	December, 2025	- Minor edits throughout the document. - Updated the evaluation board part number on first page. - Updated the storage element types in the “Simplified schematic view” figure. - Removed R_{TH} minimum value. - Updated $I_{Q,SHIP}$ and $I_{Q,RESET}$ typical value. - Updated 5 V charger sections with clearer explanations of CC and CV behavior. - Updated P_{APM} formula for the APM power meter mode in the “APM Data Summary” section.
2.5	January, 2026	Minor editorial update in the Electrical Characteristics table.

Table 80: Revision history (part 5)